



ACADEMIC REGULATIONS & SYLLABUS 2025 -26

**Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute
of Computer Applications
Bachelor of Computer Applications (B.C.A.) Programme
(as per NEP-2020)**



Charotar University of Science and Technology (CHARUSAT)
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ACADEMIC RULES

To ensure uniform system of education, duration of under graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Bachelor's levels. Each semester will be at least 90 working days' duration. The Curriculum and Credit Framework for Undergraduate Programmes (CCFUP) is based on the University Grants Commission's (UGC) guidelines, with a student-centric focus, flexible choice-based credit system, and a multidisciplinary approach. This framework also provides multiple entry and exit options, facilitating students to align their education with their career aspirations by choosing subjects/fields of their interest. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Certification and Duration of Study with Multiple Entry Multiple Exit (MEME)

The undergraduate degree will have a flexible duration of either three or four years, with multiple exit options during this period. The table below outlines the various certifications a student can earn at different stages of their undergraduate study:

Table-1 Certification and Duration on MEME

Duration of Study	Semesters Completed	Certification Earned
4 Years	Eight Semesters (4 Academic Years)	Bachelor's Degree (Honours)
		Bachelor's Degree (Honours with Research)
3 Years	Six Semesters (3 Academic Years)	Bachelor's Degree
2 Years	Four Semesters (2 Academic Years)	Undergraduate Diploma
1 Year	Two Semesters (1 Academic Year)	Undergraduate Certificate

In consonance with the National Education Policy (NEP) 2020 and the guidelines of the University Grants Commission (UGC), Charotar University of Science and Technology (CHARUSAT) implements the Multiple Entry and Multiple Exit (MEME) scheme in their Bachelor of Computer Applications (BCA) programme. This structure provides students with the flexibility to enter, exit, and re-enter the programme at various stages, each with its corresponding certification mentioned above in Table-1.

Table 2: Multiple Entry Multiple Exit options

NCrF Credit Levels	Qualification Title	Credit Requirement	Programme Exit	Additional Requirement	Re-entry to Degree Programme
4.5	UG Certificate	48	After Year 1	Completion of one 4-credit of summer internship in core specific NSQF defined course during summer	Within three years of exit

				vacation of Year 1	
5	UG Diploma	92	After Year 2	Completion of one 4-credit of summer internship in core specific NSQF defined course during summer vacation of Year 2	Within three years of exit
5.5	Three Year Bachelor Degree	132	After Year 3	Compliance with minimum credit requirements specified in regulations	Within three years of exit
6	Bachelor's Degree with Honors	176	After Year 4	Compliance with minimum credit requirements specified in regulations	NA
6	Bachelor's Degree with Honors with Research	176	After Year 4	Compliance with credit requirements as specified in regulations	NA

Students who are opting for exit at any level, shall re-enter the institution to complete the UG Degree, where they had left off. They can re-enter within three years of exit and complete the degree programme within the stipulated maximum period of seven years from the date of admission to UG programme.

3. CREDIT FRAMEWORK FOR BCA (Bachelor of Computer Applications)

Arrangement of Credit Distribution Framework for three/four years Honors/Honors with Research Degree Programme with Multiple Entry and Exit Options.

(As per GR No: KCG/admin/2023-24/0607/kh.1, Sachivalaya, Gandhinagar, Date-11/07/2023)

Sr.No .	Categories of Courses	Credit Requirement for each Category				
		Certificate (1 Year)	Diploma (2 Years)	3-Year UG	4-Year UG (Honors)	4-Year UG (Honors+Research)
1	Major - Core Courses	16	40	64	88	88
2	Minor-Discipline Specific Electives	8	12	24	32	32
3	Multidisciplinary Courses Open Electives	8	10	12	12	12
4	Ability Enhancement Courses(AEC)	4	8	10	10	10
5	Skill Enhancement Courses(SEC)	4	10	14	14	14
6	Value Added Courses (VAC)	4	8	8	8	8
7	Summer Internship/ Research Project /Dissertation	-	-	-	12	12
8	Exit Courses	4	4	-	-	
	Total	48	92	132	176	176

4. Structure of BCA (Bachelor of Computer Applications)

NCrF Credit Level	Semester	Major Core	Minor	Multidisciplinary	Ability Enhancement Courses	Skill Enhancement Courses (SEC)	Value Added Courses(VAC)/IKS	Research Project/Dissertation	Total	Qualification/Certificate
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				lin ary	es(AE C)					
4.5 First Year	I	8	4	4	2	2	2	-	22	UG Certifica te
	II	8	4	4	2	2	2	-	22	
1 st Year Credit Total		16	8	8	4	4	4	-	44	
Exit 1: Award of UG certificate in Major course with 44 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level										
NCrF Credit Level	Sem ester	Majo r Core	Min or	Mu lti/ Int er Dis cip lin ary	Abilit y Enha ncem ents Cours es(AE C)	Skill Enhanc ement Courses (SEC)	Value Added Courses(V AC)/IKS	Research Project/Dis sertation	Tota l	Qualificatio n/Certificate
5.0 Secon d Year	III	12	-	2	2	4	2	-	22	UG Diploma
	IV	12	4	2	-	2	2	-	22	
2 nd Year Credit Total		24	4	6	2	6	4	-	44	
Grand Total 1 st and 2 nd Year Credit		40	12	12	6	10	8	-	88	
Exit 2: Award of UG Diploma in Major course with 88 credits with additional 4 credits of Summer Internship in core specific NSQF defined course OR continue with Major and Minor course for t h e next NCrF credit level										
NCrF Credit Level	Sem ester	Majo r Core	Min or	Mu lti/ Int er Dis cip lin ary	Abilit y Enha ncem ents Cours es(AE C)	Skill Enhanc ement Courses (SEC)	Value Added Courses(V AC)/IKS	Research Project/Dis sertation	Tota l	Qualificatio n/Certificate
5.5 Third Year	V	12	8	-	-	2	-	-	22	UG Degree
	VI	12	4	-	2	4	-	-	22	
3 rd Year Credit Total		24	12	-	2	6	-	-	44	
Grand Total 1 st to 3 rd Year Credit		64	24	12	08	16	8	-	132	
Exit 3: Award of UG Degree in Major course with 132 credits.										
NCrF Credit Level	Sem ester	Majo r Core	Min or	Mu lti/ Int er Dis cip lin ary	Abilit y Enha ncem ents Cours es(AE C)	Skill Enhanc ement Courses (SEC)	Value Added Courses(V AC)/IKS	Research Project/Dis sertation	Tota l	Qualificatio n/Certificate
6	VII	12	4	-	-	-	-	6(OJT)	22	UG

Fourth Year	VII I	12	4	-	-	-	-	6(OJT)	22	Honors Degree
4th Year Credit Total		24	8	-	-	-	-	12	44	
Award of UG Honors Degree in Major course with 176 credits.										
6 Fourth Year	VII I	12	4	-	-	-	-	6(RP)	22	UG Honors With Research Degree
	VII I	12	4	-	-	-	-	6(RP)	22	
4th Year Credit Total		24	8	-	-	-	-	12	44	
Grand Total 1st to 4th Year Credit		88	32	12	08	16	8	12	176	
Award of UG Honors with Research Degree in Major course with 176 credits.										
*OJT – On the Job Training * RP – Research Project With										
Major Core Courses Only										

2 credit course on Community Service/NSS/NCC/Sports as a compulsory course to be undertaken by students during the initial two years of the study. This course will not affect the overall SGPA/CGPA of the students.

5. Eligibility & Mode of Admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

6. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

7. Attendance

7.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few

sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

7.2 Student attendance in a course should be 80%.

8. Course Evaluation

8.1 The performance of every student in each course will be evaluated as follows:

8.1.1 Continuous and Comprehensive Evaluation using various components such as Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc. by the course faculty member(s). The weightage of these components is 50% of the marks for the course; and

8.1.2 Semester-End-Evaluation by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 50% of the marks for the course.

8.2 University Examination

8.2.1 The final examination by the University for 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

8.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

8.2.3 Performance of Continuous and Comprehensive Evaluation(CCE) and Semester-End-Examination(SEE) Examination will be done on the grading system.

9. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Grading Scheme Range of Marks (%)	96.0-100	86.0-95.9	76.0-85.9	66.0-75.9	56.0-65.9	46.0 – 55.9	36.0 – 45.9	Below 36.0	Absent
Letter Grade	O (Outstanding)	A+ (Excellent)	A (Very Good)	B+ (Good)	B (Above Average)	C (Average)	P (Pass)	F (Fail)	Ab (Absent)
Grade Point	10	9	8	7	6	5	4	0	0

$SGPA = \sum CiGi / \sum Ci$ where Ci is the number of credits of course i Gi is the Grade Point for the course i and $i = 1$ to n , n = number of courses in the semester.

$CGPA = \sum CiGi / \sum Ci$ where Ci is the number of credits of course i Gi is the Grade Point for the course i and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

The minimum passing marks for each pattern of evaluation are 36% .

10. Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in. Awards of Degree.

10.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

10.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

10.1.2 He should have cleared all evaluation components in every course;

11. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥ 7.0 & ≤ 10.0
First class:	CGPA ≥ 6.0 & < 7.0
Second Class:	CGPA ≥ 5.0 & < 6.0
Pass Class:	CGPA ≥ 3.60 & < 5.0

12. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Choice of Courses

Undergraduate students will make course selections from the various lists and that will fall into categories as outlined below:

Major Course:

This is the primary subject area a student wishes to delve into during their undergraduate course. It is the area of study that the student is most passionate about and may wish to pursue as a career.

Minor Course:

This is a secondary area of study that complements the student's chosen Major. Students who complete a sufficient number of courses in a discipline or an interdisciplinary area of study other than their chosen Major will qualify for a Minor in that discipline or interdisciplinary area.

Multidisciplinary Course:

All undergraduate students are required to complete three introductory-level courses relating to any of the broad disciplines. These courses aim to broaden the student's intellectual experience and form part of liberal arts and science education.

Ability Enhancement Course (AEC):

These are Modern Indian Language (MIL) & English language courses focused on enhancing language and communication skills. The courses aim to enable students to acquire and demonstrate core linguistic skills, including critical reading and expository and academic writing skills. The courses also emphasize the development and enhancement of skills such as communication, and the ability to participate in/conduct discussion and debate.

Skill Enhancement Courses (SEC):

These courses aim to impart practical skills, hands-on training, soft skills, etc., to enhance the employability of students. The institution may design courses as per the students' needs and available institutional resources.

Value Added Course (VAC):

These courses are designed to provide extra skills or knowledge beyond the standard curriculum, often tailored towards enhancing employability, promoting entrepreneurship, or developing personal and professional skills.

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary

approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates

proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses (PE):

The programme specific pool of elective courses offered by respective programme.

Credit Hour Allocation for Different Course Types

Definition and Standardization

The workload associated with a course will be measured in terms of credit hours. One credit hour is equivalent to one hour of instructional time per week over the duration of a semester (minimum 15 weeks).

Types of Course Components

Courses can be comprised of one or more of the following components:

1. Lecture
2. Practicum
3. Seminar/ Internship/ Studio Activities/ Field / Practice /Projects / Community Engagement and Service

Credit Hours per Course Component

The weekly and semester-wise instructional hours associated with each credit for different components are as follows:

Credit Hours per Course Component

Course Component	Weekly Hours per Credit	Total Hours in a 15-week Semester
Lecture	1 Hour	15 Hours
Practicum	2 Hours	30 Hours
Internship, Field Practice/Projects, Community Engagement and Service/NCC/NSS/Sports	2 Hours	30 Hours

Course Categories

Courses in the study program are categorised based on the nature of the learning activities. These include:

Lecture courses: These involve expert-led sessions related to a specific field.

Practicum or Laboratory work: These apply previously learned principles/theories in practical projects or lab activities.

Internship: These involve professional activities or work experiences, typically supervised by experts from the relevant external entities.

Field practice/projects: These involve field-based learning or projects supervised by external experts.

Community engagement and service/NCC/NSS/Sports: These involve activities exposing students to societal socio-economic issues, facilitating the application of theoretical and practical knowledge to real-life problems. It also involves skills and physical fitness among students through indoor & outdoor sports, field & track events.

Vision, Mission, PEOs, POs and PSOs

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will:

PEO1: Be able to understand the requirement of computing problem and implement an effective solution.

PEO2: Be able to successfully take up various available career options.

PEO3: Be able to continuously learn in their preferred domains.

PEO4: Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to:

PO1: Computational Knowledge: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge to provide effective solution in the area of computing.

PO2: Problem Analysis: Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

PO3: Design /Development of Solutions: Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.

PO4: Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.

PO5: Professional Ethics: Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

PO6: Life-long Learning: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

PO7: Project management: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO8: Communication Efficacy: Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

PO9: Societal and Environmental Concern: Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.

P10: Individual and Team Work: Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments.

P11: Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

PROGRAM SPECIFIC OUTCOMES

At the end of the programme, the student should be able to

PSO1: Analyze, evaluate and formulate workable computing solutions to real world problems using computing methods.

PSO2: Apply proficiencies of communication, teamwork and management skills with spirit of entrepreneurship.

SALIENT FEATURES

- Four Years unique programme as per National Education Policy-2020.
- Have a flexible duration of either three or four years, with multiple exit options during this period.
- Provides students with the flexibility to enter, exit, and re-enter the programme at various stages, each with its corresponding certification.
- Award of Degree:
 - **UG Certificate** (Candidate wish to Exit after the First Year with additional requirements of 4 credits summer internship)
 - **UG Diploma** (Candidate wish to Exit after the Second Year additional requirements of 4 credits summer internship)
 - **UG Degree** (Candidate wish to Exit after the Third Year)
 - **UG Honors Degree or UG Honors with Research Degree** (Candidate finishes all Four Years)
- Programme is designed with various categories of courses such as Major, Minor, Multidisciplinary, Ability Enhancement, Skill Enhancement, Value Added, Community engagement and service/NCC/NSS/Sports based on Choice Based Credit System(CBCS).
- Focus on student-centered learning.
- Problem-based learning and industry-relevant curriculum.
- Industry visits, seminars, workshops, and guest lectures by eminent researchers and industry practitioners.
- Active student clubs and bodies.
- The multi-cultural and vibrant teaching environment.

Total Credits of Programme (8 Semesters): 186

TEACHING SCHEME AND EXAMINATION SCHEME

FOR

**BCA PROGRAMME
AS PER NEP-2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-I

Course Code	Course Name	Theory/ Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP - 2020 Classification
			CCE	SEE	CCE	SEE		
CAUC101	Foundation of Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC102	Fundamentals of Web Designing	Practical	-	-	25	25	02	Major Core
CAUE101 CAUE102 CAME107 CAME108	(Select any one) 1. Computer Essentials and Information Technology 2. Business Organization and Principles of Management 3. Privacy and Security in Online Social Media 4. Computer Architecture and Organization	Theory	50	50	-	-	04	Minor
CAUD101	Digital Electronics	Theory	50	50	-	-	04	MDC
CAUS101	Mathematics for Computer Science	Theory	25	25			02	SEC
CLUV101	Environmental Sciences	Practical	-	-	25	25	02	VAC
HSUA101	Communicative English	Practical	-	-	25	25	02	AEC
CUUV101 CUUV102	(Select any one) Community Engagement and Sustainable Development Physical Education and Sports	Practical	-	-	25	25	02	VAC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

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BCA Semester-II

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CAUC103	Fundamentals of Object Oriented Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC104	Database Fundamentals	Practical	-	-	25	25	02	Major Core
CAUE103 CAUE104 CAME103 CAME104	(Select any one) 1. Fundamentals of E-Commerce 2. Business Economics 3. E-Business 4. Business Development: From Start to Scale	Theory	50	50	-	-	04	Minor
CAUD102	Data Communications and Networks	Theory	50	50	-	-	04	MDC
CAUV101	Fundamentals of Accounting	Theory	25	25	-	-	02	VAC
HSUS101-117	(Select any one) A course on Liberal Arts HSUS101 – Painting HSUS102– Photography HSUS103– Sculpting HSUS104– Pottery and Ceramic Art HSUS105-Media and Graphic Design HSUS106-Art and Craft HSUS109- Dramatics HSUS110-Contemporary Dance HSUS111-Music (Vocal) HSUS112-Music (Instrumental)-Tabla HSUS113-Music (Instrumental)-Guitar HSUS114-Music (Instrumental)-Harmonium HSUS115-Music (Instrumental)- Flute HSUS116- Indian Classical Dance-Kathak HSUS117- Indian Classical Dance-Bharatnatyam	Practical	-	-	25	25	02	SEC
CAUA101	Presentation Skills	Practical	-	-	25	25	02	AEC
CAUI101	Summer Internship and Viva - I	VIVA	-	-	50	50	04	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

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BCA Semester-III

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SSE		
CAUC201	Fundamentals of Data Structures and Algorithms	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC202	Fundamentals of Operating Systems	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUS201 CAUS202	(Select any one) 1. Computer Oriented Management System 2. Computer Oriented Numerical Methods	Theory	50	50	-	-	04	SEC
CAUA201	Life Management	Theory	25	25	-	-	02	AEC
	University Elective-I	Practical	-	-	25	25	02	MDC
HSUV201	Creativity, Problem Solving and Innovation	Practical	-	-	25	25	02	VAC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

University Elective - I			
Sr. No.	Course Code	Course Name	Department/Faculty
1	MEUE201	Engineering Graphics and Design	Engineering
2	EEUE201	Fundamentals Of Electrical Engineering	Engineering
3	CLUE201	Environment and Development	Engineering
4	ECUE201	Scientific Computing Using Matlab	Engineering
5	CEUE201	The Joy of Computing using Python	Engineering
6	ITUE201	Introduction to Quanta Computing	Engineering
7	CSUE201	Python For Data Science	Engineering
8	AIUE201	An Introduction to Artificial Intelligence	Engineering
9	BMUD201	Money And Banking	Management
10	CAUD203	Human Computer Interactions	Computer Science
11	PTUD191	Basics of Health Promotion and Education Intervention	Physiotherapy
12	NRMD251	First Aid Masterclass- A complete guide to first aid	Nursing

BCA Semester-IV

Course Code	Course Name	Theory/ Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SSE		
CAUC203	System Analysis and Design	Theory	50	50	-	-	04	Major Core
CAUC204	Programming the Internet	Practical	-	-	25	25	02	Major Core
CAUC205	Open Source Technology	Theory and Practical	50	50	25	25	06 (04-Theory) (02- Practical)	Major Core
CAUE201 CAUE202	(Select any one) 1. Advanced Networking 2. Artificial Intelligence (AI) Fundamentals	Theory	50	50	-	-	04	Minor
CAUA202	Social Media and Blog Writing	Practical	-	-	25	25	02	AEC
	University Elective-II	Practical	-	-	25	25	02	MDC
HSUV203	Indian Knowledge System	Practical	-	-	25	25	02	VAC
CAUI201	Summer Internship and Viva - II	VIVA	-	-	50	50	04	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

University Elective - II			
Sr. No.	Course Code	Course Name	Department/Faculty
1	MEUE202	Nature and Properties of Materials	Engineering
2	EEUE202	Solar Energy Engineering and Technology	Engineering
3	CLUE202	Ecology and Environment	Engineering
4	ECUE202	Introduction to Internet of Things	Engineering
5	CEUE202	Software Conceptual Design	Engineering
6	ITUE202	Ethical Hacking	Engineering
7	CSUE202	Google Cloud Computing Foundations	Engineering
8	AIUE202	Social Network Analysis	Engineering
9	BMUD251	Economics Of Health And Health Care	Management
10	CAUD204	Modern Application Development	Computer Science
11	PTUD192	Ergonomics Workplace Analysis	Physiotherapy
12	NRMD261	Mindfulness and Well-being: Living with Balance and Ease	Nursing

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BCA Semester-V

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP - 2020 Classification
			CCE	SEE	CCE	SEE		
CAUC301	Object Oriented Programming Using JAVA	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC302	Fundamentals of Visual Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE301	Software Engineering	Theory	50	50	-	-	04	Minor
CAUE302	Python Programming	Practical	-	-	25	25	02	Minor
CAUE303 CAUE304	(Select any one) 1. Introduction to Data warehousing 2. Geographical Information System	Theory	25	25	-	-	02	Minor
HSUS301	French	Practical	-	-	25	25	02	SEC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

BCA Semester-VI

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CC E	SEE	CCE	SSE		
CAUC303	Project Work	VIVA	150	150	-	-	12	Major Core
CAUE305 CAUE306	(Select any one) 1. Introduction to Data Science 2. Fundamentals of Digital Image Processing	Theory	50	50	-	-	04	Minor
CAUS301	Mobile Application Development	Theory and Practical	25	25	25	25	04 (02-Theory) (02-Practical)	SEC
HSUA302	Professional Communication, Soft Skills and Personality Development	Practical	-	-	25	25	02	AEC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

BCA Semester-VII

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP - 2020 Classification
			CCE	SEE	CCE	SEE		
CAUC401	Enterprise Computing using Java EE	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC402	Advanced Database Technologies	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE401 CAUE402 CAUE403	(Select any one) 1. Cloud Computing 2. Blockchain Essentials 3. Cryptography and Network Security	Theory	50	50	-	-	04	Minor
CAUP401 CAUR401	(Select any one) 1. OnJob Training-I 2. Research Project-I	VIVA	-	-	75	75	06	Research Project / Dissertation
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

BCA Semester-VIII

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CAUC403	Full Stack Web Development	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC404	Advanced Mobile Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE404	Web Development Using Open-Source Technologies	Practical	-	-	25	25	02	Minor
CAUE405 CAUE406 CAUE407	(Select any one) 1. Python Framework 2. HTTP Web service for Enterprise Applications 3. Game Development using Unity	Practical	-	-	25	25	02	Minor
CAUP402 CAUR402	(Select any one) 1. On Job Training-II 2. Research Project-II	VIVA	-	-	75	75	06	Research Project/Dissertation
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								



DETAILED SYLLABUS

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications

BCA PROGRAMME
AS PER NEP-2020



TEACHING SCHEME & DETAILED SYLLABUS

FOR

**BCA PROGRAMME
(1st SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

BCA Semester-I

Course Code	Course Name	Theory/ Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP - 2020 Classifica tion
			CCE	SEE	CCE	SEE		
CAUC101	Foundation of Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02- Practical)	Major Core
CAUC102	Fundamentals of Web Designing	Practical	-	-	25	25	02	Major Core
CAUE101 CAUE102 CAME107 CAME108	(Select any one) 1. Computer Essentials and Information Technology 2. Business Organization and Principles of Management 3. Privacy and Security in Online Social Media 4. Computer Architecture and Organization	Theory	50	50	-	-	04	Minor
CAUD101	Digital Electronics	Theory	50	50	-	-	04	MDC
CAUS101	Mathematics for Computer Science	Theory	25	25			02	SEC
CLUV101	Environmental Sciences	Practical	-	-	25	25	02	VAC
HSUA101	Communicative English	Practical	-	-	25	25	02	AEC
CUUV101 CUUV102	(Select any one) Community Engagement and Sustainable Development Physical Education and Sports	Practical	-	-	25	25	02	VAC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CAUC101: Foundation of Programming

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA , certification courses, development of projects etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: In theory lectures, students will develop problem-solving skills through algorithms and flowcharts, understanding C programming's structure to create basic programs. The curriculum covers conditional and iterative statements, array manipulation, and user-defined functions, with advanced topics like pointers and structures. Emphasis is placed on hands-on experience, focusing on algorithmic thinking and creating flowcharts. Students will learn to write effective C code, mastering variables, input/output operations, and advanced concepts, promoting independent learning and real-world application.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Programming Basics	08
2	Introduction to C	12
3	Expressions and Control Statements	10
4	Arrays and Strings	10
5	Introduction to Functions	10
6	Pointers & Basic of the structures	10

Total Hours (Theory): 60
Total hours (Practical): 60
Total Contact Hours:120

Detail Syllabus:

Unit I: Programming Basics

Hours 08

Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages and limitation of flow chart, Introduction to programming languages, Introduction to editor, compiler, and translator.

Unit II: Introduction to C

Hours 12

Introduction, characteristics of C, Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program. Basic data types in C, User define data types, Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C, Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise, comma operator, size-of operator, operator precedence chart.

Unit III: Expressions and Control Statements

Hours 10

Arithmetic expressions, evaluation of expressions, type conversions in expressions, operator precedence and associativity, mathematical functions. Type conversion and casting; Introduction to decision control statements, Conditional and branching statements, loop, nested loop, Break, continue and go to statement.

Unit IV: Arrays and Strings

Hours 10

Arrays: One-dimensional, two-dimensional, Handling of Character Strings: Declaring and initializing string variables, reading string from terminal, writing string to screen, string handling functions, table of strings.

Unit V: Introduction to Functions

Hours 10

Need for user-defined functions, the form of c function, return values and their types, calling a function, category of functions, handling of non-integer functions, nesting of Functions, recursion, functions with arrays.

Unit VI Pointers & Basic of the structures

Hours 10

Understanding the Computer's Memory, Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers, Pointers and Arrays, passing an Array to

a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers, defining a structure, declaring structure variables, accessing structure members, structure initialization.

Outline of the Course:

Week No	Practical	Description
1	Introduction to Programming and IDE	Overview of programming languages, algorithms, flowchart, Setting up the development environment, Introduction of development environment.
2-3	Basic of C programming	Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program; Basic data types in C. Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C
4-5	Operators, Expressions evaluation & type conversions.	Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise and special, comma operator, size-of operator, operator precedence chart. Arithmetic expressions, evaluation of expressions, type conversions.
6-7	Decision-making statements and looping statements	Decision-making statements (if, else if, else) and nested decision-making statements, Switch statement, Looping statements (for, while, do-while), nested looping statements Break and continue statements.
8-9	Arrays	Understanding arrays, one-dimensional array, two-dimensional arrays, practical exercises of one-dimensional and two-dimensional array.
10-11	String and String functions	Declaring and initializing string variables, reading string from terminal, writing string to screen, string-handling functions.
12-13	User defined functions	Introduction to functions, Function declaration, definition, and calling, Parameters and return values. Scope and lifetime of variables, Recursion.
14-15	Pointers & Structures	Introduction to pointers, Pointers and arrays, Pointer arithmetic, Defining and declaring structures, Accessing structure members

Total hours (Practical): 60

Core Books:

1. ReemaThareja: Computer Fundamentals and Programming in C, 2nd Edition, Oxford University press, 2017
2. PradipDey and Manas Ghosh. "Programming in C", Second Edition, Oxford University Press.
3. E. Balagurusamy: Programming in C, 7th Edition.

Reference Books:

1. Brian W. Kerighan and Dennis M. Ritchie : The C Programming Language, Prentice Hall.
2. PradipDey, Manas Ghosh: Programming in C, 4th Edition, Oxford University press.
3. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill
4. YashwantKanetkar: Let us C, 13th Edition, BPB publication.

Web References:

1. <https://cprogrampracticals.blogspot.com/p/basic-concepts.html> [For basic c programming practical]
2. <https://www.programtopia.net/c-programming/docs/operators-expressions> [For operators and expressions in C]
3. <https://www.javatpoint.com/functions-in-c> [For functions]
4. <https://www.programiz.com/c-programming> [For basic c programming practical]
5. <https://www.javatpoint.com/c-pointers> [pointers in c]
6. <https://www.programiz.com/c-programming/c-structures> [structures in c]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be able to solve problems using algorithms and flowcharts.
CO2 :	Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants.
CO3 :	Gain the knowledge of different conditional and iterative statements and expressions.
CO4 :	Explore the utilization and manipulation of numbers and characters arrays both.
CO5 :	Be able to understand user define function. Also, learn Programing-using Array with Pointer and Structure.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Programming Basics	√				
2	Introduction to C		√			
3	Expressions and Control Statements			√		
4	Arrays and Strings		√	√	√	
5	Introduction to Functions		√	√		√
6	Pointers & Basic of the structures		√			√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC102: Fundamentals of Web Designing

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic concepts of web.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for design and develop entire web sites using several web designing tools and HTML scripting language.

Outline of the Course:

Week No	Practical	Description
1	Basic Construction of an HTML & HTML5 Page, HTML Text Editors, HTML Building blocks.	Learn basic concepts of web page design.
2	Basic HTML tags – Text formatting tags, Heading tags, paragraph tags, division tag, Span tags, Break line tag, Binding space in HTML.	Get familiar with different tags and attributes which is used to add the text in a web page.
3	Controlling Font size and color, Adding Table – Table tag.	Learn user interface designing using Table Layout, change font style, color etc.
4	Grouping and merging table rows and columns. Adding List – ordered, unordered and definition list.	Able to display the data in various formats and design different types of list in the table.

5	Adding Image, Adding links - Changing link colors, Email link.	Learn to add images and link in the web page.
6	Embedding audio and video, Adding Field sets.	Able to add multimedia in web page using internal and external methods.
7	Adding Form: Form tag, Action, Method. Basic form controls.	Learn basic form elements and when and where to apply it.
8	Textbox, Password field, Radio Button, Checkbox, Multiple line text area, file upload.	Use various types basic form controls with types of form input elements.
9	Various types of buttons – submit, reset, image, simple button.	Learn various types of buttons to make it more interactive.
10	HTML5 input elements - Color, Number, Date, Month.	Learn new input types in HTML5.
11	Week, Time, date time-local, Email, url, range, search.	Learn to add defining inputs and minimum and maximum range to make the form attractive.
12	Data list, output, progress, meter.	Learn to add list and provide progress horizontal view and range with output.
13	Regular expression, HTML5 new attributes – Placeholder, Required.	Learn HTML 5 form element to provide user side web form validation.
14	Pattern, Autocomplete, Autofocus, novalidate, form novalidate.	Learn HTML 5 new form properties.
15	Form action, form method, spellcheck, content editable.	Learn HTML 5 to user submit the form and redirect the form data to selective method.

Total hours: 60

Core Books:

1. Powel Thomas: HTML - The Complete Reference, 3rd Edition, McGraw-Hill Education – Europe, 2018.
2. DT Editorial Services: HTML 5 Black Book, 2nd Edition, Dreamtech Press, 2016
3. Matt West: HTML 5 Foundations, Wiley publication, 2013
4. Farrar: HTML Example book, BPB, 2007.

Reference Books:

1. David DuRocher : HTML and CSS QuickStart Guide, Paperback, 2021.
2. Laura Lemay, Rafe Colburn, Jennifer Kyrnin : Mastering Html, CSS & JavaScript Web Publishing, Paperback, 2016.
3. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
4. Shelly Woods: HTML introductory concepts and techniques, 5th Edition, Course Technology, 2009.
5. Jon Duket: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.

Web References:

1. www.w3schools.com/html [For HTML tutorials]
2. html.net/tutorials/html/ [For HTML tutorials]
3. www.htmlgoodies.com/ [For HTML resources]
4. <https://developer.mozilla.org/en-US/learn/html> [For HTML best practices]

Course Outcome: Upon successful completion of the course, student will:

CO1:	Able to understand basic construction of HTML & HTML5 document and started using basic tags.
CO2:	Able to Add content using various tags and format the content in table & list.
CO3:	Able to design links and multimedia elements such as image, audio, video.
CO4:	Able to design a web page form using HTML & HTML5.
CO5:	Able to apply validation using HTML5 attributes.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	1	3	2	2	1	1	-	1	1	1	1
CO2	-	1	2	3	1	3	1	-	-	2	2	2	2
CO3	1	1	2	3	1	3	1	-	-	2	2	2	2
CO4	1	1	2	3	1	1	1	1	-	1	-	1	1
CO5	2	2	3	3	2	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE101: Computer Essentials and Information Technology

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: Operating System Concepts and Network Technology

Methodology, Pedagogy & Andragogy: During lectures covering these topics, students will: Understand computing's historical evolution and key figures. Grasp computer organization, architecture, and processor functions. Learn storage technologies and memory management for optimization. Acquire basics of input/output devices for computer interaction. Explore Internet and Web fundamentals and emerging trends. Facilitate discussions, integrate real-world examples, and encourage self-directed exploration of advanced computing concepts. Adapt teaching strategies to cater to diverse learning styles and levels of expertise in the rapidly evolving field of computing.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Computers: History and Evolution of Computing	10
2	Understanding of Computer Organization, Architecture and Processor	10
3	Computers Storage and Memory Management	11
4	Fundamentals of Input and Output Devices	11
5	Concepts of Internet and World Wide Web	10
6	Emerging Fundamentals of Computing Environment	08

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to Computers: History and Evolution of Computing Hours 10

Introduction to computer, Simple model of a computer, Computer Hardware, Computer Software, Characteristics of computers, Evolution of computers, Generation of computers, Classification of computers, computing concepts, The computer System, Applications of computers.

Unit II: Understanding of Computer organization, Architecture and Processor Hours 10

Introduction to computer organization, Central Processing Unit, Introduction to processor, Internal communications, Inter connection of units, Processor to memory communication, I/O Devices to processor communication, The Bus, Instruction set.

Unit III: Computers Storage and Memory Management Hours 11

Introduction to computer storage, Memory cell, Memory organization and Memory Representation, Random access memory, Read only memory, Storage system, Magnetic storage system, Optical storage system, Magneto optical system, Solid-state storage system, Storage evaluation criteria, Memory Hierarchy.

Unit IV: Fundamentals of Input and Output Devices Hours 11

Introduction to Input device, Traditional computer I/O devices: Keyboard, Computer mouse, Display unit. Other Input devices: Touch pad, Touch Screen, Magnetic Ink Character Recognition (MICR), Optical Mark Reading and Recognition (OMR), Flatbed Scanner, Optical Character Recognizer, Bar Code. Computer Output Devices: Flat Panel Display Technology, E-ink Display, Printers, Inkjet Printers, Laser Printers, Dot Matrix Printers, Line Printers, Plotters. Digital Camera, Voice Recognition System.

Unit V: Concepts of Internet and World Wide Web Hours 10

Introduction to Internet, History of Internet, Internet applications, Understanding the World Wide Web, Browsing Internet, Using a Search Engine, Email Service. Web Concepts: Web Browser, HTML, Web Page, Website, Web Server, ISP, Hyperlink, DNS, URL. Internet Protocols: TCP/IP, SMTP, PPP, FTP, SFTP, HTTP and HTTPS, TELNET, POP3, IPV4, IPV6, ICMP, UDP, IMAP, SSH, Gopher.

Unit VI: Emerging Fundamentals of Computing Environment Hours 08

Overview of computing environment, Current computing scenario, Types of computing environment: Personal computing environment, Time-sharing computing environment, Client-Server computing environment, Distributed computing environment, Grid

computing environment, Cloud computing environment, Cluster computing environment. Advantages and Disadvantage of various computing environment.

Core Books:

1. E Balagurusamy: Fundamentals of Computers: Tata McGraw Hill Education Private Limited
2. V.Rajaraman, Neeharika Adabala: Fundamentals of Computers: PHI, 2015.
3. Margaret Levine Young: The Complete Reference- Internet: Tata McGraw Hill Education Private Limited

Reference Books:

1. Computer Fundamentals: Pradeep K. Sinha & Priti Sinha" by Pradeep K. Sinha and Priti Sinha
2. Computers Made Easy: From Dummy to Geek" by James Bernstein
3. Computer Organization and Design: The Hardware/Software Interface" by David A. Patterson and John L. Hennessy

Web References:

1. <https://artoftesting.com/computer-fundamentals-tutorial>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will gain an understanding of the historical progression and evolutionary milestones in computing.
CO2 :	Students will comprehend computer organization, architecture, and processor functionalities.
CO3 :	Students will learn about computer storage mechanisms and memory management techniques.
CO4 :	Students will grasp the fundamentals of various input and output devices utilized in computing.
CO5 :	Students will acquire knowledge about the concepts underlying the Internet and the World Wide Web. Explore the Emerging trends and principles shaping the contemporary computing environment.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Computers: History and Evolution of Computing	√				
2	Understanding of Computer Organization, Architecture and Processor		√			
3	Computers Storage and Memory Management			√		
4	Fundamentals of Input and Output Devices				√	
5	Concepts of Internet and World Wide Web					√
6	Emerging Fundamentals of Computing Environment					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	2	1	-	3	3	2	2	-	-	1	-
CO2	3	1	1	-	3	1	3	1	2	2	1	3	2	-
CO3	3	2	1	2	1	3	3	2	2	2	-	3	2	-
CO4	3	3	2	3	1	3	3	2	2	2	3	-	1	-
CO5	3	3	3	3	1	3	3	3	2	2	2	-	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE102 : Business Organization and Principles of Management

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of business organization and principles of management. Students will be introduced to business ownership, business services, emerging models of business, business ethics and its social responsibilities, and entrepreneurship development. Students will give practical exposure in form of case study and by visit to Entrepreneurship and Innovation Centre.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Forms of Business Ownership	09
2	Business Services	11
3	Emerging Modes of Business	09
4	Principles of Management	11
5	Business Ethics and its Social Responsibilities	09
6	Entrepreneurship Development	11

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I : Forms of Business Ownership

Hours 09

Sole proprietorship, Joint Hindu Family Business – meaning, features, merits and demerits, Partnership – meaning, types, registration, types of Partners, Co – Operative societies – types, merits and limitations, Company – Private Ltd, Public Ltd – merits, limitations, Starting a business – Basic factors, Choice of forms of business organizations.

Unit II : Business Services

Hours 11

Nature and types of Business services – Banking, Insurance, Transportation, Warehousing, communication. Meaning, role, Functions of Banking, Insurance, Postal and Telecom services, Warehousing, Transport, Hospitality and Tourism Services.

Unit III : Emerging Modes of Business

Hours 09

E – business – Meaning, Scope and benefits, Resource required for successful E – Business implementation, On – line transactions, payment mechanism, Security and safety of business transactions, Outsourcing – Concept, need and scope.

Unit IV : Principles of Management

Hours 11

Definition and nature of management, purpose of management, managerial functions at different levels, management- art or science, planning, planning process & purpose, steps of planning, goal setting, decision making, organizing function, organizing process, formal and informal organizations, centralization and decentralization organization, staffing function, staffing process, recruitment, selection, training, performance appraisal, directing function, concepts of leading, motivating, communicating, qualities of leaders, controlling, control process, control techniques.

Unit V : Business Ethics and its Social Responsibilities

Hours 09

Concept of social responsibility, cases for social responsibility, responsibility towards different interest groups, owners, investors, employees, consumers, government, community, public in general, business ethics – concept and elements, business and environmental protection.

Unit VI : Entrepreneurship Development

Hours 11

The concept and introduction, personality and mindset of an entrepreneur, difference between an entrepreneur, intrapreneur and manager/executive, entrepreneurial ecosystem, types and functions of an entrepreneur- entrepreneurial motivation, entrepreneurship development programs, business idea generation, business plan and detailed project report, funding and support aspects.

Core Books:

1. Harold Koontz & Heinz Weihrich: Essentials of Management, Ninth Edition, Tata McGraw Hill Publishing, 2012.
2. Dr. F. C. Sharma, Dr. D. Chandra, Anju Agarwal: Business Organization, SBPD Publications, 2020.

Reference Books:

1. Tripathy & Reddy: Principles of Management, fifth edition, Tata McGraw Hill Publishing, 2015.
2. Robbins , Decenzo & Coulter: Fundamentals of Management, eighth edition, Pearson Education, 2013.
3. Y.K. Bhushan: Fundamentals of Business Organisation, Sultan Chand Publication, 2018.

Web References:

1. <https://www.dynamictutorialsandservices.org/2018/10/entrepreneurship-developmentnotes.html> [Entrepreneurship Development]
2. <https://leverageedu.com/blog/emerging-modes-of-business/> [Emerging Modes of Business]
3. https://www.managementstudyguide.com/management_functions.htm [Functions of Management]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will be able to understand different types of business proprietorship.
CO2 :	Students will get the knowledge of functionality of various online business services
CO3 :	Students will learn the important concepts of management.
CO4 :	To be able to understand the ethics and social responsibilities of business towards various entities
CO5 :	Students will learn the concepts required to be a good entrepreneur.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Forms of Business Ownership	√				
2	Business Services		√			
3	Emerging Modes of Business		√			
4	Principles of Management			√		
5	Business Ethics and its Social Responsibilities				√	
6	Entrepreneurship Development					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	1	3	-	-	-	2	2	-	1
CO2	-	2	-	2	1	3	2	-	-	2	2	-	2
CO3	-	2	-	-	1	3	-	2	1	3	2	-	2
CO4	-	-	-	-	2	2	-	-	3	3	-	-	2
CO5	-	-	-	-	-	3	-	-	-	2	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAME107: Privacy and Security in Online Social Media

Description:

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

With increase in the usage of the Internet, there has been an exponential increase in the use of online social media and networks on the Internet. Websites like Facebook, YouTube, LinkedIn, Twitter, Flickr, Instagram, Google+, FourSquare, Pinterest, Tinder, and the likes have changed the way the Internet is being used. However, widely used, there is a lack of understanding of privacy and security issues on online social media. Privacy and security of online social media need to be investigated, studied and characterized from various perspectives (computational, cultural, psychological, etc.). Student completing the course will be able to appreciate various privacy and security concerns (spam, phishing, fraud nodes, identity theft) on Online Social Media and Student will be able to clearly articulate one or two concerns comprehensively on one Online Social Media, this will be achieved by homework.

Pre-requisites:

Basic / Intermediate programming course. Understanding of Python will be necessary for the course. Should be able to quickly learn APIs, and to collect data from social networks.

Industry support:

Any company which is interested in social media / networks data will be interested in recruiting the students finishing the course.

Course layout

Week 1: What is Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs .

Week 2: Collecting data from Online Social Media.

Week 3: Trust, credibility, and reputations in social systems.

Week 4: Trust, credibility, and reputations in social systems.

Week 5: Online social Media and Policing.

Week 6: Information privacy disclosure, revelation and its effects in OSM and online social networks.

Week 7: Phishing in OSM & Identifying fraudulent entities in online social networks.

Week 8: Refresher for all topics.

Week 9 to 12: Research paper discussion.

Books and references

1. Michael E. Whitman, Herbert J. Mattord, (2018). Principles of Information Security, 6th edition, Cenage Learning, N. Delhi.
2. Darktrace, "Technology"
<https://www.darktrace.com/en/technology/#machine-learning>, accessed November 2018.
3. Van Kessel, P. Is cyber security about more than protection? EY Global Information Security Survey 2018-2019.
4. Johnston, A.C. and Warkentin, M. Fear appeals and information security behaviors: An empirical study. MIS Quarterly, 2010.
5. Arce I. et al. Avoiding the top 10 software security design flaws. IEEE Computer Society Center for Secure Design (CSD), 2014.
6. Smith, H. J., Dinev, T., & Xu, H. Information privacy research: an interdisciplinary review. MIS Quarterly, 2011.
7. Subramanian R. Security, privacy and politics in India: a historical review. Journal of Information Systems Security (JISSec), 2010.
8. Acquisti, A., John, L. K., & Loewenstein, G. **What** is privacy worth? The Journal of Legal Studies, 2013
9. Xu H., Luo X.R., Carroll J.M., Rosson M.B. The personalization privacy paradox: An exploratory study of decision making process for location-aware marketing. Decision Support Systems, 2011.

Criteria to get a certificate:

If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

CAME108: Computer Architecture and Organization

Description:

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

This course will discuss the basic concepts of computer architecture and organization that can help the participants to have a clear view as to how a computer system works. Examples and illustrations will be mostly based on a popular Reduced Instruction Set Computer (RISC) platform. Illustrative examples and illustrations will be provided to convey the concepts and challenges to the participants. Starting from the basics, the participants will be introduced to the state-of-the-art in this field.

Pre-requisites:

Basic concepts in digital circuit design, Familiarity with a programming language like C or C++.

Industry support:

TCS ; Wipro ; CTS ; Google ; Microsoft ; HP ; Intel

Course layout:

Week 1: Evolution of Computer Systems.

Week 2: Instruction Set Architecture.

Week 3: Quantitative Principles of Computer Design.

Week 4: Control Unit Design.

Week 5: Memory System Design.

Week 6: Design of Cache Memory Systems.

Week 7: Design of Arithmetic Unit.

Week 8: Design of Arithmetic Unit (contd.)

Week 9: Input-Output System Design.

Week 10: Input-Output System Design (contd.)

Week 11: Instruction Set Pipelining.

Week 12: Parallel Processing Architectures.

Books and References:

1. D.A. Patterson and J.L. Hennessy, "Computer Architecture: A Quantitative Approach, 5/E", Morgan Koffman, 2011.
2. D.A. Patterson and J.L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface, 5/E", Elsevier India, 2016.
3. W. Stallings, "Computer Organization and Architecture: Designing for Performance", Pearson, 2015.
4. C. Hamacher, Z. Vranesic and S. Zaky, "Computer Organization, 5/E", McGraw Hill, 2011.
5. J.P. Hayes, "Computer Architecture and Organization, 3/E", McGraw Hill, 1998.

Criteria to get a certificate:

If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

CAUD101 : Digital Electronics

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: In order to achieve the course objectives, students will be introduced to digital technologies. Various digital modules used to create digital computer devices like gates, flip flops, decoder, encoder etc. are to be studied. Mathematical base is created to understand the organization of digital computer.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Digital Systems and Binary Numbers	12
2	Boolean Algebra and Logic Gates	12
3	Combinational Circuit Designing	10
4	Sequential Circuit Designing	10
5	Memory Interfaces and Data Representation	08
6	The Arithmetic Logic Unit and Control Unit	08

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Digital Systems and Binary Numbers

Hours 12

Introduction and Evolution of Digital computers, Major components of digital computer, Interfaces and Buses, compiler, interpreter, Assembler, Introduction to various number systems (Decimal, Binary, Octal, Hexadecimal, BCD), Conversion of number system, Binary operations, signed binary numbers, 1's complement and 2's complement.

Unit II: Boolean Algebra and Logic Gates

Hours 12

Fundamental concepts and rules of Boolean algebra, , Boolean Algebra Laws, Boolean expressions and its simplification with and without Truth Tables, Canonical form and standard form, D-Morgan's Theorem, Basic duality laws, Derivation of Boolean expressions, Introduction to logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Implementation of Boolean expressions using logic gates, Sum of product and Product of Sum forms, Universal gates, Implementation of other gates using universal gates, Karnaugh Map method for two, three, four and five variables, K-map with don't care conditions, Tabulation method, determination and selection of prime applicants.

Unit III: Combinational Circuit Designing

Hours 10

Introduction to combinational circuits, Design procedures, Adders, Subtractors, code conversion circuits, Analysis procedures, Multilevel NAND circuits, Multilevel NOR circuits, EX-OR functions, Binary adder and Subtractors, Decimal adder and Subtractors, Magnitude comparator, Decoder and encoders, Multiplexer and de-multiplexer.

Unit IV: Sequential Circuit Designing

Hours 10

Introduction to Sequential circuits, Concepts of Flip-flops, triggering flip-flops, Clock signals, Analysis of clock sequential circuits, State Reduction and assignments, Flip flop excitation table, Design procedure of asynchronous and synchronous counters (Ripple counter, Binary counter, BCD counter), Design of registers, Shift registers, Timing sequences.

Unit V: Memory Interfaces and Data Representation

Hours 08

Memory, Types of ROM, Design of ROM, Types of RAM, Magnetic disc memory, Magnetic tape, Digital recording techniques, Linear-select memory organization, Types of digital codes (Gray code, 8421 code, Alphanumeric codes), Parity checking codes, Floating point representation, Fixed point representation.

Unit VI: The Arithmetic Logic Unit and Control Unit Hours 08

Basic operations of Arithmetic Logic Unit, Construction of the Arithmetic Logic Unit, Representation of Instruction word, Control registers, Instruction and execution cycles of Control registers, Sequence operation of Control Registers.

Core Books:

1. M. Morris Mano: Digital Logic and Computer Design, Third Edition, Pearson Education, 2016.
2. Thomas C. Bartee: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.
3. M. Morris Mano: Computer System Architecture, Third Edition, Pearson Education, 2011.

Reference Books:

1. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
2. Albert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.

Web References:

1. https://www.tutorialspoint.com/digital_circuits/index.htm [Digital Circuits Tutorial]
2. <https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/> [Digital Electronics and Logic Design Tutorials]
3. <https://studymaterialz.in/digital-logic-and-computer-design-by-morris-mano/> [Digital Logic and Computer Design book]
4. <https://cupola.gettysburg.edu/oer/1/> [Digital Circuit Projects]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to get the idea about digital system and numbering system.
CO2 :	Able to study logic gates for digital circuit designing.
CO3 :	Able to learn to design various combinational circuits.
CO4 :	Able to learn to design various sequential circuits.
CO5 :	Able to learn memory design concepts and to study computer Arithmetic Logic Unit and Control Unit.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Digital Systems and Binary Numbers	√				
2	Boolean Algebra and Logic Gates		√			
3	Combinational Circuit Designing			√		
4	Sequential Circuit Designing				√	
5	Memory Interfaces and Data Representation					√
6	The Arithmetic Logic Unit and Control Unit					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	2	-	3	2	-	-	2	1	3	2
CO2	3	2	2	2	-	3	2	-	-	2	1	3	2
CO3	3	3	3	2	-	2	2	-	-	2	2	3	2
CO4	3	3	3	2	-	2	2	-	-	2	2	3	2
CO5	3	3	3	2	-	3	2	-	-	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUS101 : Mathematics for Computer Science

Credit	Component	Instruction/ Contact Hours(Per Week)	Instruction/ Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Theory	02	30	25	55
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: Basic algebra and geometry, as well as a good understanding of logic and proofs, mathematical statistics.

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of mathematics and statistics. Problem solving: Students will demonstrate the ability to solve problems, including applications outside of mathematics. Mathematical communication: Students will demonstrate the ability to communicate mathematical ideas clearly. They will use correct mathematical terminology and proper mathematical notation.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Sets and functions	04
2	Vectors and Matrices	04
3	Measure of Central Tendency & Dispersion	06
4	Permutation & Combination	05
5	Probability	06
6	Graph Theory	05

Total Hours (Theory): 30
Total Contact Hours:30

Detail Syllabus:

Unit I : Sets and Functions

Hours 04

Introduction to set theory, methods of representation of a set, operations on set, algebra of sets, De 'Morgan's law, function definition, domain, range, one-to-one function, onto function.

Unit II : Vectors and Matrices

Hours 04

Definition of vector, addition and subtraction of vector, magnitude of a vector, unit vectors, dot product and cross product, definition of matrix, equal matrices, diagonal elements of matrix, row matrix, column matrix, Symmetric matrix, skew symmetric matrix, orthogonal matrix, diagonal matrix, identity matrix, operations on matrix.

Unit III : Measure of central tendency and dispersion Hours 06

Definition, Ungrouped Data, Grouped Data (Discrete and Continuous Grouped data). Mean: Arithmetic Mean, Geometric Mean and Harmonic Mean for ungrouped data, Combined Mean, Weighted Mean. Median, Quartiles, Deciles, Percentiles and Mode. Different measure of dispersion. Quartile Deviation, Mean Deviation, Standard Deviation, Combined Standard Deviation, Coefficient of Variation.

Unit IV : Permutation and Combination

Hours 05

Meaning of permutation, Formula of permutation, Permutation of n different things, Permutation of similar things, Permutation of repeated things, Circular Permutation, Combination: Meaning of Combination, Formula of Combination.

Unit V : Probability

Hours 06

Probability: Random Experiment, Sample Space, Event, Mutually exclusive event, Exhaustive event, Equally likely event, Probability Classical definition. (Simple examples of Probability).

Unit VI : Graph Theory

Hours 05

Introduction to Graph, Graph Definition, Vertices, Edges, Loops, Parallel Edges, Simple Graph, Finite Graph, Adjacent vertices, Incidence between vertex and edge, Degree of a vertex, Isolated Vertex, Pendent Vertex, Null Graph. Isomorphism, Labeled Graph

Core Books:

1. D. C. Sancheti, V. K. Kapoor: Business Mathematics, Sultan Chand & sons.
2. Lipschutz & Marc Lipson: DISCRETE MATHEMATICS, Tata McGraw Hill
3. Narsingh Deo: Graph Theory with application to engineering and computer science, Prentice Hall of India Pvt. Ltd

Reference Books:

1. Gupta and Gupta: Business Statistics, Sultan Chand and Sons.

Web References:

1. <https://www.statlect.com/matrix-algebra/vectors-and-matrices>
2. <https://www.geeksforgeeks.org/permutations-and-combinations/>
3. <https://www.khanacademy.org/math/statistics-probability/probability-library/basic-theoretical-probability/a/probability-the-basics>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will learn basics of sets and functions
CO2 :	The students will be familiar with various linear algebra.
CO3 :	Students will be well equipped to measure accurate description of data
CO4 :	Students will be able to define a permutation and explain how to calculate probability
CO5 :	Students will be able to understand and apply the fundamental concepts in graph theory

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Sets and Functions	√				
2	Vectors and Matrices		√			
3	Measure of central tendency and dispersion			√		
4	Permutation and Combination				√	
5	Probability				√	
6	Graph Theory					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO 2
CO1	3	3	3	2	-	2	1	-	-	-	-	2	-
CO 2	3	3	3	2	-	2	1	-	-	-	-	2	-
CO3	3	3	3	2	-	2	1	-	-	-	-	2	-
CO 4	3	3	3	2	-	2	1	-	-	-	-	2	-
CO5	3	3	3	2	-	2	1	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**BCA PROGRAMME
(2nd SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-II

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CAUC103	Fundamentals of Object Oriented Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC104	Database Fundamentals	Practical	-	-	25	25	02	Major Core
CAUE103 CAUE104 CAME103 CAME104	(Select any one) 1. Fundamentals of E-Commerce 2. Business Economics 3. E-Business 4. Business Development: From Start to Scale	Theory	50	50	-	-	04	Minor
CAUD102	Data Communications and Networks	Theory	50	50	-	-	04	MDC
CAUV101	Fundamentals of Accounting	Theory	25	25	-	-	02	VAC
HSUS101-117	(Select any one) A course on Liberal Arts HSUS101 – Painting HSUS102– Photography HSUS103– Sculpting HSUS104– Pottery and Ceramic Art HSUS105-Media and Graphic Design HSUS106-Art and Craft HSUS109- Dramatics HSUS110-Contemporary Dance HSUS111-Music (Vocal) HSUS112-Music (Instrumental)-Tabla HSUS113-Music (Instrumental)-Guitar HSUS114-Music (Instrumental)-Harmonium HSUS115-Music (Instrumental)- Flute HSUS116- Indian Classical Dance-Kathak HSUS117- Indian Classical Dance-Bharatnatyam	Practical	-	-	25	25	02	SEC
CAUA101	Presentation Skills	Practical	-	-	25	25	02	AEC
CAUI101	Summer Internship and Viva - I	VIVA	-	-	50	50	04	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CAUC103: Fundamentals of Object Oriented Programming

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	50	110
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: Foundation of Programming

Methodology, Pedagogy & Andragogy: The theory sessions will use real-world examples to demonstrate how object-oriented programming (OOP) enhances problem-solving, covering concepts such as classes, objects, inheritance, polymorphism, and exception handling. Students will engage in self-directed learning through case studies, assignments, and unit tests to assess their progress. During practical sessions, students will apply these OOP concepts through various assignments and case studies, further developing their understanding of how OOP efficiently addresses complex problems. Independent learning will be emphasized to ensure students grasp and progress in their comprehension of OOP principles.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Object Oriented Programming	08
2	Classes and Objects	11
3	Polymorphism	10
4	Inheritance	13
5	Exception Handling and Type cast operators	10
6	Managing I/O formats and operations	08

Total Hours (Theory): 60
Total Contact Hours: 60

Detailed Syllabus:

Unit - I: Introduction to Object Oriented Programming Hours: 08

Structured programming vs. object oriented programming, Introduction to C++: structure of a C++ program Tokens, inline function, pass by reference, default arguments, OOP characteristics.

Unit - II: Classes and Objects Hours: 11

Define class, objects, visibility modes, static members, friend function, Constructors and Destructors, Default Constructor, Copy constructors, Parameterized Constructor, this keyword.

Unit - III: Polymorphism Hours: 10

Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading, Operator overloading through member function and friend functions, type conversions from primitive to class, class to primitive and class to class.

Unit - IV: Inheritance Hours: 13

Objective of Inheritance, types of inheritance, impact of access modes during inheritance, Virtual base classes, Dynamic Binding, Virtual functions, Pure virtual functions, Abstract class, Early vs. Late binding.

Unit - V: Exception Handling and Type Cast Operators Hours: 10

Introduction to Exception handling, catching class types, using multiple catch statements, exception handling options, creating custom exceptions, introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast.

Unit - VI: Managing I/O Formats and Operations Hours: 08

Understand and apply concepts from various header files like iostream, iomanip, fstream, etc. opening and closing file, reading and writing text files and binary files, performing random access on files, handle command line arguments.

Outline of the Course:

Week No	Practical	Description
1	Introduction to Object Oriented Programming and Setting up IDE	Introduction to C++, Setting up a development environment and its introduction, structure of a C++ program Tokens, inline function, pass by reference, default arguments
2	Classes and Objects	Define class, objects, visibility modes, static members
3	Friend Functions	Introduction and hands on friend functions
4	Constructors and Destructors	Constructors and Destructors, Default Constructor, Types of constructs
5	Inheritance	Exposure on Inheritance programs and their types
6	Virtual Function	Programs on Virtual Function, Pure Virtual functions, Abstract Class, Early Binding and Late Binding
7	Function Overloading	Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading
8	Operator Overloading	Operator overloading through member function and friend functions
9	Exception Handling	Introduction to Exception handling, catching class types,
10	Multiple Catch Statements	Using multiple catch statements, exception handling options, creating custom exceptions
11	Type Cast Operators	Introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast
12	Managing I/O Formats	Understand and apply concepts from various header files like iostream. Iomanip ,fstream, etc
13	File Handling	Opening and closing file, reading and writing text files and binary files
14	File Handling Operations	opening and closing file, reading and writing text files and binary files, performing random access on files
15	Command Line Arguments	handle command line arguments.

Total hours: 60

Core Books:

1. Herbert Schildt: C++: A Beginner's Guide, 2nd Edition Paperback, McGraw-Hill, 2003.
2. E Balagurusamy: Object Oriented Programming with C++ Paperback, McGraw Hill India, 2017.

Reference Books:

1. Yashavant Kanetkar: Let Us C++, Paperback, BPB Publications, 2020.
2. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications, 2002.
3. Bruce Eckel: Thinking in C++, Volume 1, 2nd Edition, Pearson Education, 2006.

Web References:

1. <https://www.geeksforgeeks.org/c-plus-plus/> [For Entire Syllabus]
2. <https://google.github.io/styleguide/cppguide.html> [For coding standards and concepts]
3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm [For Practicals]

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	Understand the basic structure of C++ program and understand the concept of object oriented programming characteristics.
CO2 :	Understand the concept of Constructor and Destructor and able to create class and objects.
CO3 :	Understand the need of Polymorphism and implement static polymorphism
CO4:	Understand the purpose of Inheritance, identify relations between classes, implement inheritance and perform dynamic binding.
CO5 :	Able to handle the runtime errors or exceptions in C++ and use type cast operators. Able to read input from device, file, command line arguments and apply formats to input and output.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		Co1	Co2	Co3	Co4	Co5
1	Introduction to Object Oriented Programming	√				
2	Classes and Objects		√			
3	Polymorphism			√		
4	Inheritance				√	
5	Exception Handling and Type cast operators					√
6	Managing I/O formats and operations					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	2	2	2	2	2	2	2	2	2	3	2
CO2	3	3	3	2	2	2	3	2	2	2	2	3	2
CO3	3	2	3	2	2	2	3	2	2	2	2	3	2
CO4	3	3	3	2	2	2	3	2	2	2	2	3	2
CO5	3	2	3	2	2	2	2	2	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC104 : Database Fundamentals (Practical)

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, VIVA , certification courses, development of projects etc.					

Prerequisites: Basic knowledge of working with computer

Methodology, Pedagogy & Andragogy: Learn about computer data. Understand what data is, learn its meaning, examine the types of data sources, and see the difference between data and information. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours' students' will implement the concepts. Apply SQL commands to mine data from databases and data visualization techniques to address organization information needs.

Outline of the Course:

Week No	Practical	Description
1	Introduction to Database System	Overview of data, information, file, database, database systems, DBMS, Purpose of DBMS over file system.
2-3	Data modelling and architecture	Various data models: ER model and Relational model, Three level architecture, structure of DBMS and database actors and workers
4-5	Database Design Methodology	CODD Rules, Functional Dependency and Normalization for Database Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF)

6-7	Schema Definition, Constraints, Queries – I	Basic Data types, Create Table Command, Modifying the structure of tables, renaming table, truncating table, destroying table, Insert, Delete and Update Statements in SQL
8	Schema Definition, Constraints, Queries – II	Data Constraints and Functions: - Pseudo columns, Null values, TAB table, DUAL table Operators, Data constraints, Type of data constraints, modifying constraints, working with data dictionary and use of USER_CONSTRAINTS Functions introduction
9	Inbuilt functions	Merits and demerits of functions, Types of functions: Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions
10-11	Querying the data	Different operators: logical range, searching and pattern matching, Viewing Data in the tables, sorting
12-13	Advanced Query Processing-I	Co-related Nested Queries, Group By clause, having clause
14-15	Advanced Query Processing-II	Joins (Inner Join, Outer Join, Self-Join, Equi Join, Cross Join), Creation and manipulation of database objects indexes, views, sequences and synonym.

Total hours: 60

Core Books:

1. Ivan Bayross : SQL, PL/SQL The programming Language Oracle.
2. Ramkrishnan, Gehrke : Database Management Systems, 3 rd Edition, McGrawHill Publication.
3. RamezElmasri, Shamkant B. Navathe : Fundamentals of Database Systems , 5th Edition, , Pearson Publication.

Reference Books:

1. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.
2. C.J.Date, a Kannan, S Swaminathan : An Introduction to Database Systems, 8th Edition, Pearson Education,(Equivalent Reading).

3. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
4. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
5. Peter Rob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, Cengage Learning, 2007.
6. Leon and Leon : Database management Systems, Vikas Publication.

Web References:

1. <https://www.javatpoint.com/er-model-vs-relational-model> [For difference of ER and Relational model]
2. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals#?fbid=tbZ92pOp_Tt [For overall subject]
3. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html [for relational database design]
4. http://docs.oracle.com/cd/A97335_02/apps.102/a81358/05_dev1.htm [For ER Diagram]

Course Outcome: Upon successful completion of the course, student will:

CO1:	Understand basic concepts regarding data, database systems and various data models.
CO2:	Understand various data modeling techniques and entity relationship models.
CO3:	Learn about relational data model and related concepts.
CO4:	Gain insights into database design, SQL and normalization concepts.
CO5:	Advanced query processing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	1	1	-	3	-	-	-	-	-	2	-
CO2	1	3	3	3	-	3	1	-	-	-	-	1	3
CO3	3	3	3	3	-	3	2	-	-	2	-	2	
CO4	-	3	3	3	-	3	3	-	-	3	3	-	3
CO5	2	3	3	3	2	3	3	-	-	-	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE103 : Fundamentals of E-Commerce

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of E-commerce. Students will be introduced to E-Commerce, E-Marketing, E-Security and E-Payment, Mobile Commerce, CRM and ERP, and E-Commerce Applications. Students will give practical exposure in form of case study and by visit to E-commerce / Digital Marketing company and/or supply chain Centre.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to E-Commerce	09
2	E-Marketing	11
3	E-Security and E-Payment	09
4	Mobile Commerce	11
5	CRM and ERP	09
6	Quick Commerce	11

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I : Introduction to E-Commerce

Hours 09

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of E-commerce, Features of E-Commerce, Electronic Commerce over the Internet, Types of Ecommerce.

Unit II : E-Marketing

Hours 11

Traditional Marketing, Online Marketing, E- advertising, Internet Marketing Trends and strategies, E-branding.

Unit III : E-Security and E-Payment

Hours 09

Security on the Internet, E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash, Digital Signature.

Unit IV : Mobile Commerce

Hours 11

definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of M-Commerce, Areas of M-Commerce Applications, Payment Systems and Models in M-Commerce.

Unit V : CRM and ERP

Hours 09

Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

Unit VI : Quick Commerce

Hours 11

Introduction to Quick Commerce, Key features of Quick Commerce, Business models in Quick Commerce, Technology backbone of Quick Commerce, Examples of Quick Commerce (India and Global), Benefits of Quick Commerce, and Future Trends in Quick Commerce.

Core Books:

1. P.T.Joseph, S.J.: E-Commerce- An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.

Reference Books:

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business Managers, 2nd Edition, Amacom.
2. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful Web-based Business, 2nd Edition, CRC Press.
3. By Philippe Humeau&Matthieu Jung : The White Book of Ecommerce Solutions, NBS System.
4. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce

Web References:

1. <http://www.ecommercetutorial.net/> [For Tutorial]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand basic of Electronic Commerce and its Application
CO2 :	Getting knowledge of E-marketing and its Strategies.
CO3 :	Learn E-security and E-Payment in Detail and its term. Also, learn Encryption and decryption in basic
CO4 :	To gain understanding of Mobile Commerce and its Application through real life examples.
CO5 :	To learn concepts of Customer Relationship Management and various solutions of E-Commerce (Software and Framework).

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to E-Commerce	√				
2	E-Marketing	√	√			
3	E-Security and E-Payment		√	√		
4	Mobile Commerce		√		√	
5	CRM and ERP					√
6	Quick Commerce				√	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE104 : Business Economics

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theoretical lessons, real-world examples will be used to discuss economic topics such as demand, supply, production, cost function, market structure, factor pricing, and microeconomics. Students will be offered appropriate case studies in order to provide actual exposure to theoretical economic ideas.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Fundamentals of Economics	09
2	Basic Elements of Demand and Supply	10
3	Theory of Production and Cost	11
4	Analysis of Revenue	11
5	Market Failures and Price Regulation	09
6	Macro-Economic Environment and National Income Accounting	10

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Fundamentals of Economics

Hours 09

Fundamentals of Economics, Needs, wants, nature of economics, scope of economics, business economics, laws of economics, business cycle, difference between Micro and Macro Economics.

Unit II: Basic Elements of Demand and Supply

Hours 10

Demand and supply, types of demand, determinants of demand and supply, law of demand and supply, balance between demand and supply, demand and supply schedule, demand function.

Unit III: Theory of Production and Cost

Hours 11

Theory of Production- Factors of Production, Basic Concepts, Production Function, Producer's Equilibrium Least-Cost Factor Combination and Output Maximization for a given Level of Outlay. Theory of Costs- Basic Concepts, Short-run Total Cost, Short-run Average and Marginal Cost, Relationship between Average and Marginal Cost, Average and Marginal Cost in the Long-run.

Unit IV: Analysis of Revenue

Hours 11

Basic Concepts of Revenue, Revenue, Relationship between Average and Marginal Revenue, Concept of Market and Main Forms of Market; Equilibrium of the Firm-Meaning, Objectives of the Firm, Monopoly and its features, Monopolistic Competition and Oligopoly.

Unit V: Market Failures and Price Regulation

Hours 09

Market failures and need for regulation, Regulations and market structure, Firm behaviour, Price regulation.

Unit VI: Macro-Economic Environment and National Income Accounting

Hours 10

Basic macro- economic concepts –aggregate, demand aggregate supply, money, income employment consumption savings and investment. National Income Accounting-

concepts and methods of national income, different methods of national income accounting.

Core Books:

1. P.L Mehta: Managerial Economics, Sultan Chand and Sons Publication, 2014
2. Keat Paul, K Young Philip , Erfle Steve, Banerjee Sreejata: Managerial Economics, seventh edition, Pearson Education, 2017.
3. N. Gregory Mankiw: Principles of Economics, sixth edition, South-Western, Cengage Learning, 2012.

Reference Books:

1. Geetika, Ghosh & Choudhury: Managerial Economics, third edition, Mc Graw Hill Education, 2005.
2. Mote V.L., Samuel Paul and G.S. Gupta: Managerial Economics Concepts and Cases, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2001.
3. G. S. Gupta: Macro Economics Theory and Applications, Second Edition, Tata McGraw Hill, 2006.

Web References:

1. <https://www.sciencedirect.com/topics/economics-econometrics-and-finance/economic-law> [Economic Law]
2. <https://www.toppr.com/guides/business-economics-cs/basic-elements-of-demand-and-supply> [Basic Elements of Demand and Supply]
3. <https://corporatefinanceinstitute.com/resources/knowledge/economics/national-incomeaccounting/> [National Income Accounting]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to Interpret the various basic economic principles.
CO2 :	Able to understand theory of demand and supply.
CO3 :	Able to deal with balancing between cost and revenue.
CO4 :	Able to learn the causes and solution of market failures.
CO5 :	Able to understand the basics of macroeconomics and accounting.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Fundamentals of Economics	√				
2	Basic Elements of Demand and Supply		√			
3	Theory of Production and Cost			√		
4	Analysis of Revenue			√		
5	Market Failures and Price Regulation				√	
6	Macro-Economic Environment and National Income Accounting					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	1	-	-	2	-	-	1	-	2	2	1
CO2	3	1	2	-	-	2	-	-	1	-	2	2	1
CO3	3	1	2	-	-	2	-	-	1	-	2	2	1
CO4	3	1	2	-	-	2	-	-	1	-	2	2	1
CO5	3	1	1	2	-	2	-	-	1	-	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAME103 : E-Business

Description:

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	4

About the course:

The Internet has changed the way companies carry out their businesses. The primary objective of this course is to introduce concepts, tools and approaches to electronic business to the post- graduate and undergraduate students. Further, the subject will help the students to develop skills to manage businesses in the digital world. The course will cover following aspects of E-Business Systems.

- Part 1: Foundations of E-Business systems
- Part 2: Infrastructure
- Part 3: Functional Areas
- Part 4: Decision Support for E-Business Systems

The course provides a balance approach including concepts from technology and management.

Pre-requisites: None

Industry support: All the companies

Course layout

Week 1: Introduction to E-Business

Week 2: Making Functional Areas E-Business Enabled : Value chain and supply chain, inter and intra organizational business processes, ERP

Week 3: Making Functional Areas E-Business Enabled : E-Procurement

Week 4: Making Functional Areas E-Business Enabled : E-marketing, E-Selling, E-Supply Chain Management

Week 5: Technologies for E-Business: Internet and Web based system

Week 6: Technologies for E-Business: Security and payment systems

Week 7: Technologies for E-Business: Supply chain integration technologies (EDI, RFID, Sensors, IoT, GPS, GIS)

Week 8: Technologies for E-Business: Supply chain integration technologies (Web services and cloud)

Week 9: Decision Support in E-Business: Web analytics

Week 10: Decision Support in E-Business: Customer behavior modeling

Week 11: Decision Support in E-Business: Auctions

Week 12: Decision Support in E-Business: Recommender systems

Books and references

1. Management Information Systems: Managing the Digital Firm, Laudon and Laudon, Pearson
2. Scaling for E-Business, Menasce & Almeida, PHI
3. eBusiness & eCommerce – Managing the Digital Value Chain, Meier & Stormer, Springer
4. eBook is available in springerlink.com
5. Some reference books, Internet Resources, and Research Papers

Criteria to get a certificate:

If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

CAME104 : Business Development: From Start to Scale

Description:

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	4

About the course:

This 12-week, 60 lecture course titled ""Business Development: From Start to Scale"" equips the learners with various concepts and frameworks for establishing and growing businesses. Focusing on customers and markets, the course covers the foundational as well as advanced constructs of business development. Multiple practical examples and case studies are provided. This versatile course will be useful for students and working professionals, and relevant for startups and entrepreneurial firms as well as established small, medium, and large companies for crafting and executing their growth journey. This course, in addition, will be an ideal next-step course for those who would complete the Entrepreneurship course.

Pre-requisites: Graduation of any discipline; No requirement to do any other course as a pre-requisite to taking this course

Industry support: Likely to be recognized by all types of industries and all types of firms as starting up and scaling up of business is a universal and perpetual need.

Course layout

Week 1: Business Fundamentals

- Understanding Business Development
- Marketing and Business Development
- Markets and Marketing
- Strategy Formulation
- Business Development Cases

Week 2: Business Development Strategies

- Successful Businesses
- Industry and Market
- Vision, Mission and Strategy

- Goals
- Case Studies of Business Development Excellence

Week 3: Industry Structure and Company Analysis

- Industry and Business
- Porter's Five Forces Theory
- Industrial Transformations
- Competitive Strategies
- Company Analysis

Week 4: Market and Competitor Analysis

- Industry, Market and Business
- Industry and Market Analysis
- Market Structures
- Demand Forecasting
- Competitor Analysis

Week 5: Connecting with Customers

- Customer Characteristics
- Customer Typologies
- Market Research and Design Thinking
- Customer bonding
- Customer Relationship Management

Week 6: Business and Market Segments

- Market and Market Descriptors
- Market and Product Segmentation
- Product-Market Segmentation
- Segmentation Deep Dive
- Market Attractiveness and Competitive Positioning

Week 7: Branding and Pricing

- Branding
- Brand Organisation
- Advertising and Communication
- Servitization
- Pricing

Week 8: Corporate Development

- A New IT Start-up

- An FMCG Start-up
- A Logistics Start-up
- A Nutraceuticals Start-up
- A Telecom Fightback

Week 9: Business Development Structures

- Collaborations
- Strategic Alliances
- Joint Ventures
- Subsidiaries
- Mergers and Acquisitions

Week 10: Business Development Competencies

- Value Chain Competencies
- Functional Competencies
- Negotiation Skills
- Cultural Skills
- Leadership Attributes

Week 11: Strategies for Markets and Industries

- Growth Strategies
- Growth Examples
- Fragmented Industries and Emerging Industries
- Mature Industries and Declining Industries
- Global Industries and New Businesses

Week 12: Business Development Case Studies

- Business Transformation
- Strategic Alliances for Growth
- Business Turbulence
- Creating Value
- From Start to Scale
- In Closing

Books and references

1. Michael E. Porter, Competitive Strategy, New York: Free Press, 1980
2. Orville C. Walker, Jr., and John W. Mullins, Marketing Strategy: A Decision- Focused Approach, McGraw Hill, 2011
3. Philip Kotler and Gary Armstrong, Principles of Marketing, Englewood Cliffs, NJ, Prentice Hall, 1989
4. Philip Kotler, Leven Lane Keller, Marketing Management, Pearson,

2017W. Chan Kim and Renee Mauborgne, Blue Ocean Strategy, Harvard Business Review Press, 2017

5. Charles W L Hill, Gareth R Jones and Melissa A Schilling, Strategic Management, Cengage Learning, 2015
6. Thomas L. Wheelen, J. David Hunger, Alan N. Hoffman, and Charles E. Bamford, Strategic Management and Business Policy: Globalization, Innovation, and Sustainability, Pearson, 2015
7. Rao, C B, Strategic Management, Practice and Philosophy for India Inc, Notion Press, 2021

Criteria to get a certificate:

If You wish to get certified on this course you must register and write the proctored exam after payment of exam fee.

30 Marks will be allocated for Internal Assessment and 70 Marks will be allocated for end term proctored examination.

Securing 40% in both separately is mandatory to pass the course and get Credit Certificate.

CAUD102 : Data Communications and Networking

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study, Certification courses, etc.					

Pre-requisite: Basic knowledge of computers and their working.

Methodology, Pedagogy & Andragogy: During the lecture sessions, the students will learn about role of computer network in modern communication systems, underlying principles of network communication, need for standardization through network reference model. The teacher will also discuss communication issues at different layers and concepts related to network security. The students will adopt the latest certification courses from reputed and recognized platforms. Furthermore, the students will be engaged in problem-solving, emphasizing real-world applications and relevance.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Computer Networks	10
2	Basics of network communication	08
3	Network reference model	11
4	Data Link Layer Functions and Protocol	12
5	Network Layer Functions and Protocol	10
6	Network security	09

Total Hours (Theory):60

Total Contact Hours:60

Detail Syllabus:**Unit I: Introduction to computer network****Hours 10**

Definition of computer network, aim and objective of computer network: applications of computer network, merits & demerits of computer network, examples of computer network, components of computer network, types of networks, network topologies, network devices: hub, switch, repeater, router, gateway, bridge.

Unit II: Basics of network communication

Hours 08

Parallel & serial communication, synchronous & asynchronous communication, modes of communication: simple, half-duplex, full-duplex, multiplexing, demultiplexing: frequency division multiplexing (FDM), time division multiplexing (TDM), communication media: guided media, unguided media, switching techniques: circuit switching, message switching, packet switching.

Unit III: Network reference model

Hours 11

Need for network reference model, history of TCP/IP and OSI model, layers of OSI and TCP/IP model, comparison between TCP/IP and OSI model, Protocol Hierarchy, Network Protocols: Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Multiple Internet Mail Extension (MIME), Post Office Protocol (POP), Telnet, Domain Name Service (DNS), Dynamic Host Configuration Protocol (DHCP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP).

Unit IV: Data Link Layer Functions and Protocol

Hours 12

Definition of Framing, framing methods, Error detection techniques, Error correction techniques, Flow control mechanisms- Simplex protocol, Stop and Wait ARQ, Go-Back-N ARQ, Point to Point protocol.

Unit V: Network Layer Functions and Protocol

Hours 10

Connection oriented vs Connectionless services, Definition of Routing, Routing algorithms, IP protocol, IP addresses, ARP, RARP

Unit VI: Network security

Hours 09

Introduction to cryptography, symmetric key and public key encryption systems, digital signature and key management concepts, communication security.

Core Books:

1. Andrew S. Tanenbaum: Computer Networks, 6th edition, Pearson Publication: 2021.
2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill, 2018.
3. William Stallings: Data and computer communication: 5th Edition, Pearson Publication, 2019.

Reference Books:

1. James F. Kurose, Keith W. Ross: Computer networking – A top-down approach: 3rd Edition: Pearson Publication: 2017
2. Narsimha Karumanchi: Elements of computer networking: An Integrated Approach: 1st Edition: Careermonk Publication: 2014
3. Ed Tittel: Computer networking: 1st Edition, Schum's Publication: 2020

Web References:

1. <https://computer.howstuffworks.com/computer-networking-channel.htm>
[How Stuff Works]
2. [https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012/dn313100\(v=ws.11\)?redirectedfrom=MSDN](https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012/dn313100(v=ws.11)?redirectedfrom=MSDN) [Microsoft Networking]
3. https://help.sap.com/docs/SAP_COMMERCE/do224eca81e249cb821f2cdf45a82ace/8c74877b866910148cc9ba5f39a2fd28.html?version=6.7.0.0 [For multicast and unicast]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Student will be able to understand role of computer network in digital communication
CO2 :	Student will grasp underlying concepts related with network communication
CO3 :	Student will understand role of reference models in standardizing communication
CO4 :	Students will learn about intra and inter network communication and technology behind it.
CO5 :	Students will learn about different aspects of network security.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Computer Networks	√				
2	Basics of network communication		√			
3	Network reference model			√		
4	Data Link Layer Functions and Protocol				√	
5	Network Layer Functions and Protocol				√	
6	Network security					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	2	1	-	1	1	1	-	-	1	2	-
CO2	2	3	3	2	-	1	1	1	-	-	1	3	1
CO3	3	3	3	2	-	1	1	1	1	-	1	3	3
CO4	1	3	3	2	-	1	1	1	1	-	1	3	2
CO5	3	3	3	2	-	1	1	1	1	-	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUV101 : Fundamentals of Accounting

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Theory	02	30	20	50
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During the classroom sessions, the emphasis will be given on the fundamentals of accounting. Students will be introduced to book keeping systems in modern accounting, financial accounting, and cost accounting. Students will prepare financial statements such as journal, ledger, trial balance, trading and profit and loss account, and balance sheet. Students will also get hands on exposure in make or buy decisions in cost accounting. Same concepts will be made concrete through assignments, case studies, and unit tests.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Accounting	05
2	Book Keeping System	05
3	Financial Accounting - I	05
4	Financial Accounting - II	07
5	Cost Accounting - I	04
6	Cost Accounting - II	04

Total Hours (Theory): 30
Total Contact Hours:30

Detail Syllabus:

Unit I : Introduction to Accounting

Hours 05

Introduction and users of Accounting, Branches of Accounting, Objectives, Accounting Concepts and Conventions, Accounting Terms.

Unit II : Book Keeping System

Hours 05

Single and Double Entry Book Keeping Systems, Advantages of Double Entry Book Keeping System, Limitations of Double Entry Book Keeping System, Types of Accounts: Personal, Real, and Nominal, Terms of Book Keeping Systems.

Unit III : Financial Accounting – I

Hours 05

Stages of Accounting Cycle, Types of Transactions, Forms of Economic Transactions, Cash and Credit Transactions, Introduction to Journal, Journal Entries.

Unit IV : Financial Accounting – II

Hours 07

Introduction to Ledger, Relationship Between Journal and Ledger, Rules of Ledger Posting, Balancing of the Accounts in Ledger, Trial Balance, Objectives of Preparing Trial Balance, Trading and Profit and Loss Account, Balance Sheet, Difference between Trial Balance and Balance Sheet.

Unit V : Cost Accounting – I

Hours 04

Introduction to Cost Accounting: Cost, Costing, and Cost Accounting, Limitations of Financial Accounting, Difference Between Cost Accounting and Financial Accounting, Objectives of Cost Accounting.

Unit VI : Cost Accounting – II

Hours 04

Elements of Cost: Material Cost, Labor Cost, and Expenses, Techniques of Costing, Marginal Costing, Make or Buy Decision, Breakeven Analysis

Core Books:

1. Dr. S. N. Maheshwari: Principles of Management Accounting, Sultan Chand Publication.
 - a. Eighteenth Edition, 2021
2. M. Pande: Financial Management, Vikas Publishing House, 2015
3. M.N. Arora: Cost Accounting, Vikas Publishing House, 2013

Reference Books:

1. Dr. S.N. Maheshwari: Advanced Accountancy, Volume 1, Eleventh & Revised Edition, Vikas Publishing House, 2017

Web References:

1. <https://www.accountingtools.com/articles/basics-of-accounting.html>

[For Basics of Accounting]

2. https://www.youtube.com/watch?v=vuetn_PQOvM

[For Journal Entries – 1]

3. <https://www.youtube.com/watch?v=3xCzh3-bm4o&list=RDCMUCNh1egMomGI3hjJoDHExdUg&index=3>

[For Journal Entries – 2]

4. https://www.youtube.com/watch?v=z_KO49Pk3DM

[For Ledger Posting & Trial Balance]

5. <https://www.youtube.com/watch?v=Y4azRCTTWoU>

[For Trading A/c & Profit & Loss A/c]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will get basic understanding of the field of Accounting
CO2 :	Students will be familiar with Accounting Book Keeping Systems
CO3 :	Students will learn concepts of financial accounting and different types of accounts with rules of debit and credit
CO4 :	Students will be able to generate Journal, Ledger, Trial Balance, and Final Accounts
CO5 :	Students will get basic understanding of the field of Cost Accounting

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Accounting	√				
2	Book Keeping System		√			
3	Financial Accounting – I			√		
4	Financial Accounting – II				√	
5	Cost Accounting – I					√
6	Cost Accounting – II					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	1	-	-	2	2	3	1	2	1	1	1	1
CO2	1	2	-	-	2	2	3	1	2	1	1	-	-
CO3	3	3	2	1	1	2	3	-	2	2	1	2	-
CO4	3	3	3	1	1	2	3	-	2	2	1	2	-
CO5	2	3	2	1	3	2	3	1	2	3	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUA101 : PRESENTATION SKILLS

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	50	110
Experiential Learning Components: Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic computer skills and knowledge of English.

Methodology, Pedagogy & Andragogy: The aim of this course is to develop effective presentation skills in students. Students will be taken systematically through the key stages of giving presentations, from planning and introducing to concluding and handling questions. The topics for the presentation will range from academic presentations to research presentations.

Outline of the Course:

Week No	Practical	Description
1	Introduction	Knowing the Purpose, Knowing the Audience, Internal and External Presentation, Presentation to Heterogenic Group, Greetings, Introducing self and peers, Asking and sharing information
2	ICT tools for presentation – Brief introduction	Microsoft PowerPoint, Canva, Prezi, AI-powered presentation tools
3	Presentation using Canva	Introduction to Canva, Creating, presenting, sharing and downloading presentation
4	Presentation design in Canva	Templates, layouts, Presenter Notes, Duration, Timer
5	Basic design elements in Canva	Shapes, Graphics, Charts, Tables
6	Advanced design elements in Canva	Photos, Videos, Audio, Frames, Grids, Stickers

7	Intermediate features of PowerPoint	Using Screenshot, Inserting objects, video and audio in presentation, Screen recording
8	Advanced features of PowerPoint	Using Hyperlink and Action, Slide Show Set-up, Presenter View, Master views
9	Cloud-based presentation using Google Slides	Creating a new presentation, importing slides, Sharing, Printing, Downloading in different formats, Version History
10	Basic features of Google Slides	Slideshow, Motion, Theme Builder, Inserting image, audio, video, shape
11	Advanced features of Google Slides	Spell-check, Linked objects, Q-A History, Notification Settings, Accessibility, Activity Dashboard
12	Engaging audience using feedback-gathering tools	Use of menitimeter for Word Clouds, Live Polls, Scales, Ranking and Pin It
13	Ethical practices	Using Citations/References for texts, images and other type of information, Plagiarism check
14	AI and Presentation	Use of Generative AI tools such as ChatGPT, CoPilot etc. in presentations
15	Select case-study	Examples of best presentations.
Total hours: 60		

Core Books:

1. The Presentation Book : How to Create It, Shape It and Deliver It! Improve Your Presentation Skills Now, 2nd Edition, Pearson Education.
2. Beginners Guide for Canva 2024, by Karla J Kane. ISBN: 979-8875801853

Reference Books:

1. Microsoft PowerPoint Guide for Success by Kevin Pitch, Top Notch International. ISBN:978-1915331489
2. Getting Started With Google Slides: A Practical Guide to Cloud-Based Presentations by Scott La Counte, SL Editions ISBN:978-1629179513

Web References:

1. <https://www.canva.com/education/students/>
2. <https://support.microsoft.com/en-us/powerpoint>
3. <https://support.google.com/a/users/answer/9282488>
4. <https://www.mentimeter.com/features>

Course Outcome: Upon successful completion of the course, student will be able to:

CO1:	understand key elements of an effective presentation and ICT tools for presentation.
CO2:	acquire in-depth knowledge of one of the popular presentation tools.
CO3:	develop intermediate and advanced skills in alternative presentation tool.
CO4:	familiarize with cloud-based presentation tool.
CO5:	create effective presentation with ethical practices and enhance it with AI tools.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	3	-	2	1	-	-	2	-	-
CO2	2	-	3	-	2	1	-	-	2	2	3	1	1
CO3	-	3	-	2	-	1	3	2	-	-	-	3	2
CO4	2	1	2	2	2	1	2	-	2	-	2	3	2
CO5	-	2	2	3	1	-	2	2	2	2	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUI101 : Summer Internship and Viva – I

Objective and Scope:

Students who choose to exit the programme after the completion of the first two semesters must undertake internships, community engagement programmes and field based learning/minor projects during the summer term. This internship, worth 04 credits, is a requirement for the award of the Undergraduate Certificate. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. This provision ensures that students gain valuable practical experience and industry exposure, complementing their academic learning and enhancing their career prospects. Following are the intended objectives of internship training:

- **Practical Experience:** Provide hands-on experience with real-world projects to apply theoretical knowledge gained during the course.
- **Skill Development:** Enhance technical skills, such as programming, software development, database management, and other relevant IT skills.
- **Industry Exposure:** Familiarize students with the working environment of the IT industry, including understanding company culture, workflows, and professional practices.
- **Professionalism:** Instil a sense of professionalism and work ethic, including adhering to deadlines, following instructions, and working collaboratively.
- **Transition to Employment:** Facilitate a smoother transition from academic life to professional careers by bridging the gap between theory and practice.

Guidelines of Internship:

The Summer Internship shall be of 15 (Minimum 60 hours) duration and will be undertaken during the Summer Vacation. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. It is mandatory for the students to seek written approval from the coordinator about the topic and the organization before commencing the Summer Internship. During Summer Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or phone. Students must maintain regular reports detailing the work completed, challenges faced, and learning experiences. Students must submit a final report and presentation summarizing the internship experience, project outcomes, and key learnings to guide. The technical specifications for report preparation is as under:

Technical details:

1. The report shall be printed on A-4 size white paper.
2. 12 pt. Times New Roman font shall be used with 1.5-line spacing for typing the report.
3. 1" margin shall be left from all the sides.
4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be spiral bound as per the standard format of the cover page given by the Institute.
6. The report should include a Certificate (on company's letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Principal of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.

Evaluation:

The evaluation of Internship will be done as per the following criteria:

Sr. No.	Evaluation Criteria	Marks
1	Summer Internship Report	50
2	Presentation and Viva-voce examination	50
TOTAL MARKS		100

Students must secure 36% passing marks in both components to qualify for the Certificate.

Internship and Exit:

Any student intending to exit the course must complete the summer internship (Including scoring at least 36% marks in the evaluation) before exit. A student intending to exit shall have to submit an application to Principal through Counsellor before even semester university theory examination. However, any such student desiring to withdraw the exit option may do that before the last paper of theoretical examinations of even semester.

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**BCA PROGRAMME
(3rd SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-III

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SSE		
CAUC201	Fundamentals of Data Structures and Algorithms	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC202	Fundamentals of Operating Systems	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUS201 CAUS202	(Select any one) 1. Computer Oriented Management System 2. Computer Oriented Numerical Methods	Theory	50	50	-	-	04	SEC
CAUA201	Life Management	Theory	25	25	-	-	02	AEC
	University Elective-I	Practical	-	-	25	25	02	MDC
HSUV201	Creativity, Problem Solving and Innovation	Practical	-	-	25	25	02	VAC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

University Elective - I			
Sr. No.	Course Code	Course Name	Department/Faculty
1	MEUE201	Engineering Graphics and Design	Engineering
2	EEUE201	Fundamentals Of Electrical Engineering	Engineering
3	CLUE201	Environment and Development	Engineering
4	ECUE201	Scientific Computing Using Matlab	Engineering
5	CEUE201	The Joy of Computing using Python	Engineering
6	ITUE201	Introduction to Quantum Computing	Engineering
7	CSUE201	Python For Data Science	Engineering
8	AIUE201	An Introduction to Artificial Intelligence	Engineering
9	BMUD201	Money And Banking	Management
10	CAUD203	Human Computer Interactions	Computer Science
11	PTUD191	Basics of Health Promotion and Education Intervention	Physiotherapy
12	NRMD251	First Aid Masterclass- A complete guide to first aid	Nursing

CAUC201: Fundamentals of Data Structures and Algorithms

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: Foundation of C and C++ Programming, Fundamentals of Object Oriented Programming.

Methodology, Pedagogy & Andragogy: Theory lectures focus on fundamental concepts of data structures and algorithms, emphasizing the selection of the most efficient structures for various applications. Andragogical principles are integrated to promote self-directed and practical learning for learners. For practical teaching of Data Structures and Algorithms, the approach combines hands-on programming exercises with real-world case studies to reinforce the selection of efficient structures for various applications. This blended method enhances both theoretical understanding and practical skills, ensuring learners can effectively apply concepts in real-world scenarios.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1.	Fundamentals of Data Structures and Algorithms	06
2.	Arrays and Linear Data Structures	10
3.	Advanced Dynamic Data Structures	12
4.	Tree-Based Data Structures	11
5.	Graph Theory and Applications	11
6.	Sorting and Searching Techniques	10

Total Hours (Theory) : 60
Total hours (Practical) : 60
Total Contact Hours : 120

Detail Syllabus:

Unit I: Fundamentals of Data Structures and Algorithms Hours 06

Atomic and composite data, Data type, Data object, Data Structure, Abstract Data type, Types of Data Structures, Introduction to Algorithms, Relationship among Data, Data Structures and Algorithms, Analysis of Algorithms, space and time complexity algorithm

Unit II: Arrays and Linear Data Structures Hours 10

Linear Data Structures using sequential organization, Representation of Stacks using sequential organization, Applications of Stack, Application of Recursion, Concepts of Queues, Realization of Queues using sequential organization, circular queue, Multiqueue, Deque, Priority Queue, Applications of Queue.

Unit III: Advanced Dynamic Data Structures Hours 12

Linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, types of link list , Applications of linked list, operation on link list.

Unit IV: Tree-Based Data Structures Hours 11

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree and algorithms, Types of Binary Trees , AVL trees,

Unit V: Graph Theory and Applications Hours 11

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure, minimum spanning tree.

Unit VI: Sorting and Searching Techniques Hours 10

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Search Trees.

Outline of the Course (Practical):

Week No	Practical	Description
1	Basic of C and C++	foundational concepts in C and C++.
2	Two dimension array	Understand and utilize arrays effectively.
3	Stack	Implement stack operations including push, pop, and peek.
4	Stack and its application	Cover infix, prefix, and postfix notations.
5	Queue and Circular queue	Implement queue operations such as insert, delete, and display.
6	Linear and Binary Search	Apply various searching techniques on arrays.
7	Merge sort, Bubble Sort	Implement and compare different sorting techniques to sort data.
8	Selection sort , Insertion Sort,	
9	Quick sort	
10	Single Link List	Perform insert and delete operations at the beginning, end, and middle of link list data structures.
11		
12	Double Link List	
13		
14	Circular Single Link List	
15	Circular Double Link List	

Total hours (Practical) : 60

Core Books:

1. Jean-Paul Tremblay, Paul G. Sorenson: An introduction to data structures with applications, 2nd Edition, Tata McGraw Hill Publications, 1991.
2. D. Samanta: Classic Data Structures, 2 Edition, PHI Publications, 2002.

Reference Books:

1. Varsha h. Patil : Data Structures Using ndC++, 1st Edition, Oxford University Press, 2012.
2. Yashvant Kanethkar : Data structures through C++, 2nd Edition, BPB Publications, 2003.

3. Mark Allen Weiss: Data Structures and Algorithm Analysis in C++, 3rd Edition, Pearson Education, 2009.
4. Reema Thareja: Data Structures using C, 1st Edition, Oxford University Press, 2012.

Web References:

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html [For materials of Data Structures and Algorithms]
2. <http://www.cs.usfca.edu/~galles/visualization/> [For Data Structures Visualization]
3. <https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html> [For Graph Data Structures]
4. <http://interactivepython.org/runestone/static/pythonds/Trees/trees.html> [For Tree Data Structures]
5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and Searching Cookbook]
6. <http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf> [For Hashing Materials]
7. <http://nptel.ac.in/video.php?subjectId=106102064> [For Data Structures videos]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand and explain the basic concepts and importance of data structures and algorithms.
CO2 :	Implement and utilize array-based and linear data structures effectively.
CO3 :	Analyze and apply advanced dynamic data structures such as linked lists.
CO4 :	Construct and manipulate various tree data structures including binary trees and AVL trees.
CO5 :	Apply graph theory concepts to solve complex problems and understand real-world applications.
CO6:	Implement and compare different sorting and searching algorithms for optimal performance.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		CO1	CO2	CO3	CO4	CO5	CO6
1	Fundamentals of Data Structures and Algorithms	√					
2	Arrays and Linear Data Structures		√				
3	Advanced Dynamic Data Structures			√			
4	Tree-Based Data Structures				√		
5	Graph Theory and Applications					√	
6	Sorting and Searching Techniques						√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	-	1	3	1	-	-	1	1	2	1
CO2	3	2	3	1	1	3	1	-	-	1	2	2	1
CO3	3	2	3	1	1	3	1	-	-	1	2	2	1
CO4	3	2	3	1	1	3	1	-	-	1	2	2	1
CO5	3	2	3	1	1	3	1	-	-	1	2	2	1
CO6	3	2	3	1	1	3	1			1	3	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC202 : Fundamentals of Operating Systems

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Component (Theory): Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: In order to achieve the course objectives, students will be introduced to the basic operating system concepts and basic functions. Students will be offered appropriate case studies in order to provide actual exposure to theoretical operating system ideas. During the practical session, student will commence the learning from MS-DOS commands in windows environment. The student will also get hands on working with basic Linux commands. Finally using base of Linux commands, shell scripts would be written to demonstrate operating system programming.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Operating System Fundamentals	09
2	Memory Management	11
3	Processor Management	11
4	Device Management	10
5	File Management	10
6	Introduction to Shell Script	09

Total Hours (Theory): 60
Total hours (Practical): 60
Total Contact Hours:120

Detail Syllabus:

Unit I: Operating System Fundamentals

Hours 09

What is an operating system, Brief history of an operating system, Pyramid structure of an operating system, Operating system hardware and software, Evolution of an operating system, types of operating system, Batch system, Interactive system, Real time system, Hybrid system, Shell, compiler, assembler, interpreter.

Unit II: Memory Management

Hours 11

Memory management of early system, Fixed partition, Dynamic partition, Relocatable dynamic partition, Best fit and first fit allocation method, De-allocation, Paged memory allocation, Demand paging allocation, segmented memory allocation, Virtual memory, cache memory, Associative memory, Page replacement policies.

Unit III: Processor Management

Hours 11

What is job and Process, Multiprogramming concept, Process scheduler, Job scheduler, various process states, Thread control, Process Control Block (PCB), Processor scheduling policies, FCFS, SJN, SRT, RR, EDF, Multiple level queues.

Unit IV: Device Management

Hours 10

Types of devices, dedicated device, Virtual device, shared device, I/O devices in cloud, Sequential Access Storage Media, Direct access storage media, Device handler seek strategies, FCFS, SSTF, SCAN and LOOK strategies, disk storage.

Unit V: File Management

Hours 10

Fundamentals of File management, File naming conventions, Fixed length and variable length records, Physical file organization, Physical storage allocation, File access methods, File control matrix, File compression.

Unit VI: Introduction to Shell Script

Hours 09

Shell Script, Shell environment, Making Script interactive, Command Line Argument, Evaluating Expression, Operator, Control Statement, Looping statement.

Outline of the Course:

Week No	Practical	Description
1	Disk operating system -Commands	Introduction to MS DOS environment with basic commands
2	Introduction to Linux Editor	Introduction to Cygwin as Linux Editor along with its usage
3	Linux Operating System-Commands	Using basic commands of Linux
4	Linux Operating System-Advanced Commands	Implementation of advanced Linux commands used in industry
5	Vi editor-Create Text file	Practice of writing and modifying various files in Vi editor
6	Introduction of Shell Programming-Create .sh file and Execute file	Creating simple shell script to get introduced with operating system programming
7	Variable and Rules for variable	Creating and using variables, and constants in shell scripts
8	Interactive Shell Script, Working with Command Line Arguments	Practicing shell scripts which are responsive to the user input
9	Working with expression	Arithmetic expressions in shell script
10-11	Conditional statement-IF, Working with Operator	Learning conditional statements such as if, elif, case. Using arithmetic, relational, and logical operators.
12-13	Looping Statement-While	Writing shell scripts with while loops with real life examples
14-15	Looping Statement-Until, For	Writing shell scripts with until and for loops with real life examples

Total hours (Practical) : 60

Core Books:

1. Ann McIver McHoes, Ida M Flynn: Understanding Operating System, 7 Edition, Cengage Learning, 2014.
2. William Stallings: Operating Systems Internals and Design Principles, 5th Edition, PHI, 2005.
3. Sumitabha Das: Unix concepts & application, 4th Edition, Tata McGraw Hill, 2010.
4. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.

Reference Books:

1. Silberschatz: Operating System Concepts, 5th Edition, John Wiley & Sons (ASIA) Ltd., 2008.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. KJ.George, Operating System Concepts and Principles, Sroff Publishers, 2003.

Web References:

1. <https://study.com/learn/operating-system.html> [Operating System Video Lessons]
2. <https://www.studytonight.com/operating-system/> [Operating System Tutorial]
3. <http://www.ics.uciedu/~ics43/lectures.html> [Lecture notes of OS]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand the fundamental concepts of an operating System
CO2 :	To understand memory allocation and de-allocation methods
CO3 :	To study fundamentals of processor and its scheduling policies
CO4 :	To study various supported device and file access technologies
CO5 :	Learning basic shell programming

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Operating System Fundamentals	√				
2	Memory Management		√			
3	Processor Management			√		
4	Device Management				√	
5	File Management				√	
6	Introduction to Shell Script					√

Course Articulation Matrix:

	PO1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2
CO1	1	1	2	2	-	2	2	-	1	1	-	3	2
CO 2	3	2	2	2	-	2	2	-	1	1	-	3	2
CO 3	3	2	2	2	-	2	2	-	1	1	-	3	2
CO 4	3	2	2	2	-	2	2	-	1	1	-	3	2
CO 5	2	2	3	3	-	2	3	-	1	1	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUS201 : Computer Oriented Management System

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The theory sessions will emphasize real-time systems. To enhance comprehension, we will provide illustrations using actual data across various systems. The sessions will meticulously explain the system's functionality through detailed flow descriptions. To bridge the gap between theoretical knowledge and practical application, we have to schedule industrial visits. These visits are designed to offer students a tangible correlation between their studies and real-world practices. Students should be assigned case studies to facilitate the practical application of their acquired knowledge. These case studies aim to deepen understanding and foster the ability to apply theoretical concepts to real-world scenario.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Basics of Information System	09
2	Architecture and Design of Information System	08
3	Transaction Processing System	09
4	Management Information System	06
5	Decision Support System & Executive Support System	18
6	Introduction to ERP and trending technologies	10

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I : Basics of Information System

Hours 09

Role of data and information, Types of Systems, Types of information, Quality of Information, what is an Information System, Functions of Information System, classification of an Information System, Major Types of Information Systems, Interrelationship of Information Systems

Unit II : Architecture and Design of Information System

Hours 08

Meaning of Architecture, development and maintenance of information system, centralized and decentralized systems, factors of success and failure in information system and risk involved with information system

Unit III : Transaction Processing System

Hours 09

Introduction of Transaction Processing System (TPS), Transaction Processing System (TPS) Architecture, Transaction Processing System (TPS) Life cycle, Methods of Transaction Processing System (TPS), Information System Availability Control, Illustrations and live demonstration of TPS such as Banking System, Railway reservation system etc.

Unit IV : Management Information System

Hours 06

Introduction to Management: Approaches of Management, Function of Management, System from Functional Perspective, Reports of Management Information System, A business perspective of MIS, Dimensions of MIS, contemporary Approaches to Information System.

Unit V : Decision Support System and Executive Support System

Hours 18

Business value of Improved Decision making, Types of Decision, Decision making Process, The difference between MIS and DSS, Components of DSS, System for Decision Support Group Decision Support System, Business value of GDSS. Important Dimensions of knowledge, Organizational learning and Knowledge Management, The Knowledge Management value chain, Overview of different types of Knowledge Management Systems, Characteristics of ESS, The Role of ESS in the Firm, Business value of ESS, Monitoring Corporate Performance, Working Examples of ESS.

Unit VI : Introduction to ERP and trending technologies

Hours 10

An overview of ERP, Basic ERP Concepts, Risk and Benefits of ERP, ERP and Related Technologies, Business Intelligence, Some case studies on ERP or Information system

Core Books:

1. K. C. Laudon and J. P. Laudon: Management Information Systems, 12th Edition, Pearson Education.
2. Alexis Leon: ERP Demystified, Second Edition, Tata McGraw-Hill.

Reference Books:

1. W.S. Jawadekar: Management Information Systems, 2nd Edition, Tata McGraw-Hill.

Web References:

1. <http://www.slideshare.net/NorazilaMat1/laudon-mis12-ppt01-16595885>
[Function of Information system, A business perspective of MIS, Dimensions of MIS, Contemporary Approaches to Information System]
2. <http://www.uh.edu/~mrana/try.htm> [Types and Functions of Information system]
3. <http://bisom.uncc.edu/courses/info2130/Topics/istypes.htm> [Types of Information system]
4. <http://kalyan-city.blogspot.com/2011/05/levels-of-management-top-middle-and.html> [classification of information system]
5. http://my.safaribooksonline.com/book/management/9780470916803/operational-planning-and-control-systems/management_levels_comma_functions_comm [classification of information system]
6. <http://ambarwati.dosen.narotama.ac.id/files/2011/05/FIS-2011-w2.pdf>
[Transaction processing system]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand basic of data & information, types of information system, function of information system and its quality.
CO2 :	To understand designing and architecture of information system
CO3 :	To understand transaction processing concepts and its working.
CO4 :	Learning Management Information system, its types and its implementation. To gain understanding of Decision Support Systems.
CO5 :	To learn Knowledge Management, Executive Support Systems, ERP and its technology.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basics of Information System	√				
2	Architecture and Design of Information System		√			
3	Transaction Processing System			√		
4	Management Information System				√	
5	Decision Support System & Executive Support System				√	
6	Introduction to ERP and trending technologies					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	3	1	-	3	-	1	2	3	3	-	1	1
CO2	1	3	1	-	2	1	-	2	3	3	2	-	1
CO3	1	3	1	1	3	3	-	2	3	3	-	-	1
CO4	1	2	1	-	2	3	1	2	2	2	1	1	1
CO5	1	3	1	-	3	3	-	1	3	3	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUS202 : Computer Oriented Numerical Methods

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The course covers key topics like finding roots of equations, solving linear equations, interpolation, and solving differential equations. Teaching includes lectures, interactive problem-solving, and project-based assessments with continuous feedback. The focus is on practical applications and real-world problems, using flexible and engaging teaching methods. The aim is to make the content relevant and useful for students' careers and interests.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Numerical Methods	08
2	Interpolation	09
3	Numerical Integration and Differentiation	10
4	Solution of Linear Equation	12
5	Solution of Differential Equations	11
6	Solution of Partial Differential Equations	10

Total Hours (Theory): 60**Total Contact Hours: 60**

Detailed Syllabus:

Unit - I: Numerical Methods

Hours: 08

Introduction to Numerical Methods, Various methods such as Bisection, False-Position and Newton Raphson methods

Unit - II: Interpolation

Hours:09

What is an interpolation, Difference tables, Newton forward and backward Interpolation formula, Lagrange's formula, Newton's Divided Difference Formula?

Unit - III: Numerical Integration and Differentiation

Hours:10

Numerical Integration, Trapezoidal rule, Differentiating Continuous function (Two point and Three-point formula), Tabulated function and its maxima and minima.

Unit - IV: Solution of Linear Equation

Hours:12

Solution of Linear Equation, Gauss-Jordan Elimination methods, Gauss - Seidal and Gauss Jacobi Interactive methods

Unit - V: D Solution of Ordinary Differential Equations

Hours:11

Solution of Ordinary Differential Equations, Taylor Series and Euler Methods, Runge-Kutta Methods.

Unit - VI: Solution of Partial Differential Equations:

Hours:10

Review and examples of partial differential equations, classification of partial differential equations, Difference equation, Laplace's equation and Poisson's equation

Core Books:

1. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2012.
2. S.S. Sastry : Introductory methods of Numerical Analysis, Prentice Hall of India, 4th edition 2006

Reference Books:

1. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt.Ltd, 3rd edition, 2016
2. M. Ray, Har Swarup Sharma, Sanjay Chaudhary: Mathematical Statistics, Ram Prasad and sons, 11th edition
3. B S Grewal: Numerical Methods In Engineering & Science (With Programs In C, C++ And MATLAB), Khanna Publishers, 10th edition, 2014

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to use Numerical methods in computer technology
CO2 :	To learn Interpolation
CO3 :	To Learn the concepts of Numerical Integration and Numerical Differentiation
CO4 :	To understand linear equations
CO5 :	To learn techniques to solve ordinary differential and partial differential equations

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Numerical Methods	√				
2	Interpolation		√			
3	Numerical Integration and Differentiation			√		
4	Solution of Linear Equation				√	
5	Solution of Differential Equations					√
6	Solution of Partial Differential Equations					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO 2
CO1	3	3	2	1	-	2	-	-	-	2	-	2	1
CO 2	3	3	2	1	-	2	-	-	-	2	-	2	1
CO3	3	3	2	1	-	2	-	-	-	2	-	2	1
CO 4	3	3	2	1	-	2	-	-	-	2	-	2	1
CO5	3	3	2	1	-	2	-	-	-	2	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUA201 : Life Management

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Theory	02	30	25	55
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of life management. Students will be introduced life styles and its basic, time management, work efficiency, yoga, physical health management, stress and mental health management, and personality skills. Students will give experimental learning exposure in form of assignment and case study.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Life Style and its Basics	06
2	Work Efficiency	04
3	Personality Skills	06
4	Time Management for students	05
5	Physical health management	04
6	Stress and mental health management	05

Total Hours (Theory):30
Total Contact Hours:30

Detail Syllabus:

Unit I : Life Style and its Basics

Hours 06

Purpose of life and its dimensions, Importance of Self-Evaluation; (Daily routine, Food habits, Dressing Sense, Habit formation, Company, Etiquettes), Duties and Commitment of Self, Family and Society, Adjustment with Self and Environment

Unit II : Work Efficiency

Hours 04

Positive way of thinking, Tools & techniques for Positive thinking, Karma and Karma Phal Sidhanta, Behavioral Skill

Unit III : Personality Skills

Hours 06

Self-Assessment Techniques, Adjustment Skills, Art of Positive Thinking, Developing Self Confidence,

Unit IV : Time Management for students

Hours 05

Importance, Steps, Barriers and solutions of Time Management, Reading and Writing Skill, Making Notes and Conceptual Clarity of Subject topic.

Unit V : Physical health management

Hours 04

Importance of physical health and exercise, importance of yoga, physical problems, exercise for health improvements

Unit VI : Stress and mental health management

Hours 05

Importance of mental health, spirituality, improving concentration, importance of Dhyan, Memory problem and Memory Boosting techniques, causes and effects of stress, importance of Prathana, Upasna, Sadhna, and Aradhna

Core Books:

1. Time Management, McGraw-Hill, 2003
2. How To Manage Your Life, Vishal Tatwaved, 2012.

Reference Books:

1. Living Stress Free - The Secret of How to Manage Stress and Live Life Fully, Sonali Perera, SP MEDIA MARKETING.
2. TIME MANAGEMENT, BRIAN TRACY, American Management Association, 2013.

Web References:

1. Life management: a holistic approach to make the most of your life, Dr. Hannah Rose <https://nesslabs.com/life-management>
2. Life Management, <https://www.sciencedirect.com/topics/computer-science/life-management>
3. <https://edynamiclearning.com/course/health-1-life-management-skills/>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will learn basics of life managements
CO2 :	The students will be familiar with various styles
CO3 :	Students will get the knowledge of time management and stress management
CO4 :	Students will get various aspects of physical and mental health
CO5 :	They will be able to develop personality and good habits

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Life Style and its Basics	√				
2	Work Efficiency		√			
3	Personality Skills					√
4	Time Management for students			√		
5	Physical health management				√	
6	Stress and mental health management				√	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	3	1	-	3	-	1	2	3	3	-	1	1
CO2	1	3	1	-	3	1	-	1	3	3	1	-	1
CO3	1	3	1	1	3	3	-	2	3	3	-	-	1
CO4	1	2	1	-	2	3	1	2	2	2	1	1	1
CO5	1	3	1	-	3	3	-	1	3	3	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUD201 : Introduction to Web Designing

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic concepts of web.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for design and develop entire web sites using several web designing tools and HTML scripting language.

B. Outline of the Course:

Week No	Practical	Description
1	Basic Construction of an HTML & HTML5 Page, HTML Text Editors, HTML Building blocks.	Learn basic concepts of web page design.
2	Basic HTML tags – Text formatting tags, Heading tags, paragraph tags, division tag, Span tags, Break line tag, Binding space in HTML	Get familiar with different tags and attributes which is used to add the text in a web page.
3	Controlling Font size and color, Adding Table – Table tag.	Learn user interface designing using Table Layout, change font style, color etc.
4	Grouping and merging table rows and columns. Adding List – ordered, unordered and definition list.	Able to display the data in various formats and design different types of list in the table.
5	Adding Image, Adding links - Changing link colors, Email link.	Learn to add images and link in the web page.

6	Embedding audio and video, Adding Field sets.	Able to add multimedia in web page using internal and external methods.
7	Adding Form: Form tag, Action, Method. Basic form controls.	Learn basic form elements and when and where to apply it.
8	Textbox, Password field, Radio Button, Checkbox, Multiple line text area, file upload.	Use various types basic form controls with types of form input elements.
9	Various types of buttons – submit, reset, image, simple button.	Learn various types of buttons to make it more interactive.
10	HTML5 input elements - Color, Number, Date, Month.	Learn new input types in HTML5.
11	Week, Time, date time-local, Email, url, range, search.	Learn to add defining inputs and minimum and maximum range to make the form attractive.
12	Data list, output, progress, meter.	Learn to add list and provide progress horizontal view and range with output.
13	Regular expression, HTML5 new attributes – Placeholder, Required.	Learn HTML 5 form element to provide user side web form validation.
14	Pattern, Autocomplete, Autofocus, novalidate, formnovalidate.	Learn HTML 5 new form properties.
15	formaction, formmethod, spellcheck, contenteditable.	Learn HTML 5 to user submit the form and redirect the form data to selective method.

Total hours: 60

Core Books:

1. Powel Thomas: HTML - The Complete Reference, 3rd Edition, McGraw-Hill Education – Europe, 2018.
2. DT Editorial Services: HTML 5 Black Book, 2nd Edition, Dreamtech Press, 2016
3. Matt West: HTML 5 Foundations, Wiley publication, 2013
4. Farrar: HTML Example book, BPB, 2007.

Reference Books:

1. David DuRocher : HTML and CSS QuickStart Guide, Paperback, 2021
2. Laura Lemay, Rafe Colburn, Jennifer Kyrnin : Mastering Html, CSS &

- JavaScript Web Publishing, Paperback, 2016.
3. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
 4. Shelly Woods: HTML introductory concepts and techniques, 5th Edition, Course Technology, 2009.
 5. Jon Duket: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.

Web References:

1. www.w3schools.com/html [For HTML tutorials]
2. html.net/tutorials/html/ [For HTML tutorials]
3. www.htmlgoodies.com/ [For HTML resources]
4. <https://developer.mozilla.org/en-US/learn/html> [For HTML best practices]

Course Outcome: Upon successful completion of the course, student will:

CO1:	Able to understand basic construction of HTML & HTML5 document and started using basic tags.
CO2:	Able to Add content using various tags and format the content in table & list.
CO3:	Able to design links and multimedia elements such as image, audio, video.
CO4:	Able to design a web page form using HTML & HTML5.
CO5:	Able to apply validation using HTML5 attributes.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	
CO1	1	2	1	3	2	2	1	1	-	1	1	1	1	CO1
CO2	-	1	2	3	1	3	1	-	-	2	2	2	2	CO2
CO3	1	1	2	3	1	3	1	-	-	2	2	2	2	CO3
CO4	1	1	2	3	1	1	1	1	-	1	-	1	1	CO4
CO5	2	2	3	3	2	3	3	2	1	2	2	3	2	CO5

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**TEACHING SCHEME &
DETAILED SYLLABUS

FOR

BCA PROGRAMME
(4th SEMESTER)
AS PER NEP 2020

EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-IV

Course Code	Course Name	Theory/ Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SSE		
CAUC203	System Analysis and Design	Theory	50	50	-	-	04	Major Core
CAUC204	Programming the Internet	Practical	-	-	25	25	02	Major Core
CAUC205	Open Source Technology	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE201 CAUE202	(Select any one) 1. Advanced Networking 2. Artificial Intelligence (AI) Fundamentals	Theory	50	50	-	-	04	Minor
CAUA202	Social Media and Blog Writing	Practical	-	-	25	25	02	AEC
	University Elective-II	Practical	-	-	25	25	02	MDC
HSUV203	Indian Knowledge System	Practical	-	-	25	25	02	VAC
CAUI201	Summer Internship and Viva - II	VIVA	50	50			04	Exit Course
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

University Elective - II			
Sr. No.	Course Code	Course Name	Department/Faculty
1	MEUE202	Nature and Properties of Materials	Engineering
2	EEUE202	Solar Energy Engineering and Technology	Engineering
3	CLUE202	Ecology and Environment	Engineering
4	ECUE202	Introduction to Internet of Things	Engineering
5	CEUE202	Software Conceptual Design	Engineering
6	ITUE202	Ethical Hacking	Engineering
7	CSUE202	Google Cloud Computing Foundations	Engineering
8	AIUE202	Social Network Analysis	Engineering
9	BMUD251	Economics Of Health And Health Care	Management
10	CAUD204	Modern Application Development	Computer Science
11	PTUD192	Ergonomics Workplace Analysis	Physiotherapy
12	NRMD261	Mindfulness and Well-being: Living with Balance and Ease	Nursing

CAUC203 : System Analysis and Design

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding how Information Systems are designed and constructed and how major modifications are made to existing systems, analyze business needs for information and develop an appropriate strategy to solve a particular business problem, understand the issues associated with physical design, utilize some of the tools used by systems analysts in the design of Information Systems, a significant portion of the course will deal with computer aided system engineering (CASE) tools and techniques in the context of a case study.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to System Development	08
2	Classical Systems Development Methodology	11
3	Structured Systems Development Methodology	11
4	Prototype Development Methodology and Tools	11
5	Systems Design	10
6	Systems Implementation & Maintenance	09

Total Hours (Theory): 60
Total Contact Hours:60

Detailed Syllabus:

Unit I: Introduction to System Development

Hours: 08

Concept of system, Basic components, Information systems categories, Business Information Systems, Examples of real-life systems, System analyst – meaning, role and need, Users, Need of information system development, Application development portfolio, Integration of portfolio, Controlling and monitoring committees.

Unit II: Classical Systems Development Methodology

Hours: 11

Classical Systems Development Life Cycle (SDLC) method, Requirement determination, Fact finding techniques, Modern methods for determining systems requirements, Joint Application Development Program, Tools for documenting procedures and decisions, Tools for systems development, understanding the format of system requirement specification.

Unit III: Structured Systems Development Methodology

Hours: 11

Structured Analysis Development Strategy, Data Flow Diagrams (DFDs) for analysis and design, Physical and logical DFDs, Data Dictionary.

Unit IV: Prototype Development Methodology and Tools

Hours: 11

Purpose of Application Prototyping, steps in prototype method, Use of prototypes, Tools for prototyping, Strategies for prototype development, misconceptions about prototyping, Role of tools, Categories and use of automated tools, using a CASE tool

Unit V: Systems Design

Hours: 10

Analysis to design transition, Design of output system, Design of input system, Design of security and control, Design of user interface and Design of Database Interaction.

Unit VI: Systems Implementation & Maintenance

Hours: 09

Designing reliable and maintainable systems, design of software, software design and documentation tools, managing quality assurance, user training, training methods and conversion.

Core Books:

1. James A Senn : Analysis and Design of Information System, McGraw Hill International, 2003.
2. Kendall and Kendall: Systems analysis and Design, 5th Edition, Prentice-Hall of India Private Limited, 2003.

Reference Books:

1. V. Rajaraman: Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
2. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
3. Tuthill and Leavy: Knowledge Based Systems: Mangers Perspectives: Tab professional and Reference Books, 1991.

Web references:

1. <http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-and-design-sad.html>
2. <http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys>
3. <http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWIeo>
4. <http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html>
5. <http://faculty.washington.edu/ytan/is460/notes/LN11.pdf>

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the basics of system, type of information systems, system analyst- meaning, role and need, portfolio development of the system, the methods or ways to review and select the project.
CO2	Understand the concept of SDLC, the fact finding techniques, detailed study of modern methods for collecting the requirements of system to be developed, Decision and procedure documentation and a brief introduction to the SRS document.
CO3	Learn the concept of structured analysis, detail study of DFD and its types and the study of data dictionary.
CO4	Gain understanding of prototypes, its uses and tools for development, clarification of the misconception about the prototype and categories of tools including the use of automated tools like case tools.
CO5	Be able to learn objective of collecting data, methods to collect data for designing the input and output interface, the concept and objective of database design as well as designing of reliable and maintainable systems with quality assurance

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1.	Introduction to System Development	√				
2.	Classical Systems Development Methodology		√			
3.	Structured Systems Development Methodology			√	√	
4.	Prototype Development Methodology and Tools					√
5.	Systems Design					√
6.	Systems Implementation & Maintenance					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	2	2	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	3	3	3	3	3	3	3	3	3	2	3	3	2
CO4	3	3	3	2	3	2	3	2	2	2	2	3	2
CO5	1	2	2	2	3	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC204 : Programming the Internet

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, Case Study					

Prerequisites: CAUC103 – Fundamentals of Web Designing

Methodology, Pedagogy & Andragogy: This subject focuses on hands-on experiments to create dynamic and interactive web sites using HTML5, CSS3 and JavaScript.

Outline of the Course:

Week No	Practical	Description
1	CSS Styling the web - Applying CSS Building blocks –CSS Selectors, Cascade & inheritance, CSS Properties, CSS Box model.	Student can apply various types of CSS with its basic properties.
2	Creating Layout - Header, footer, navigation menu, and sub-navigation. Floats.	Student can create different layout using semantic elements of HTML5 & CSS classification properties.
3	Creating Layout - Flexbox, Introduction to Flexbox, Flex Container and Flex Items, Flexbox Properties and Layout Techniques.	Student can create flexible and responsive web layouts. Its ability to distribute space and align items effectively makes it a vital part of modern web design.
4	Creating Layout - Introduction to CSS Grid Layout- Grid Container and Grid Items, multi column layout	Student can Understand its properties and axes can significantly improve the adaptability and aesthetics of your web pages.
5	Creating Image Gallery and Animation - 2D/3D Transformation, Transitions and Animation,	Student can create layout image gallery & animation.
6	CSS Media Query	<i>Student can create different layouts depending on the size of the viewport.</i>

7	JavaScript – Adding Internal & External JavaScript	Student can write script for basic Programs.
8	,Working with variables, datatype – Number, String, Boolean.	Student can write script for basic programs using different datatypes.
9	datatype –Array, Object, Operators, Conditional statements, Looping statements	Student can write script for basic programs using array and object, condition, Loop.
10	Adding Dialog boxes in HTML page – Alert, Confirm and Prompt.	Student can able to use dialog boxes to make website alive.
11	Creating user defined function.	Student can able to use in-build and user defined functions of JavaScript.
12	Adding Mouse and Keyboard Event in HTML page, Manipulating String and Date.	Students can make user interaction using various Events.
13	JavaScript Document Object Model hierarchy – Create find and manipulate HTML Element using Objects and methods	Student can explore the DOM Hierarchy.
14	Form validation, Applying Style using JavaScript.	Student can able to use form validations.
15	Creating New window, Accessing & manipulating History of HTML Pages.	Student can apply various objects to make web page dynamic with user interactions.

Total hours: 60

Core Books:

1. Julie C. Meloni, Jennifer Kyrnin : HTML, CSS, and JavaScript All in One, 3rd Edition, Pearson, 2018.
2. Joseph R. Lewis, Meitar Moscovitz : AdvancED CSS, W3C CSS Working Group, Apress, 2020.
3. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
4. MArijin Haverbeke : Eloquent Javascript – A modern Introduction to JavaScript, 3rd Edition, 2018
5. Michael Morrison: Head First JavaScript, O'Reilly, 2014

Reference Books:

1. Andy Budd, Emil Björklund: CSS Mastery 3rd Edition, Apress - Advance Web Standard Solutions.
2. Karl Barksdale, Shane Turner: HTML and Java Scripts Basics, 4th Edition, Course Technology, 2006.
3. Jon Duckett: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.
4. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

Web References:

1. https://www.tutorialspoint.com/css/css3_tutorial.htm [For CSS Basics]
2. <https://www.tutorialrepublic.com/css-tutorial/> [For CSS3 BOX Model & Animation]
3. <https://javascript.info/> [For modern JavaScript tutorials]
4. <https://www.tutorialsteacher.com/javascript/javascript-hoisting> [JavaScript Basics]

Course Outcome: Upon successful completion of the course, student will:

CO1:	Able to apply CSS & its basic properties in web pages.
CO2:	Able to create HTML, CSS layout.
CO3:	Able to apply the transition, transformation and animation in web page.
CO4:	Able to write client side script – JavaScript to perform basic program
CO5:	Able to use DOM objects for dynamic web pages

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	
CO1	1	2	3	3	2	1	2	1	1	2	1	1	1	CO1
CO2	1	3	3	3	2	2	1	1	1	2	3	2	1	CO2
CO3	1	2	2	3	2	1	1	1	1	2	2	1	1	CO3
CO4	2	2	3	3	2	1	1	1	2	1	2	1	1	CO4
CO5	2	2	3	3	2	1	2	1	2	1	2	2	2	CO5

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC205 : Open Source Technology

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: Working knowledge of HTML and basic knowledge of MYSQL.

Methodology, Pedagogy & Andragogy: During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, students will be emphasized to develop dynamic web applications.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Web Development and Open Source Technology	08
2	PHP Basics	10
3	Array and Functions	10
4	Handling Form Data	10
5	Database Connectivity with MYSQL	12
6	PHP Utilities	10

Total Hours (Theory): 60
Total hours (Practical): 60
Total Contact Hours:120

Detail Syllabus:

Unit I: Introduction to Web Development and Open Source Technology

Hours 8

Introduction of open source software, Development philosophy of open source software, pros and cons of open source software, open source vs. close source software, Community support in open source, Version Control, Introduction to Webpage and Website, Static and Dynamic Website, Client & Server Side Scripting, Web Hosting: Shared / Cloud / VPS / Dedicated.

Unit II: PHP Basics

Hours 10

Introduction to PHP, Installation of Apache, MySQL and PHP (OOP approach), How PHP code is parsed, Embedding PHP and HTML, Executing PHP and viewing in Browser, Comments in PHP, PHP variables: static and global variables, Data types, Operators, Conditional Statements, Looping Structure.

Unit III: Array and Functions

Hours 10

Array, Types of Array, Array Functions, Miscellaneous Functions: define, constant, include, require, header, die, exit, Overview of built in functions of PHP: String functions, Date and Time, Math functions, File handling functions, User Defined Functions.

Unit IV: Handling Form Data

Hours 10

HTML Form element & its attributes, Send Form data using GET Method & POST Method, Receive Form data using \$_GET, \$_POST & \$_REQUEST variables, Super Global Variables, PHP form validations.

Unit V: Database Connectivity with MYSQL

Hours 12

OOPs Concepts, PHP Data Objects (PDO), Introduction to MYSQL DB, CRUD operations, Handling Errors, State Management Techniques: PHP Sessions, starting session, modifying session variables, Un-registering and deleting session variable, PHP Cookies, Types of Cookies.

Unit VI: PHP Utilities

Hours 10

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server, Sending mails.

Outline of the Course:

Week No	Practical	Description
1	Student registration page using HTML and CSS	Revised the concepts of HTML and CSS.
2	Installation of XAMPP & generating basic PHP scripts	Get familiar with echo() and print(). Able to create basic business logic.
3	Variables, Data Types and Casting	Able to work with variables, data types and able perform type casting.
4	User Inputs and Conditional Statements	Deal with user inputs. Use of conditional statements.
5	Looping Structures	Able to work with Looping structures like: for, while, do..while and foreach
6	Inbuilt Functions: string, math and date & time	Use various built-in functions available in PHP.
7	PHP Forms	Creating Forms using GET and POST methods. Also use the super global variables \$_GET, \$_POST and \$_REQUEST.
8	PHP Forms & Validations	Creating Forms with server side validations
9	PHP Array	How to create an array? Array Functions
10	Database Connectivity	Understand MYSQL Dashboard. Implement Database connectivity using PDO (PHP Data Object).
11	Database Connectivity	Implement Database connectivity using PDO (PHP Data Object).
12	File Upload and Download	Able to upload, download and view various kind of files and store path into database using \$_FILES.
13	State Management: Session	Creating Session, modifying session variables, Un-registering and deleting session variable.
14	State Management: Cookies. Super Global Variables	Creating Cookies and learning various super global variables like \$_SERVER, \$GLOBALS and \$_ENV
15	PHP Mail	Sending mail using PHP mail() & PHPMailer

Total hours (Practical) : 60

Core Books:

1. Josh Lockhart, Modern PHP, O’Rielly, 2015.
2. Kevin Tatroe, Peter Macintyre, Programming PHP, 4e: Creating Dynamic Web Pages, 4th Edition, O’Rielly, 2020.

Reference Books:

1. Urva Patel, Kaushal Gor, Sanskruti Patel, PHP Hand Book: Crafting Dynamic Web Application, Notion Press Publication, 2024.
2. Matt Zandstra, PHP 8 Objects, Patterns, and Practice: Mastering OO Enhancements, Design Patterns, and Essential Development Tools, Apress, 2021.
3. Luke Welling, Laura Thomson, PHP and MYSQL Web Development, 5th Edition, Pearson Education, 2016.

Web References:

1. <https://www.php.net/> [Official website of PHP]
2. <https://www.geeksforgeeks.org/php-tutorials/> [Lecture notes of PHP]
3. <https://www.w3schools.com/php/default.asp> [Lecture notes of PHP]
4. <https://github.com/PHPMailer/PHPMailer> [PHPMailer Code]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Gain the skills and knowledge for Open Source and Web Hosting.
CO2 :	To utilize knowledge and skills for basics of PHP and configuration of MYSQL Server.
CO3 :	Learn the array and built-In functions in PHP.
CO4 :	Gain understanding of Form handing and utilities in PHP.
CO5 :	Be able to develop dynamic web based application using PHP and MySQL with state management techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Web Development and Open Source Technology	√				
2	PHP Basics		√			
3	Array and Functions			√		
4	Handling Form Data				√	
5	Database Connectivity with MYSQL					√
6	PHP Utilities				√	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	3	-	1	-	1	2	-	-	2	-
CO2	3	3	3	3	-	1	-	1	1	-	-	3	-
CO3	3	3	3	3	-	1	-	1	1	-	-	3	-
CO4	3	3	3	3	-	1	-	1	1	-	-	3	-
CO5	3	3	3	3	2	1	1	1	2	1	1	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE201 : Advanced Networking

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study, Certification courses, etc.					

Pre-requisite: Basics of Computer Networks, Topologies.

Methodology, Pedagogy & Andragogy: The methodology for teaching Advanced Networking integrates a blend of theoretical instruction and practical application, employing both pedagogy and andragogy to cater to diverse learning needs. Pedagogically, the course leverages structured lectures, and formative assessments to provide foundational knowledge and skills to less experienced learners. Concurrently, andragogical techniques are employed to engage adult learners, emphasizing experiential learning, problem-solving, and critical thinking. This includes case studies, real-world simulations, collaborative projects, and research-based assignments that draw on the students' existing knowledge and professional experiences. The dual approach ensures a comprehensive understanding of advanced networking concepts, fostering both technical proficiency and strategic thinking necessary for complex network management and innovation.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Basic Network Concepts	07
2	Networking Device, media connection and network model	11
3	Network Virtualization	12
4	Adhoc Networking	11
5	Wireless Sensor networks	10
6	Fundamental of Network Security	09

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Basic Network Concepts

Hours 07

Introduction to Computer Networks, Element of Network, Type of Network: LAN, MAN, WAN, Network Topologies: Bus, Star, Mesh, Ring, etc, Data communication & Representation, Network Operating System.

Unit II: Networking Device, media connection and Network Model

Hours 11

Common LAN Media: STP, UTP, Coaxial Cable, Optical Fiber, Making and Testing Cable, straight thru cable, Cross over Cable, Connector, Jack, Patch Panels, NIC, Repeater and Hub & its type, Bridges and its Types, Switch and Router. Description of Seven Layers of OSI Model, TCP/IP Model, Comparison of OSI & TCP/IP Model, Physical and Data link Layer, Network and Transport Layer, Presentation and Session Layer, Application Layer

Unit III: Network Virtualization

Hours 12

Need for Virtualization, The Virtual Enterprise, Transport Virtualization- VNs, Central Services Access: Virtual Network Perimeter, A Virtualization Technologies primer: theory, Network Device Virtualization, Data-Path Virtualization, Control-Plane Virtualization, Routing Protocols.

Unit IV: Adhoc Networking

Hours 11

Introduction, application of MANET, challenges, Routing in Ad hoc networks, topology & position-based approaches, Routing protocols: topology based, position based, Broadcasting, Multicasting, & Geocasting, Wireless LAN, Transmission techniques, MAC protocol issues, Wireless PANs, The Bluetooth technology

Unit IV: Wireless Sensor networks

Hours 10

Need and application of sensor networks, sensor networks design considerations, empirical energy consumption, sensing and communication range, design issues, localization scheme, clustering of SNs, Routing layer, Sensor networks in controlled environment and actuators, regularly placed sensors, network issues, RFID as passive sensors.

Unit VI: Market Failures and Price Regulations

Hours 09

Market failures and need for regulation, Regulations and market structure, Firm behavior, Price regulation. More on Externalities, and role of government, importance of competition in market failure and price regulations.

Core Books:

1. Computer Networking: A Top-Down Approach 6th edition, James F. Kurose, Keith W. Ross, Pearson (2012).
2. Network Virtualization, Victor Moreno, Kumar Reddy, Cisco Press (2006).
3. Ad Hoc and Sensor Networks: Theory and Applications 2nd edition; Carlos de Moraes Cordeiro, Dharma Prakash Agrawal, World Scientific Publishing Company; 2 edition (2011)

Reference Books:

1. TCP/IP Protocol Suite 4 edition, Behrouz Forouzan, McGraw-Hill Science (2009)
2. Mobile Ad Hoc Networks: Current Status and Future Trends, Jonathan Loo, Jaime Lloret Mauri, Jesús Hamilton Ortiz, CRC Press(2011)
3. Fundamentals of Sensor Network Programming: Applications and Technology, S. Sitharama Iyengar, Nandan Parameshwaran, Vir V. Phoha, N. Balakrishnan, Chuka D. Okoye, Wiley-IEEE Press (2010).

Web References:

1. <https://www.ping.fm/ip/>
2. <https://www.edynamiclearning.com/course/advanced-networking-1a-introduction/>
3. <https://aws.amazon.com/training/learn-about/advanced-networking/>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Demonstrate comprehensive understanding of basic network concepts, including networking devices, media connections, and network models, and their roles in network architecture.
CO2 :	Analyze and implement network virtualization techniques, assessing their benefits and challenges in various networking environments.
CO3 :	Design and manage ad hoc networking solutions, addressing the unique requirements and challenges of decentralized network configurations.
CO4:	Develop and optimize wireless sensor networks, leveraging their capabilities for data collection and transmission in diverse applications.
CO5 :	Evaluate and apply fundamental network security principles to protect network infrastructures from potential threats and vulnerabilities, ensuring robust and secure communication channels.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basic Network Concepts	√				
2	Networking Device, media connection and network model		√			
3	Network Virtualization			√		
4	Adhoc Networking				√	
5	Wireless Sensor networks					√
6	Fundamental of Network Security					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	1	2	2	-	-	-	1	-	2	1	1
CO2	-	2	2	-	3	2	-	-	1	1	2	2	2
CO3	-	2	3	1	2	2	-	-	-	1	2	3	3
CO4	1	2	2	2	3	3	1	-	1	-	3	2	3
CO5	1	3	3	2	2	3	-	1	-	2	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE202 : Artificial Intelligence (AI) Fundamentals

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The theory sessions will be focused on basics of Artificial Intelligence, problem solving paradigms, and search strategies. Areas of application such as Expert Systems, Software Agents, knowledge representation, natural language processing, expert systems, robotics will be explored.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Artificial Intelligence	08
2	Problem-solving by Search and constraint satisfaction	11
3	Knowledge representation and reasoning	11
4	Expert Systems	11
5	Agents	10
6	Introduction to AI Techniques and Application Areas	09

Total Hours (Theory): 60

Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to Artificial Intelligence **Hours 08**

Concepts and definitions of Artificial Intelligence (AI) – Brief history of AI – AI and related fields – The AI problems and underlying assumptions, AI approaches: Cognitive Science, Turing Test.

Unit II: Problem solving by Search and constraint satisfaction **Hours 11**

Problem spaces; brute-force search; best-first search; two-player games; constraint satisfaction

Unit III: Knowledge representation and reasoning **Hours 11**

Review of propositional and predicate logic; resolution and theorem proving; non-monotonic inference; probabilistic reasoning; Bayes theorem, reasoning under uncertainty.

Unit IV: Expert Systems **Hours 11**

Introduction – Representing and using domain knowledge – Knowledge acquisition and representation – General structure of Expert Systems – Expert System Shell – Advantages and disadvantages of Expert Systems, Applications of Expert Systems

Unit V: Agents **Hours 10**

Definition of agents; successful applications and state-of-the-art agent-based systems; software agents, personal assistants, and information access; multi-agent systems.

Unit VI: Introduction to AI Techniques and Application Areas **Hours 09**

Introduction to fuzzy logic – Introduction to various application areas of AI like Natural Language Processing (NLP), Learning, Speech Recognition, Computer Vision, Game Playing, Robotics.

Core Books:

1. Norvig, Peter, Paradigms of AI Programming, Morgan Kauffman, 1992.
2. S. Russell and P. Norvig, Modern Approach to Artificial Intelligence, Prentice Hall of India Ltd., 2006.
3. Rich & Knight, Artificial Intelligence, 2nd edition, TMH, 1991.

Reference Books:

1. R.Akerker and P. S. Sajja, Knowledge-Based Systems, Jones and Bartlett, MIT, 2010.
2. George Luger, Artificial Intelligence, 5th Edition, Addison Wesley, 2004.

Web References:

1. http://www.tutorialspoint.com/artificial_intelligence/ [Basics of AI]
2. <http://intelligence.worldofcomputing.net/ai-branches/expert-systems.html> [Expert Systems]
3. <http://pages.cs.wisc.edu/~bolo/shipyard/neural/local.html> [Neural Network]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand basic problems of AI and history of AI.
CO2 :	Getting deep knowledge of searching algorithms and its applications in AI.
CO3 :	Learning Knowledge Representation and reasoning techniques.
CO4 :	To gain understanding Expert System and Agent Based Systems.
CO5 :	To explore various application areas of AI.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Artificial Intelligence	√				
2	Problem solving Search and constraint satisfaction		√			
3	Knowledge representation and reasoning			√		
4	Expert Systems				√	
5	Agents				√	
6	Introduction to AI Techniques and Application Areas					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO 8	PO9	P10	P11	PSO1	PSO2
CO1	-	2	2	2	1	-	-	-	-	-	-	1	2
CO2	3	2	2	3	2	2	-	3	-	-	-	2	2
CO3	-	-	3	3	2	-	-	-	-	-	-	1	-
CO4	3	2	2	-	2	2	-	3	-	-	-	1	2
CO5	2	-	-	-	2	-	3	3	-	-	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUA202 : Social Media and BLOG Writing

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components : Lab Assignments, Practical Test, VIVA					

Prerequisites: None.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for understanding social media marketing strategies and BLOG writing skills.

Outline of the Course:

Week No	Practical	Description
1	Introduction to Social Media - Platforms for social media marketing - Roles of social media in digital marketing	Learn basics of Social Media
2	Optimizing for Social Media Platforms - Facebook - Instagram - X (formerly Twitter)	Learn about various social media platforms and optimizing for them.
3	Content Calendar for Social Media Creating content calendar with different focus	
4	Content creation and Management in Meta Business Suite Understanding and Managing Meta Business Suite	Understanding Search Engine Result Page, and learn about On-the-Page and Off-the-Page Optimization
5	Understanding audience and insights in Meta Business	-
6	Boosting posts from Meta Business	-
7	Introduction to BLOG - What is BLOG? - Importance of BLOG	-
8	Popular Blogging Platforms - WordPress - Blogger	-

9	Using SEO in Blogging • Using AI in Blogging • Understanding SEO in Blogging	-
10	Using Blogger for Blogging	-
11	Using WordPress for Blogging – I • Installation of WordPress • Dashboard management	-
12	Using WordPress for Blogging – II • Creating and Publishing posts • Managing comments and subscription	
13	Using WordPress for Blogging – III • Exploration of themes for blogging • Exploration of plugins for blogging	
14	Case Study – I (Social Media)	
15	Case Study – II (Blogging)	

Total hours: 60

Core Books:

1. Damian Ryan and Calvin Jones: Understanding Digital Marketing, Kogan Page, 2014
2. Stephanie Diamond: Digital Marketing All-in-One for Dummies, Wiley Publication, 2019
3. Amy Lupold Bair, Blogging for Dummies, 7th Edition, Wiley Publication, 2020

Reference Books:

1. Philip Kotler, Hermawan Kartajaya, and Iwan Setiawan: Marketing 4.0: Moving from Traditional to Digital, Wiley Publication, 2016
2. Jan Zimmerman and Deborah Ng: Social Media Marketing All-in-One for Dummies, Wiley Publication, 2017

Web References:

1. <https://www.simplilearn.com/tutorials/digital-marketing-tutorial>
2. https://www.tutorialspoint.com/digital_marketing/index.htm
3. <https://intellipaat.com/blog/digital-marketing-tutorial/>
4. <https://www.javatpoint.com/digital-marketing>
5. <https://www.javatpoint.com/blog>

Course Outcome: Upon successful completion of the course, student will:

CO1:	Able to understand digital marketing and its importance.
CO2:	Able to create and manage content using social media platforms
CO3:	Able to optimize social media platforms
CO4:	Able to blogging and its importance in digital marketing.
CO5:	Able to use generative AI in Digital Marketing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO 5	PO 6	PO7	PO8	PO 9	PO10	PO11	PSO1	PSO2
CO1	1	2	1	3	2	2	1	1	-	1	1	1	1
CO2	-	1	2	3	1	3	1	-	-	2	2	2	2
CO3	1	1	2	3	1	3	1	-	-	2	2	2	2
CO4	1	1	2	3	1	1	1	1	-	1	-	1	1
CO5	2	2	3	3	2	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUD202 : Programming the Internet

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, Case Study					

Prerequisites: CAUD201 – Introduction to Web Designing

Methodology, Pedagogy & Andragogy: This subject focuses on hands-on experiments to create dynamic and interactive web sites using HTML5, CSS3 and JavaScript.

Outline of the Course:

Week No	Practical	Description
1	CSS Styling the web - Applying CSS Building blocks –CSS Selectors, Cascade & inheritance, CSS Properties, CSS Box model.	Student can apply various types of CSS with its basic properties.
2	Creating Layout - Header, footer, navigation menu, and sub-navigation. Floats.	Student can create different layout using semantic elements of HTML5 & CSS classification properties.
3	Creating Layout - Flexbox, Introduction to Flexbox, Flex Container and Flex Items, Flexbox Properties and Layout Techniques.	Student can create flexible and responsive web layouts. Its ability to distribute space and align items effectively makes it a vital part of modern web design.
4	Creating Layout - Introduction to CSS Grid Layout- Grid Container and Grid Items, multi column layout.	Student can Understand its properties and axes can significantly improve the adaptability and aesthetics of your web pages.
5	Creating Image Gallery and Animation - 2D/3D Transformation, Transitions and Animation.	Student can create layout image gallery & animation.
6	CSS Media Query.	<i>Student can create different layouts depending on the size of the viewport.</i>

7	JavaScript – Adding Internal & External JavaScript	Student can write script for basic Programs.
8	Working with variables, datatype – Number, String, Boolean.	Student can write script for basic programs using different datatypes.
9	datatype –Array, Object, Operators, Conditional statements, Looping statements.	Student can write script for basic programs using array and object, condition, Loop.
10	Adding Dialog boxes in HTML page – Alert, Confirm and Prompt.	Student can able to use dialog boxes to make website alive.
11	Creating user defined function.	Student can able to use in-build and user defined functions of JavaScript.
12	Adding Mouse and Keyboard Event in HTML page, Manipulating String and Date.	Students can make user interaction using various Events.
13	JavaScript Document Object Model hierarchy – Create find and manipulate HTML Element using Objects and methods	Student can explore the DOM Hierarchy.
14	Form validation, Applying Style using JavaScript.	Student can able to use form validations.
15	Creating New window, Accessing & manipulating History of HTML Pages.	Student can apply various objects to make web page dynamic with user interactions.

Total hours: 60

Core Books:

1. Julie C. Meloni, Jennifer Kyrnin : HTML, CSS, and JavaScript All in One, 3rd Edition, Pearson, 2018.
2. Joseph R. Lewis, Meitar Moscovitz : AdvancED CSS, W3C CSS Working Group, Apress, 2020.
3. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
4. MArijin Haverbeke : Eloquent Javascript – A modern Introduction to JavaScript, 3rd Edition, 2018
5. Michael Morrison: Head First JavaScript, O'Reilly, 2014

Reference Books:

1. Andy Budd, Emil Björklund: CSS Mastery 3rd Edition, Apress - Advance Web Standard Solutions.
2. Karl Barksdale, Shane Turner: HTML and Java Scripts Basics, 4th Edition, Course Technology, 2006.
3. Jon Duket: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.
4. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

Web References:

1. https://www.tutorialspoint.com/css/css3_tutorial.htm [For CSS Basics]
2. <https://www.tutorialrepublic.com/css-tutorial/> [For CSS3 BOX Model & Animation]
3. <https://javascript.info/> [For modern JavaScript tutorials]
4. <https://www.tutorialsteacher.com/javascript/javascript-hoisting> [JavaScript Basics]

Course Outcome: Upon successful completion of the course, student will:

CO1:	Able to apply CSS & its basic properties in web pages.
CO2:	Able to create HTML, CSS layout.
CO3:	Able to apply the transition, transformation and animation in web page.
CO4:	Able to write client side script – JavaScript to perform basic program.
CO5:	Able to use DOM objects for dynamic web pages.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	3	3	2	1	2	1	1	2	1	1	1
CO2	1	3	3	3	2	2	1	1	1	2	3	2	1
CO3	1	2	2	3	2	1	1	1	1	2	2	1	1
CO4	2	2	3	3	2	1	1	1	2	1	2	1	1
CO5	2	2	3	3	2	1	2	1	2	1	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUI201 : Summer Internship and Viva – II

Objective and Scope:

Students who choose to exit the programme after the completion of the first two semesters must undertake internships, community engagement programmes and field based learning/minor projects during the summer term. This internship, worth 04 credits, is a requirement for the award of the Diploma. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. This provision ensures that students gain valuable practical experience and industry exposure, complementing their academic learning and enhancing their career prospects. Following are the intended objectives of internship training:

- **Practical Experience:** Provide hands-on experience with real-world projects to apply theoretical knowledge gained during the course.
- **Skill Development:** Enhance technical skills, such as programming, software development, database management, and other relevant IT skills.
- **Industry Exposure:** Familiarize students with the working environment of the IT industry, including understanding company culture, workflows, and professional practices.
- **Professionalism:** Instil a sense of professionalism and work ethic, including adhering to deadlines, following instructions, and working collaboratively.
- **Transition to Employment:** Facilitate a smoother transition from academic life to professional careers by bridging the gap between theory and practice.

Guidelines of Internship:

The Summer Internship shall be of 15 (Minimum 60 hours) duration and will be undertaken during the Summer Vacation. Students have to undergo internship in reputed Tech Companies, Software Development Firm, IT services companies, Research Institutions and Lab etc. It is mandatory for the students to seek written approval from the coordinator about the topic and the organization before commencing the Summer Internship. During Summer Internship students are expected to take necessary guidance from the faculty guide allotted by the Institute. To do it effectively they should be in touch with their guide through e-mail or phone. Students must maintain regular reports detailing the work completed, challenges faced, and learning experiences. Students must submit a final report and presentation summarizing the internship experience, project outcomes, and key learnings to guide. The technical specifications for report preparation is as under:

Technical details:

1. The report shall be printed on A-4 size white paper.
2. 12 pt. Times New Roman font shall be used with 1.5-line spacing for typing the report.

3. 1" margin shall be left from all the sides.
4. Considering the environmental issues, students are encouraged to print on both sides of the paper.
5. The report shall be spiral bound as per the standard format of the cover page given by the Institute.
6. The report should include a Certificate (on company's letter head) from the company duly signed by the competent authority with the stamp.
7. The report shall be signed by the respective guide(s) & the Principal of the Institute 10 (Ten) days before the viva-voce examinations.
8. Student should prepare two hard bound copies of the Summer Internship Project Report and submit one copy in the institute. The other copy of the report is to be kept by the student for their record and future references.

Evaluation:

The evaluation of Internship will be done as per the following criteria:

Sr. No.	Evaluation Criteria	Marks
1	Summer Internship Report	50
2	Presentation and Viva-voce examination	50
TOTAL MARKS		100

Students must secure 36% passing marks in both components to qualify for the Diploma.

Internship and Exit:

Any student intending to exit the course must complete the summer internship (Including scoring at least 36% marks in the evaluation) before exit. A student intending to exit shall have to submit an application to Principal through Counsellor before even semester university theory examination. However, any such student desiring to withdraw the exit option may do that before the last paper of theoretical examinations of even semester.

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**BCA PROGRAMME
(5th SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-V

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP - 2020 Classification
			CCE	SEE	CCE	SEE		
CAUC301	Object Oriented Programming Using JAVA	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC302	Fundamentals of Visual Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE301	Software Engineering	Theory	50	50	-	-	04	Minor
CAUE302	Python Programming	Practical	-	-	25	25	02	Minor
CAUE303 CAUE304	(Select any one) 1. Introduction to Data warehousing 2. Geographical Information System	Theory	25	25	-	-	02	Minor
HSUS301	French	Practical	-	-	25	25	02	SEC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CAUC301 : Object Oriented Programming Using JAVA

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: CAUC101 - Basic Knowledge of Introduction to Programming

Methodology, Pedagogy & Andragogy: Upon successful completion of the syllabus, students will get basic knowledge of object oriented programming. Concretely, students should be able to create appropriate classes using the Java Programming Language to solve problems using Object Oriented Approach. They should be able to write console based and multithreaded applications using the Java Programming Language. Students should be able to implement the interface and classed of collection framework.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1.	Introduction to Object Oriented Programming in Java	10
2.	Fundamental programming structures in java	09
3.	Inheritance, Abstract class, and Interface	09
4.	Packages and Wrapper class	09
5.	Exception handling and Multithreading	11
6.	Basics of Collections Framework	12

Total Hours (Theory) :60
Total hours (Practical) : 60
Total Contact Hours: 120

Week No	Practical	Description
1	Basic of Java	Fundamental concepts in Java, Class & Objects , Structure of java program.
2	Data Types & Variables	Understand and utilize arrays effectively, keywords, Identifiers.
3	Defining classes, variables and methods	Implement this keyword, static keyword, method overloading, final keyword, creating objects, accessing rules of java
4	Constructors, Basics of Inheritance & Runtime Polymorphism	Implement Default constructors, Parameterized constructors, passing object as a parameter, constructor overloading. Implement inheritance types and runtime polymorphism
5	Exeption handling	Able to implements exception handling: types of exceptions, Throwable class, keywords – try, catch, throw, throws and finally, nested try statement, java built in exceptions, user defined exceptions.
6	Creating packages	Able to apply the predefined packages and the use defines package and access protection.
7	Multithreading	Implementing methods of Thread class.
8	Creating Multiple Threads	Creating Multiple Threads, Thread class and the Runnable Interface.
9	Thread Priority	Implementing Thread priority
10	Synchronization & Inter-thread communication	Implementing thread synchronization and inter-thread communication.
11	Collections Framework	Implementation of Collections Framework, Collection,.
12	Iterator interface	Accessing a collection using an Iterator, ListIterator.
13	Interfaces of Collections	List, Set, Sorted Set
14	Collection classes	Array List, Linked List, Vector, Hash Set, Tree Set, Priority Queue
15	Map interface	Map, Map Entry, SortedMap, HashMap, LinkedHashMap, TreeMap.
		Total Hours: 60

Core Books:

1. Pravin M. Jain: The class of JAVA, Pearson Education, 2011.
2. Herbert Schildt: The Complete Reference Java J2SE, 5th Edition, TMH Publishing Company Ltd, New Delhi.

Reference Books:

1. Ivor Horton: Beginning Java JDK, 5th Edition, Wiley Press, 2008.
2. Cay Horstmann and Gary Cornell: Core Java Volume 1, 8th Edition or later, Pearson Education, 2008.
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

Web References:

1. <http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-092introductiontoprogramming-in-java-january-iap-2010/lecture-notes/> [Lecture Notes in PDF]
2. <http://www.javatpoint.com/java-tutorial> [Notes]

Course Outcomes: Upon successful completion of the course, students will be able

CO1 :	To understand basic of Object Oriented Concepts, execution environment in NetBeans
CO2 :	To understand basic syntax and semantics of Java, structure of java.
CO3 :	Learning Inheritance, Abstract class and Interface to understand how the java works ethically.
CO4 :	To introduce the implementation of Exception and Packages.
CO5 :	To introduce the concepts of multithreading & Collections framework.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basic Introduction of Java	√	√			
2	Variable & Data Types		√			
3	Class Objects and Methods			√		
4	Exception Handling & Packages			√	√	
5	Multithreading					√
6	Basics of Collections Framework					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	2	-	-	-	1	-	3	3
CO2	3	3	2	3	3	3	-	-	-	1	-	3	3
CO3	3	2	2	3	2	2	-	-	-	2	-	2	2
CO4	3	2	2	3	2	2	-	-	-	2	-	2	2
CO5	2	2	2	3	2	2	-	-	-	2	-	2	2
CO6	-	-	-	-	-	2	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC302 : Fundamentals of Visual Programming

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Component (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: CAUC101 - Basic knowledge of programming language.

Methodology, Pedagogy & Andragogy: During theory lectures, illustrations will be provided to emphasize the importance of advanced features of the .Net framework and their application in developing Windows and desktop-based solutions using the C#.Net. Practical sessions will focus on hands-on learning, where students will be tasked with creating desktop applications by applying the concepts discussed during the lectures. This approach ensures a comprehensive understanding of both theoretical aspects and their practical implementation, equipping students with the skills to develop robust Windows-based applications.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction of .NET Framework	07
2	Object-Oriented Programming with C#.NET	12
3	Introduction To Windows Forms	12
4	Introduction To ADO.NET	11
5	WPF and Advanced Features of .NET	08
6	Web Applications	10

Total Hours (Theory): 60
Total Hours (Practical): 60
Total Contact Hours:120

Detail Syllabus:

Unit I: Introduction of .NET Framework

Hours 07

Introducing .NET Framework, .NET Framework Component, .NET Framework Version Compatibility, Core of .NET Framework: Application Services, Base Class Library and CLR, Types of Applications which can be developed using MS.NET, MS.NET Base Class Library, MS.NET Namespaces, Common Language Runtime (CLR), Managed Code, MS.NET Memory Management/Garbage Collection, Common Type System (CTS), Common Language Specification (CLS), Types of JIT Compilers

Unit II: Object-Oriented Programming with C#.NET

Hours 12

Overview of Object-Oriented Programming, Principles of Object Orientation programming: Encapsulation, inheritance, polymorphism, Object, Defining Properties, Using Auto- Implemented Properties, Defining Methods, Understanding Constructors, Defining and Implementing Interfaces, Understanding the Role of Interfaces in .NET. C# Datatypes & Variables Declaration, Implicit and Explicit Casting, Checked and Unchecked Blocks – Overflow Checks, Boxing and Unboxing, Operators, Control Statements, Working with Arrays, C# Application Basics: Command line and VS.NET compilation.

Unit III: Introduction To Windows Forms

Hours 12

Building Windows Forms Applications, Setting Form Properties, Understanding the Life-cycle of a Form, Using the Windows Forms Designer, MessageBox Class and .config File, Working with Windows Forms Controls: TextBox, Button, Selection, List, DateTimePicker, Container, Image, Timer, ErrorProvider, TooltipProvider, Dialog box (ColorDialog, FolderBrowserDialog, OpenFileDialog, SaveFileDialog, FontDialog), MenuStrip, Handling Events: Understanding the Event-Driven Programming Model, Writing Event Handlers, Sharing Event Handlers, MDI, SDI.

Unit IV: Introduction To ADO.NET

Hours 11

Evolution of ADO.NET, Understanding the ADO.NET Object Model, Connected vs. Disconnected Access, Use of Connection, Command, DataReader, DataAdapter and Dataset class, Exceptional Handling.

Unit V: WPF and Advanced Features of .NET

Hours 08

Overview of WPF, Features of WPF, Windows Forms vs WPF, Introduction to XAML, WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel, WPF Controls: TextBlock, PasswordBox, ListBox, Slider, Popup, Menus and Expander, Animation using WPF, Introduction to MVC, Custom Controls, Introduction to Multithreading.

Unit VI: Web Applications

Hours 10

Basic of Http Request and Http Response. o Understand form Tag and Comparison between Get and Post. ASP.NET Page Life Cycle. Structure of an ASP.NET Page: ASPX Page, Code behind File, Web config and Machine config, Rich Web Control, Validation Controls. State Management Techniques o Client side: ViewState, Hidden Field, Cookie , Server Side: ApplicationState, Session.

Outline of the Course (Practical) :

Week No	Practical	Description
1	Console Environment without IDE	To set the path and environment variable to execute a console-based program without any IDE
2	Introduction to Visual Studio IDE and .NET	Familiarize with the Visual Studio IDE and the basic programming structure of C#.NET.
3	Developing Console Applications	Creating console-based applications using input and output statements, and understanding the role of the Main method.
4		
5	Object-Oriented Programming Concepts	Implementing encapsulation, inheritance, and polymorphism in C#.
6	Control Statements and Arrays	Using operators, control statements, and working with arrays in C#.
7	Building Windows Forms Applications and Events Handling	Working with TextBox, Button, and other common controls in Windows Forms. Understanding event-driven programming and writing event handlers.
8		
9		
10	Database Connectivity with ADO.NET	Connecting to a database, executing commands, and retrieving data using ADO.NET.
11		
12	A Complete Windows Forms Application	Designing and implementing a complete Windows Forms application, integrating various controls, handling events, and database connectivity.
13	Building a WPF Application	Creating a basic WPF application.
14	Basics of ASP.NET	Understanding HTTP requests and responses, and creating a simple ASP.NET page.
15		

Total hours (Practical) : 60

Core Books:

1. Christian Nagel : Professionals C# and .NET 2021 Edition , Wiley – India, WROX
2. Albahani Josheph: C# 4.0 Pocket Reference, Shroff publication,2010.

Reference Books:

1. Jon Skeet, C# in Depth, Managing Publication,2010
2. Drayton David: C# Language Pocket Reference, Shroff publication,2005

Web References:

1. <http://csharp.net-informations.com/> [to learn C# and .NET fundamentals]
2. <http://www.wpftutorial.net/> [to learn WPF]
3. <https://dotnet.microsoft.com/en-us/learn/csharp> [tutorial and videos to learnC#]
4. <https://dotnettutorials.net/course/csharp-dot-net-tutorials/> [to learn C# and .NET fundamentals]
5. <https://www.javatpoint.com/c-sharp-tutorial> [to learn basic fundamentals of C#]

Course Outcomes: Upon successful completion of the course, the students will:

CO1 :	Understand .NET Framework core features and Components.
CO2 :	Develop modular and user-friendly desktop applications using object-oriented programming concepts in C# and Windows Forms.
CO3 :	Perform database operations using ADO.NET for seamless data integration.
CO4 :	Create rich, graphical desktop applications using WPF.
CO5:	Design and develop dynamic web applications in the .NET framework.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction of .NET Framework	√				
2	Object-Oriented Programming with C#.NET		√			
3	Introduction To Windows Forms		√			
4	Introduction To ADO.Net			√		
5	WPF and Advanced Features of .NET				√	
6	Web Applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	P11	PSO1	PSO2
CO1	3	2	2	3	-	2	1	-	-	1	1	2	-
CO2	2	3	3	2	-	3	2	-	-	2	-	1	-
CO3	-	3	3	-	-	-	-	-	-	-	1	-	-
CO4	-	2	2	3	-	-	-	-	-	-	-	-	-
CO5	2	3	3	2	-	3	3	-	-	2	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE301 : Software Engineering

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study, Certification courses, etc.					

Pre-requisite: CAUC203 - Fundamentals of System Analysis and Design.

Methodology, Pedagogy & Andragogy: Software engineering is an art and while learning this the students will also get familiar with the concept of project management and development of SRS document. Students will get used to entire software development process by incorporating the latest tools for designing and documenting the process.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to software and Software Engineering	08
2	Software Process Models	10
3	Managing Software Projects	12
4	Software Requirement analysis and specification	12
5	Software Maintenance and Risk Management	10
6	Modern methods in Software Engineering	08

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to software and Software Engineering Hours 08

Definition of Software, Types of Software, Nature of Software, Software Applications, Definition of Software Engineering, Software Engineering-A Layered Technology, A process framework.

Unit II: Software Process Models Hours 10

Definition of Software Process, Software Process Model - The Waterfall model, Incremental Process Models: The Incremental model, The RAD model Evolutionary Process models: Prototyping, The Spiral model, The Concurrent Development model, A final comment on evolutionary Processes, Component based development, Concept of Product and Process.

Unit III: Managing Software Projects**Hours 12**

The Management Spectrum, W5HH principles, Concepts of measures, metrics and indicators. Process Metrics and Project Metrics: An overview, Software Measurements: Size oriented and function oriented metrics.

Unit IV: Software Requirement analysis and specification**Hours 12**

Introduction to system and software requirements, types of software requirements: Functional and non- functional requirements. Requirement Specification (SRS), Requirement Analysis and Requirement Elicitation, Requirement Engineering.

Unit V: Software Maintenance and Risk Management**Hours 10**

Software risks, risk identification, risk projection, risk mitigation, monitoring and management. Introduction to software maintenance, types of maintenance. Concept of software reengineering, Business Process Reengineering(BPR), BPR life cycle, reverse engineering, Applications of Software Engineering.

Unit VI: Modern methods in Software Engineering**Hours 08**

Introduction to Agile development: What is Agility? principles and values of Agile development, concept of Extreme Programming, values of XP.

Core Books:

1. Roger S.Pressman, Software Engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User
3. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.

Reference Books:

1. Ian Sommerville, Software engineering, Pearson education Asia
2. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
3. Merlin Dorfman (Editor), Richard H. Thayer (Editor), Software Engineering
4. Robert C. "Uncle Bob" Martin, Clean Architecture: A Craftsman's Guide to Software Structure and Design
5. Deepak Gaikwad, Viral Thakkar, DevOps Tools from Practitioner's ViewPoint, Wiley

Web References:

1. https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology [Introduction to Software Engineering/Process/Methodology]
2. <https://www.uml-diagrams.org/> [UMLUnits]
3. <https://nptel.ac.in/courses/106/101/106101061/> [Agile Software Development and Extreme Programming and Overview of DevOps]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand concept of Software and its types.
CO2 :	Get familiarize with software processes and related models.
CO3 :	Able to analyze requirements and manage project
CO4 :	Able to understand concepts of creating Software Requirement Specification
CO5 :	Able to understand risk management, software maintenance and modern Software engineering methods.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to software and Software Engineering	√				
2	Software Process Models		√			
3	Managing Software Projects			√		
4	Software Requirement analysis and specification				√	
5	Software Maintenance and Risk Management					√
6	Modern methods in Software Engineering					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	3	1	3	3	3	1	1	3	1	3	3
CO2	2	2	3	2	3	3	3	1	1	3	3	2	2
CO3	2	3	3	3	3	3	3	2	1	3	3	3	3
CO4	3	2	3	2	3	3	3	3	1	2	2	3	3
CO5	2	3	3	3	2	3	3	1	1	2	1	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE302 : Python Programming

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components: Lab Assignments, Practical Test, Case Study					

Prerequisites: CAUC103-Fundamentals of Object Oriented Programming

Methodology, Pedagogy & Andragogy: The practical sessions will cover the Python Programming concepts and practices that support application development using Python Programming. The practical sessions will cover Object Oriented programming using Python Programming .

Outline of the Course:

Wee k No	Practical	Description
1	Downloading and Installation: Installation of Python and Python documentation.	Student will be familiarizing with downloading and installing Python.
2	Primary and secondary prompts, basic statements and expressions, Datatypes and Operators	Students will understand executing programs using Python. They also learn various data types supported by Python.
3	Conditional statements and Iterative statements	Students will to learn conditional and looping statements in Python.
4	Introduction to sequence types supported by Python: String, List, Tuple	Students will learn how to implement String, List, and Tuple types supported by Python.
5	Mapping and other types: Dictionary, set	Students will learn how to implement Dictionary and Set types supported by Python.
6	Exposure of python built-in Functions	Student will be familiarize with built-in functions in Python.

7	Python User-Define Functions	Students will to learn to create own function for reusability of the code.
8	Python Comprehension and filter() and map()	Students will learn to optimize the conditional and looping statements.
9	Introduction to Object Oriented Programming	Student will be familiarize with the concept of Object Oriented Programming using Python.
10	Introduction to Inheritance and concept of method overriding.	Student will be familiarize with the concept of Inheritance, method overriding using Python
11	Introduction to Polymorphism, Encapsulation, Abstraction and magic methods	Student will be familiarize with the concept of Polymorphism, Encapsulation, Abstraction and magic methods.
12	Exception Handling in python	Student will be able to understand exception handling using Python.
13	Python Module and Libraries	Student can able to learn python module and libraries to use.
14	Introduction to Web Framework: Django.	Students understand MVT based web application development using Django.
15	Creating web application based on MVT and database.	Student can apply concept of MVT and develop applications for CRUD operations.

Total Hours (Practical): 60

Total Contact Hours:60

Core Books:

1. Wesley J. Chun: Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress,2009.
2. Megnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress,2009.
3. Julia Elman, Mark Lavin: Lightweight Django: Using REST, WebSockets, and Backbone, 1st edition, O'Reilly Media, Inc, 2014

Reference Books:

1. Richard L. Halterman: Learning to Program with Python, Southern Adventist University 2011.
2. Mark Lutz: Programming Python, 4th Edition,O'reilly, 2011.
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. <https://www.python.org/about/gettingstarted/> [For beginners of Python]
2. <http://ocw.mit.edu/courses/electrical-engineering-and-computerscience/6-189-agentleintroduction-to-programming-using-pythonjanuary-iap-2008/> [Python Materials]
3. http://people.cs.aau.dk/~normark/prog303/html/notes/paradigms_them.esparadigmoverview-section.html [Overview of Programming Paradigms]
4. <https://docs.python.org> [Python Documentation]
5. <https://www.tutorialspoint.com/django/index.htm> [Overview of Django Framework]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand scripting and the contributions of scripting languages.
CO2 :	Able to aware the built-in objects of Python and object-oriented concepts
CO3 :	Able to understand and implement functional and Object Oriented Programming
CO4 :	Able to learn exception handling
CO5	Able to grasp Python for web based programming

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	-	2	-	-	-	-	-	3	1
CO2	-	3	3	-	-	-	3	1	-	-	-	3	1
CO3	2	-	2	-	3	-	1	-	-	-	-	3	1
CO4	-	-	3	3	1	-	2	-	-	-	-	3	1
CO5	1	2	3	3	1	2	2	1	1	1	1	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE303 : Introduction to Datawarehouse

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Theory	02	30	20	50
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: CAUC104 - Basic concepts of DBMS

Methodology, Pedagogy & Andragogy: During the classroom sessions, the emphasis will be given on the fundamentals of data warehouse and its architectures along with the different applications. Also, the emphasizes given on using OLAP and OLTP tools for understanding the real time applications of data warehouse technologies and the broad concept of data mining techniques. Same concepts will be made concrete through assignments, case studies, and unit tests.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction Data Warehouse	5
2	Data Warehouse Architecture	6
3	Fundamentals of OLTP and OLAP	5
4	Concepts of ETL and ELT	5
5	Applications of Data Warehouse and Business Intelligence	4
6	Overview of Data Mining	5

Total Hours (Theory): 30
Total Contact Hours:30

Detail Syllabus:

Unit I: Introduction to Data Warehouse

Hours 05

Overview of data warehousing, Types of data warehouses, Comparison of databases and data warehouses. Importance and defining characteristics of data warehousing, Classifications and key components of a data warehouse, Advantages, limitations, and future trends in data warehousing.

Unit II: Data Warehouse Architecture

Hours 06

Concepts of Data Warehouse Design and Dimensional modelling, Evolution of Data Warehouse Architecture, Understanding Data Warehouse Architecture: Explore Architecture, Concepts, and Components, Exploring Components of Data Warehouses, Data Warehouse Design Methodologies: Creating Star Schema and Snowflake Schema.

Unit III: Fundamentals of OLTP and OLAP

Hours 05

Understanding Data Integration: Concepts, Processes, and Techniques, Exploring OLAP Types: ROLAP, MOLAP, and HOLAP, and Their Distinctions, Introduction to Different OLAP Tools, Executing OLAP Operations on Multi-Dimensional Data Models through SQL Queries.

Unit IV: Concepts of ETL and ELT

Hours 05

Fundamental concepts of ETL and ELT terminologies, including extraction methods, transformation techniques, key differences between ETL and ELT, tools used in ETL.

Unit V: Applications of Data Warehouse and Business Intelligence

Hours 04

Overview of applications such as Financial Services, Banking services, Consumer Goods, Retail Sector; Importance of Data Warehouse in Business Intelligence.

Unit VI: Overview of Data Mining

Hours 05

Differentiate between database, data warehouse, data mining and text mining. Core data mining tasks and the process of knowledge discovery in databases, Understanding the concepts of supervised learning and unsupervised learning.

Core Books:

1. William H.Inmon: Building the Data Warehouse, 4th edition,, wiley publication, 2005
2. Paulraj Ponniah: Data Warehousing: Fundamentals for IT Professionals, 2nd edition, 2010
3. Han J, Kamber M, Pei J. Data mining concepts and techniques third edition. University of Illinois at Urbana-Champaign Micheline Kamber Jian Pei Simon Fraser University. 2012.

Reference Books:

1. Nenad Jukic: Database Systems: Introduction to Databases and Data Warehouses 1st Edition, Prospect Press, 2017

Web References:

1. <https://www.guru99.com/>
2. <https://www.ibm.com/think/topics/data-warehouse>
3. <https://aws.amazon.com/compare/the-difference-between-olap-and-oltp/>
4. <https://panoply.io/data-warehouse-guide/>

Course Outcomes: Upon successful completion of the course, students will,

CO1 :	Have a deeper understanding of database systems and their underlying theory to be able to improve the decision-making process.
CO2 :	Be Understand the technology of data warehousing.
CO3 :	Be able to develop applications of higher order database systems.
CO4 :	Have Clear understanding of implementation of huge amount of static data.
CO5 :	Understand Data Mining concept and techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction Data Warehouse	√				
2	Data Warehouse Architecture		√			
3	Fundamentals of OLTP and OLAP			√		
4	Concepts of ETL and ELT			√		
5	Applications of Data Warehouse and Business Intelligence				√	
6	Overview of Data Mining					√

Course Articulation Matrix:

	PO 1	PO2	PO 3	PO4	PO5	PO6	PO7	PO 8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE304: Geographical Information System

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Total Notional Hours
02	Theory	02	30	30
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.				

Pre-requisite: CAOS101: Mathematics for Computer Science

Methodology, Pedagogy & Andragogy: Geographical Information System integrates theoretical approaches to provide a comprehensive understanding of GIS. It includes lecture-based learning to explain core concepts such as GIS fundamentals, data models, spatial analysis, and geostatistical techniques. Case studies and examples are incorporated to help students understand the real-world implications and applications of GIS in various fields. Assessments are designed to evaluate the depth of theoretical knowledge and conceptual understanding.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to GIS	06
2	Types and Data Models	04
3	Spatial Data Editing	06
4	Attribute Data and Data Exploration	05
5	Spatial Analysis	04
6	Geo Statistical Analysis Techniques	05

Total Hours (Theory): 30

Total Contact Hours: 30

Detail Syllabus:

Unit I : Introduction to GIS

Hours 06

Basic concepts: Definition and history, Components of GIS, Recent trends and applications of GIS; Data structure and formats, Spatial data models – Raster and vector, Data base design- editing and topology creation in GIS, Linkage between spatial and non-spatial data, Data inputting in GIS. Rectification, Transformation Methods; Root Mean Square (RMS) Error

Unit II : Data Types and Data Models

Hours 04

Data Types; Spatial Data; Non-Spatial Data, Data Input; Existing GIS Data, Metadata; Conversion of Existing Data, Creating New Data, Data Models; Vector Data Model; Raster Data Model; Integration and Comparison of Vector and Raster Data Models

Unit III : Spatial Data Editing

Hours 06

Types of Digitizing Errors, Causes for Digitizing Errors; Topological Editing and Non-topological Editing; Other Editing Operations; Editing Using Topological Rules

Unit IV : Attribute Data and Data Exploration

Hours 05

Attribute Data in GIS, Attribute Data Entry, Manipulation of Fields and Attribute Data, Data Exploration; Attribute Data Query, Raster Data Query, Map- Based Data Manipulation.

Unit V : Spatial Analysis

Hours 04

Spatial Data: Definition, Analysis, Processes & Steps, Software and Tools, Geodatabase Model, Role of Databases in GIS, Creating, Editing and Managing, Classification scheme of Vector- Based and Raster- Based GIS Operation Raster- Based Techniques: Methods of reclassification, overlay analysis, Digital Terrain Analysis and Modeling- TIN and DEM, Surface representation and analysis, Slope and Aspect, Geographic Visualization Data Classification, Map Comparison.

Unit VI : Geo Statistical Analysis Techniques

Hours 05

Introduction to Spatial Interpolation: Control Points, Global Method- Trend surface analysis, regression model, local methods- Thiessen polygons, density estimation, Inverse Distance weighted Interpolation, Kriging- Ordinary Kriging and Universal Kriging, GIS and decision support system, Introduction to AHP, basic principal of AHP. Principal and components of multiple criteria decision Making.

Core Books:

1. kang-tsung Chang, 'Introduction to Geographic Information Systems' Tata McGraw Hill, 2007, New Delhi
2. C.P.Lo and Albert K.W. Yeung, "Concepts and Techniques of Geographic information Systems" Prentice Hall of India, 2006, New Delhi.

Reference Books:

1. Burrough, Peter A. and Rachael McDonnell, 'Principles of Geographical Information Systems' Oxford University press, 1998, New York.
2. Magwire, D.J. Goodchild, M.F. and Rhind, D.M., 'Geographical Information Systems: Principles and Applications', Longman Group, 2005, U.K.

3. Siddiqui, M.A.; Introduction to Geographical Information System, Sharda Pustak Bhavan, 2006, Allahabad

Web References:

1. <https://www.usgs.gov/faqs/what-a-geographic-information-system-gis>
2. <https://education.nationalgeographic.org/resource/geographic-information-system-gis/>

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand the fundamental concepts of GIS, including its definition, history, components, data structures, and formats, as well as recent trends and applications
CO2 :	Analyze different spatial and non-spatial data types and models, including raster and vector data models, and demonstrate integration and conversion techniques.
CO3 :	Identify and rectify digitizing errors in spatial data and perform topological and non-topological editing to enhance the quality of GIS data.
CO4 :	Manipulate attribute data within GIS, perform data exploration and queries, and use map-based techniques for data analysis.
CO5 :	Apply spatial and geostatistical analysis techniques, including interpolation, reclassification, overlay analysis, and terrain modeling, to support decision-making in GIS.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to GIS	√				
2	Types and Data Models		√			
3	Spatial Data Editing			√		
4	Attribute Data and Data Exploration				√	
5	Spatial Analysis					√
6	Geo Statistical Analysis Techniques				√	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	1	-	2	1	1	2	2	3	-	1	1
CO2	1	3	1	-	2	1	-	1	1	3	1	-	1
CO3	1	2	1	-	3	3	-	2	1	3	-	2	1
CO4	1	2	1	1	2	3	1	2	1	2	1	1	1
CO5	1	3	1	1	3	3	-	1	1	3	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**TEACHING SCHEME &
DETAILED SYLLABUS
FOR
BCA PROGRAMME
(6th SEMESTER)
AS PER NEP 2020
EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-VI

Course Code	Course Name	Theory/ Practical/ Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SSE		
CAUC303	Project Work	VIVA	150	150	-	-	12	Major Core
CAUE305 CAUE306	(Select any one) 1. Introduction to Data Science 2. Fundamentals of Digital Image Processing	Theory	50	50	-	-	04	Minor
CAUS301	Mobile Application Development	Theory and Practical	25	25	25	25	04 (02-Theory) (02-Practical)	SEC
HSUA302	Professional Communication, Soft Skills and Personality Development	Practical	-	-	25	25	02	AEC
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CAUC303 : Project Work

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
12	VIVA	12	180	50	230
Experiential Learning Components (Practical): VIVA					

Pre-requisite: None

Introduction:

The Project work constitutes a major component in most Professional Programmes. It needs to be carried out with due care, and should be executed with seriousness by the students. The project work is not only a partial fulfilment of the course requirements, but also provides a mechanism to demonstrate your skills, abilities and specialization. The Project work is one that involves practical work for understanding and solving problems in the field of computing. The objective of the project is to help the student develop the ability to apply theoretical and practical tools/techniques to solve real life problems related to industry, academic institutions and research laboratories.

Methodology, Pedagogy & Andragogy: The core objective of the course is to provide project and practical based on deep learning, concept understanding, its analysis, deriving inferences and documenting it towards productive outcome of the subject. The main objective of this mini project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Objective:

Student should be able to:

- demonstrate a systematic understanding of project contents;
- understand methodologies and professional way of documentation;
- know the meaning of different project contents, and
- understand established techniques of project report development.

Importance of the project

The Project is not only a part of the coursework, but also a mechanism to demonstrate student's abilities and specialization. It provides the opportunity for them to demonstrate originality, teamwork, inspiration, planning and organization in a software project, and to put into practice some of the techniques they have been taught

throughout the previous courses. The Project is important for a number of reasons. It provides students with:

- Opportunity to specialize in specific areas of computer science;
- Future employers will most likely ask about project at interview;
- Opportunity to demonstrate a wide range of skills and knowledge learned,
- Encourages integration of knowledge gained in the previous course units.

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Student will understand the implementation of concepts of SDL and Software Engineering.
CO2 :	The programming concepts they learn during their academics, it will be converted in to the actual implementations.
CO3 :	Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
CO4 :	Students must understand the difference between a program and professional application/product/software.
CO5 :	Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Mini Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Mini Project to the institute. Each Student is required to make a copy of Mini Project and submit along with Mini Project report.

Evaluation:

The project must be evaluated in two aspects:

- a. Internal (100 Marks):
 - i. Reporting to internal project guide
 - ii. Incorporation of suggestions by project guide
 - iii. Internal Project viva examination
- b. External (200 Marks):
 - i. Project Report Preparation & Evaluation
 - ii. External Project Viva Examination

Web References:

1. <http://techwhirl.com/writing-software-requirements-specifications/> [For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. <http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf> [Software Design Principles and Guidelines]
5. <http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf> [For Effective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template.pdf [For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO 1	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO 2	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO 3	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO 4	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO 5	3	3	3	2	3	3	3	3	3	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE305 : Introduction to Data Science

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction/Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: CAUC101-Foundation of Programming, CAUC104-Database Fundamentals

Methodology, Pedagogy & Andragogy: During theory sessions topics related to Data Science will be covered with suitable examples. Students will get exposure of model building using machine learning algorithms. Student will get in-depth knowledge of Data Science based real applications using case study approach.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Fundamentals of Data Science	08
2	Statistics and Linear Algebra for Data Science	10
3	Data Preprocessing	12
4	Data Visualization	10
5	Machine Learning for Data Science	10
6	Data Science Applications and future trends	10

Total Hours (Theory):60
Total Contact Hours:60

Detailed Syllabus:

Unit I : Fundamentals of Data Science

Hours 08

Introduction to Data Science, Definition and description of Data Science, Need of Data Science, Pre-requisites of Data Scientists, Tools and Skills for Data Science, Data Science Process Life-cycle - Data Discovery, Preparation, Model Planning, Model Building, Communicate Results, Types of Data Analysis- Descriptive, Diagnostic, Predictive, Prescriptive.

Unit II : Statistics and Linear Algebra for Data Science

Hours 10

Descriptive Statistics- Measure of frequency, Measure of Central Tendency, Measure of Dispersion, Measure of Position. Overview of Inferential Statistics, Linear Algebra for Data Science using Vectors and Matrices- Basic Vector operations, Common Matrix operations.

Unit III : Data Preprocessing

Hours 12

Reading data using different functions, Introduction to Data Pre-processing, Data types and forms, Possible data error types, Consistent Data, Data Cleaning- Filling Missing Values, smoothing noisy data, Detecting and removing outliers, Data Integration, Data transformation- Rescaling, Normalizing data, Standardizing data Implementation Using Python.

Unit IV : Data Visualization

Hours 10

Introduction to Data Visualization, Visual encoding, Basic plots using Python, Data Visualization libraries in Python.

Unit V : Machine Learning for Data Science

Hours 10

Overview of Machine Learning- Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning, Regression Methods – Simple Linear Regression, Multiple Linear Regression, Classification Methods- KNN, Logistic Regression, Naïve Bayes, SVM, Clustering Method- k-means clustering, Fuzzy c-means, Introduction of Reinforcement Learning.

Unit VI : Data Science Applications and future trends

Hours 10

Applications- Manufacturing, Personalization, Medical Image Analysis, Internet Search, Genomics, Speech Recognition, Advertising, Financial Fraud Detection, Airline Route Planning, Cyber-threats, transportation, gaming, recommendation system.

Core Books:

1. Sanjeev J, Wagh, Manisha S. Bhende, Anuradha D. Thakare : Fundamentals of Data Science: First Edition: CRC Press : 2022
2. Dr. Gypsy Nandi, Dr.Rupamkumar Sharma: Data Science Fundamentals and Practical Approaches: BPB Publications:2020
3. Sandhya Arora, Latesh Malik: Data Science and Analytics with Python: Universities Press: 2023, ISBN:978-93-93330-34-5

Reference Books:

1. Parikshit Narendra Mahalle, Gitanjali Rahul Shinde, Priya Dudhale Pise, Jyoti Yogesh Deshmukh: Foundation of Data Science for Engineering Problem Solving: Springer:2022
2. K.G.Srinivasa,G.M.Siddesh,ChetanShetty,SowmyaB.J:Statistical Programming in R: First Edition: Oxford Press: 2017.

Web References:

1. <https://www.cs.cornell.edu/jeh/book.pdf> [Machine Learning Concepts]
2. <https://www.edureka.co/blog/math-and-statistics-for-data-science/> [Statistics for Data Science]
3. https://www.w3schools.com/datascience/ds_analyze_data.asp [Data Preparation]
4. <https://www.kdnuggets.com/2022/06/plotting-data-visualization-data-science.html> [Data Visualization for Data Science]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand fundamentals of Data Science.
CO2 :	Able to use statistical data analysis techniques for Data Science Applications
CO3 :	Able to apply data cleaning, visualization and exploratory data analysis techniques on data.
CO4 :	Able to build model using machine learning techniques.
CO5 :	Able to understand applications and future trends of Data Science.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Fundamentals of Data Science	√				
2	Statistics and Linear Algebra for Data Science		√			
3	Data Preprocessing			√		
4	Data Visualization			√		
5	Machine Learning for Data Science				√	
6	Data Science Applications and future trends					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	2	1	-	3	3	2	2	-	1	-
CO2	3	1	1	-	3	1	3	1	2	2	1	2	-
CO3	3	2	1	2	1	3	3	2	2	2	-	2	-
CO4	3	3	2	3	1	3	3	2	2	2	3	1	1
CO5	3	3	3	3	1	3	3	3	2	2	2	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE306 : Fundamentals of Digital Image Processing

Credit	Component	Instruction/Contact Hours(Per Week)	Instruction/Contact Hours(Per Course)	Experiential Learning Hours(Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components: Unit Tests, Assignments, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: The theory sessions will emphasize fundamentals and applications of digital image processing. The classroom sessions will make students aware of various operations such as colour image processing, image enhancement, segmentation, representation, recognition, and compression, which are performed on digital images. To enhance comprehension, we will provide illustrations of digital image processing operations using a programming language. To bridge the gap between theoretical knowledge and practical application, we have to schedule industrial visits. These visits are designed to offer students a tangible correlation between their studies and real-world practices. Students shall be assigned case studies to facilitate the practical application of their acquired knowledge. These case studies aim to deepen understanding and foster the ability to apply theoretical concepts to real-world scenario of digital image processing and analysis.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Digital Image Fundamentals	18
2	Colour Image Processing	10
3	Image Enhancement	09
4	Image Segmentation	06
5	Image Representations and Recognition	09
6	Image Compression	08

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I : Digital Image Fundamentals

Hours 18

Introduction, History, Components of image processing system, Applications of Digital Image Processing, Human visual system, Image as a 2D data, Image representation, Gray scale and Colour images, image sampling and quantization, Basic Relationships between Pixels.

Unit II : Colour Image Processing

Hours 10

Colour fundamentals, Colour models, Colour transformation, Smoothing and Sharpening, Colour segmentation, Noise Models, Noise Reduction, Inverse Filtering.

Unit III : Image Enhancement

Hours 09

Basic gray level Transformations, Histogram Processing, Spatial Filtering, Extension to functions of two variables, Image Smoothing, Image Sharpening, Frequency domain filtering, Homomorphic filtering

Unit IV : Image Segmentation

Hours 06

Edge based segmentation, Region based segmentation, Region split and merge techniques, Region growing by pixel aggregation, optimal thresholding, Hough Transform, boundary detection.

Unit V : Image Representations and Recognition

Hours 09

Boundary representation, Chain code, Polygonal Approximation, Signature, Boundary segments, Boundary description, Regional Descriptors, Topological Feature, Texture, Pattern and Patterns classes, Recognition Based on matching

Unit VI : Image Compression

Hours 08

Introduction, Image compression model, Error free compression, Variable length coding, Bit plan coding, Lossless predictive coding, Lossy compression, Compression standards.

Core Books:

1. Rafael C. Gonzalez, Richard E. Woods: Digital Image Processing, Fourth edition, Pearson education.
2. Anil K. Jain: Fundamentals of Digital Image Processing, First Edition, Pearson Education.

Reference Books:

1. Burger, Wilhelm, Burge, Mark J: Principles of Digital Image Processing, Springer.
2. Milan Sonka, Vaclav Hlavac, Roger Boyle: Image Processing, Analysis, and Machine Vision, CL Engineering
3. Nixon, Mark, and Alberto Aguado. Feature extraction and image processing for computer vision. Academic press, 2019

Web References:

1. <https://www.tutorialspoint.com/dip/> [Basics of Image Processing]
2. <https://sisu.ut.ee/imageprocessing/book/1> [All concepts of digital image processing]
3. <https://www.cs.auckland.ac.nz/courses/compsci773s1c/lectures/ImageProcessing-html/topic3.htm> [Image segmentation]
4. <http://www.elel.p.lodz.pl/mstrzeI/imageproc/enhancement1.PDF> [Image enhancement]

Course Outcomes: Upon successful completion of the course, students will,

CO1 :	Understand the concept of various types of images and analyze various techniques for intensity transformation and spatial filtering with applications.
CO2 :	Understand the concepts of color image processing and noise models.
CO3 :	Understand the concepts of image enhancement and its techniques.
CO4 :	Learn various image segmentation techniques.
CO5 :	Be able to represent and recognize images using various pattern recognition methods its compression techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Digital Image Fundamentals	√				
2	Colour Image Processing		√			
3	Image Enhancement			√		
4	Image Segmentation				√	
5	Image Representations and Recognition					√
6	Image Compression					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	1	3	-	1	3	1	2	3	1	3	2	1
CO2	2	2	1	1	-	2	-	-	2	-	3	3	-
CO3	2	2	1	1	-	2	-	-	2	-	3	3	-
CO4	2	2	1	1	-	2	-	-	2	-	2	2	-
CO5	2	2	1	1	-	2	-	-	2	-	3	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUS301 : Mobile Application Development

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Theory	02	30	30	60
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Quizzes, Assignments, Case Studies etc. Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: CTUC103-Object Oriented Programming Concepts.

Methodology, Pedagogy & Andragogy: The teaching methodology integrates theoretical and practical approaches to provide a comprehensive understanding for mobile application development subject for android application development using Kotlin language. Pedagogy involves interactive lectures, real-world examples, and guided coding sessions to build foundational knowledge and problem-solving skills. Andragogic strategies focus on self-directed learning through projects, case studies, and hands-on lab sessions, enabling students to apply theoretical concepts in developing functional mobile applications. Regular assessments, collaborative group tasks, and instructor feedback ensure an engaging and practice-oriented learning environment.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1.	Introduction Android and Android Development tools	05
2.	User Interface – I	07
3.	User Interface - II	05
4.	Material Design	04
5.	Application Data Handling	06
6.	Web Services	03

Total Hours (Theory): 30
Total hours (Practical): 60
Total Contact Hours: 90

Detail Syllabus:

Unit I: Introduction Android and Android

Hours 05

Development tools

Introduction to Mobile Applications and Android, Basics of Kotlin, setting up development environment, Dalvik Virtual Machine, Android Application Structure, DDMS, Android Manifest File, Gradle, Android permissions, APK, Basic Building blocks – Activities, Services, Broadcast Receivers & Content providers.

Unit II: User Interface – I

Hours 07

Working with Containers: LinearLayout, RelativeLayout, TableLayout, ScrollView, and FrameLayout, Fragment, Views: Toast, TextView, EditText, Button, ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, ProgressBar, AutoComplete TextView, Components for communication Intents & Intent Filters Intents: Explicit Intents, Implicit Intents, switching between activities and passing data between activities using Intents, Logging.

Unit III: User Interface - II

Hours 05

Calendar: Manage Date and Time, Adapters: ArrayAdapter, BaseAdapter ListView and ListActivity, GridView, Gallery Menus: Option menu, Context menu, Sub menu Tabs and TabActivity Alert Dialogs, Android Toolbar.

Unit IV: Material Design

Hours 04

Android Design Support Library, Vector Drawables, Basics of material design components and principles, Android styles and themes, Animations and Transitions.

Unit V: Application Data Handling

Hours 06

Basics of Storage in Android, Shared Preference: Save key-value data, Store Data in SQLite Database, store data in a firebase, save files on Device, Sharing of data.

Unit VI: Web Services

Hours 03

Working with Web Service in Android, Deploying Web Service.

Outline of the Course (Practical):

Week No	Practical	Description
1	Kotlin, Toast	Kotlin Code Programs, Toast Messages “Hello World”
2	TextView, EditText, Activity, Intent	Use of Textview, EditText and Toast with Operations. Use of Activity and Intent for values.
3	Views, EditText, ViewBinding, findViewById, layouts	Create the calculator app and use proper controls, Handle the views without using findViewById
4	Implicit, Explicit Intent, Activity Life Cycle	Manage Student Information using Intent and also create the App to learn the activity life cycle.
5	Different Activities, UI, toggle button, progress bar	Splash Screen, Login and user creation as well as create the application for addition operation of two variables.
6	EditText, Button	Create an app to perform the arithmetic operations using 4 buttons.
7	Implicate Intent, Explicit Intent, UI with functions	Open CHARUSAT website using Implicit Intent. Create an application to share data with native mobile apps for browser, calling app with optimized code.
8	Shared Preference and UI with complete application, Material design	Use shared Preferences for the session management and also create application for one college event using radio button, check boxes and spinners.
9	Images, Alert Dialog box	Display Toast message by clicking on Image Button and open CHARUSAT website. Use alert dialog for the messages in the application
10	AutoCompleteTextView	Create the application using AutocompleteTextView for Countries list.
11	Adapters, Array Adapter, Base Adapter, ListView	Use the Spinner using Array Adapter. Create the app with student photos using base adapter
12	Alert Dialog, Back Press	Manage the back press event using alert dialog.
13	Menu	Create an application with different types of menus. Context Menu Popup Menu Option Menu
14	Recyclerview, SQLite	Create app with Photos and names using recyclerview. Also create the app for CRUD operations using SQLite.
15	Firebase	Create the app for authentication using firebase. Also create the app for data management in Firestore.
		Total hours (Practical) : 60

Core Books:

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Reto Meier: Professional Android 2 Application Development, Wrox publication
3. Mark L. Murphy: The Busy Coder's Guide to Android Development.

Reference Books:

1. Dawn Griffiths, David Griffiths: Head First Android Development, O'REILLY publication
2. Mark L. Murphy: Beginning Android 2, APRESS publication
3. Mark L. Murphy: The Busy Coder's Guide to Advanced Android Development, Commons Ware, LLC.

Web References:

1. <https://kotlinlang.org/> [Kotlin Tutorials]
2. <https://developer.android.com/guide/> [Android Tutorials]
3. www.vogella.com/tutorials/android.html [Android Practical Tutorials]
4. <https://www.androidhive.info/> [Android Practical Tutorials]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be acquainted with Android OS
CO2 :	Be aware with Android Development Tools
CO3 :	Be aware with Android Development Tools
CO4 :	Be capable to create interactive mobile application that handle data
CO5 :	Be able to create mobile application that works web service

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction Android and Android Development tools	√				
2	User Interface – I		√			
3	User Interface - II			√		
4	Material Design				√	
5	Application Data Handling				√	
6	Web Services					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	3	-	2	1	-	-	2	-	-
CO2	2	-	3	-	2	1	-	-	2	2	3	2	1
CO3	-	3	-	2	-	1	3	2	-	-	-	3	3
CO4	2	1	2	2	2	1	2	-	2	-	2	2	3
CO5	-	2	2	3	1	-	2	2	2	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**TEACHING SCHEME &
DETAILED SYLLABUS**

FOR

**BCA PROGRAMME
(7th SEMESTER)
AS PER NEP 2020**

**EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-VII

Course Code	Course Name	Theory/ Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP - 2020 Classification
			CCE	SEE	CCE	SEE		
CAUC401	Enterprise Computing using Java EE	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC402	Advanced Database Technologies	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE401 CAUE402 CAUE403	(Select any one) 4. Cloud Computing 5. Blockchain Essentials 6. Cryptography and Network Security	Theory	50	50	-	-	04	Minor
CAUP401 CAUR401	(Select any one) 3. On-Job Training-I 4. Research Project-I	VIVA	-	-	75	75	06	Research Project / Dissertation
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CAUC401 : Enterprise Computing using Java EE

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: Object Oriented Programming Using Java (CAUC301)

Methodology, Pedagogy & Andragogy: Advanced Java fundamentals like JDBC, Servlet, JSP and Servlet Security provides capabilities and features for developing web application which is essential for developer in order to understand entire web application development life cycle. During the theory sessions of the syllabus, fundamentals of the mentioned technologies will be explained in detail with real time examples. Moreover, practical session will provide hands-on exposure about JDBC, Servlet, JSP and Servlet Security with variety of practical assignment. Also, MVC architecture using these technologies will be covered during theory as well as practical sessions.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1.	Fundamentals of Object Oriented Concepts	12
2.	JDBC Programming	10
3.	Fundamentals of Web Application Components	08
4.	Servlet API	12
5.	Java Server Pages	10
6.	Basics of Hibernate 4.0	08

Total Hours (Theory) : 60
Total hours (Practical) : 60
Total Contact Hours: 120

Detail Syllabus:

Unit I: Fundamentals of Object Oriented Concepts

Hours 12

Object Oriented programming concepts, Classes & Objects, Structure of Java Program, Constructors, Inheritance, Abstract Class, Interface, Exception Handling, Packages, Multithreading, Fundamentals of Collections.

Unit II: JDBC Programming

Hours 10

The JDBC Connectivity Model, Database Programming: Connecting to the Database, creating a SQL Query, Executing SQL Queries, Updating Database Data, JDBC Types, Error Checking and the SQLException Class, SQLWarning class, Statement Interface, PreparedStatement, Callable Statement, Result Set Interface, Updatable Result sets, Result SetMetaData, Transaction Management.

Unit III: Fundamentals of Web Application Components

Hours 08

Basics of web application development and web components, Introduction of MVC design pattern, Fundamentals of: Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, HTTP request methods, HTML Form Processing, HTTP request response cycle.

Unit IV: Servlet API

Hours 12

Servlet Model: Overview of Servlet, Servlet Life Cycle, HTTP Methods Structure and Deployment descriptor ServletContext and ServletConfig interface, Attributes in Servlet.

Request Dispatcher interface, The Filter API: Filter, FilterChain and Filter Config, Cookies and Session Management: Understanding state and session.

Understanding Session Tracking Techniques, Hidden Form Fields, URL Rewriting, Session & Cookies.

Unit IV: Java Server Pages

Hours 10

JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment, JSP Directives, JSP Action.

JSP Implicit Objects JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking, JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing.

Unit – VI: Basics of Hibernate 4.0

Hours 08

Overview of Hibernate, Hibernate Architecture, Hibernate Mapping Types, Hibernate O/R Mapping, Hibernate Annotation, Hibernate Query Language.

Outline of the Course (Practical):

Week No	Practical	Description
1	Fundamentals of Object Oriented Concepts	Implementing the concepts of Inheritance, Interface, Abstract class, Exception Handling and Packages.
2	Fundamentals of Object Oriented Concepts	Implementing the concepts of multithreading and fundamental of collection framework.
3	Basics JDBC and JDBC Queries	Configuration of Java SDK, NetBeans and Oracle 11 G for creating database, Understanding and executing queries such as create, Read, Update and Delete.
4	Statement types, ResultSetMetaData and Types of ResultSet	Implementing Statement, PreparedStatement, Callable Statement. Implementing ResultSetMetaData to know the more specific information about the records and to update the records even if the records are fetched in ReslultSet.
5	Web Application Components	Able to implement and deploy the applications on web server while configuring the Tomcat server. Understanding JAR Files, WAR Files, HTTP, HTTP request response cycle.
6	Web Application Components	Understanding the concept of Model View Controller(MVC) with proper code.
7	Servlet API	Able to configure the Tomcat Server, Understanding servlet, MVC architecture. Implementing ServletContext and ServletConfig, RequestDispatcher Interface and Filters in servlets.
8	JDBC Connectivity with Servlet	Creating servlet application for implementing CRUD operations using JDBC with servlets.
9	Session management using Servlet	Implementing the concepts of session techniques such as: Hidden Form Fields, URL Rewriting, Session and Cookies.
10	Java Server Pages (JSP)	Understanding JSP for web application, JSP Life cycle, MVC design with JSP, JSP directives.
11	JSP Actions	Understanding JSP implicit object of JSP, Actions in JSP and JDBC using JSP.
12	JSP Session management	Managing the session of the user using JSP session and cookies.
13	JSP Standard Tag Libraries	JSP Custom Tag, JSP Expression Language, JSP Exception Handling.
14	Hibernate mappings	Understanding the concepts Hibernate Mapping Types, Hibernate O/R Mapping,
15	Basics of Hibernate Annotations	Understanding the concepts of Hibernate Annotation, Hibernate Query Language.
		Total Hours: 60

Core Books:

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.
2. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.
3. Jeff Linwood and Dave Minter: Hibernate 2nd edition Beginning Après publication

Reference Books:

1. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition
2. James Keogh: Complete Reference J2EE, Mcgraw publication

Web References:

1. <http://courses.coreservlets.com/Course-Materials/csajsp2.html> [Servlet Basics]
2. <http://www.ceit.es/asignaturas/InteInfo/Recursos/Servlets/JavaServlets.pdf>
3. <http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf>
4. www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt

Course Outcomes:

CO1 :	To make understand the fundamental Concepts of Core Java.
CO2 :	Able to connect Java application with various RDBMS using JDBC.
CO3 :	To make understand web application architecture and terminologies.
CO4 :	Able to develop real time web application using Servlet and JSP technologies .
CO5 :	Able to use Object Relation Mapping using Hibernate to build database.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Fundamentals of Object Oriented Concepts	√				
2	JDBC Programming	√	√			
3	Fundamentals of Web Application Components	√	√	√		
4	Servlet API			√	√	
5	Java Server Pages			√	√	
6	Basics of Hibernate 4.0					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	2	1	3	1	1	1	1	1	2	1
CO2	3	2	2	3	1	3	1	1	1	2	2	2	2
CO3	3	3	3	2	2	3	2	2	1	2	2	2	2
CO4	3	3	2	2	2	3	1	2	2	1	2	2	2
CO5	3	2	2	2	1	3	2	2	2	2	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC402 : Advanced Database Technologies

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: Fundamentals of database

Methodology, Pedagogy & Andragogy: During theory lectures student will understand the essential concepts that act as the building blocks for many of the basic concepts of database, advanced concepts of database specifically PL-SQL and NoSQL products. It starts from the fundamentals of NoSQL and graduates to advanced concepts of architecture, storage internals and indexing. During Practical sessions, students will be required to connect both the databases, relational and non-relational with the application programs.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Database System	10
2	Advanced Database Systems	12
3	Introduction to NoSQL	08
4	NoSQL Stores Data Models	10
5	Storage Architecture	08
6	Working and Querying (CRUD operation) with Data using MongoDB	12

Total Hours (Theory): 60**Total Hours (Practical): 60****Total Contact Hours:120****Detail Syllabus:**

Unit I: Introduction to Database System

Hours 10

Basic Concepts: data, information, file, database, database systems, database management system Purpose and advantages of Database management system (over file systems), Applications of DBMS Various data models ER Model and Relational Model, Three level architecture, Structure of DBMS, Database actors and workers, Codd Rules.

Unit II: Advanced Database Systems

Hours 12

Basics of PL/SQL block, Usage of conditional statements and looping structures, Implicit and explicit cursor, User defined function and procedure, Packages, Introduction to row level and statement level triggers.

Unit III: Introduction to NoSQL

Hours 08

Introduction to SQL Relational Model, Overview and History of NoSQL Databases, Taxonomies by Data Model, Overview of No-SQL Storage: Key Value, Document-Oriented, Column-Family and Graph storage. Techniques and Pattern: CAP Theorem, ACID Vs. BASE properties and Comparison of relational databases to new NoSQL stores.

Unit IV: NoSQL Stores Data Models

Hours 10

Introduction, use and applications of: MongoDB, Cassandra, HBASE, Neo4j, Challenges of NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Aggregate-Oriented Databases, Replication and Sharding, Overview of MapReduce on databases. Distribution Models, Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication.

Unit V: Storage Architecture

Hours 08

Working with Document Databases, contrasting document databases with RDBMS, features of MongoDB, advantages of MongoDB over RDBMS, working with Column-Oriented Databases, Contrasting Column Databases with RDBMS, HBase Distributed Storage Architecture.

Unit VI: Working and Querying (CRUD operation) with Data using MongoDB

Hours 12

Installing and running MongoDB on your System, use of JavaScript's shell in MongoDB, navigating your databases, viewing available data and collections, inserting data, querying data, using sort, limit and skip functions, retrieving single document, using aggregation commands, working with condition, updating data, upsert and save command, rename a collection.

Outline of the practical course:

Week No	Practical	Description
1	Fundamentals of database management system	Installation and configuration of database
2	Basics of SQL	Basic commands of SQL: Data Definition Language (DDL), Data Manipulation Language (DML), Data Query Language (DQL) commands
3	Basics of PL-SQL	Understanding of PL-SQL blocks, basic datatypes used in PL-SQL, variables and constants in PL-SQL
4	Conditional statements	Understanding the conditional statements used in PL-SQL
5	Looping constructs	Understanding the looping constructs available in PL-SQL
6	Cursor and its types	Understanding the implicit and explicit cursors in PL-SQL
7	Stored procedures	Understanding the use of stored procedures in PL-SQL
8	User defined functions	Understanding the use of user defined functions in PL-SQL
9	Packages	Understanding the use of packages in PL-SQL
10	Triggers	Understanding the use of row and statement level triggers in PL-SQL
11	Basics of NoSQL	Installation and configuration of MongoDB, Understanding the use of JavaScript's shell in MongoDB
12	Basics of MongoDB	MongoDB – Database, Collection and Document MongoDB – Cursor, datatypes, creation and removal of database in MongoDB
13	MongoDB CRUD Operations	Create, Read, Update and Delete documents
14	Operators used in MongoDB	Understanding the use of Comparison, Logical, Arithmetic, Field update operators in MongoDB to work with the conditions
15	Querying MongoDB	MongoDB – Find(), FindAndModofy(), FindOne(), sort(), limit(), skip(), aggregation

Core Books:

1. Ivan Bayross : SQL, PL/SQL The programming Language Oracle.
2. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Author: Sadalage, P. & Fowler, Publication: Pearson Education.

Reference Books:

1. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.
2. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
3. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
4. Definite Guide to MongoDB by Eelco Plugge, Peter Membrey and Tim Hawkins, Apress.
5. Getting started with NOSQL by Gaurav Vaish, PACKT.
6. Name: Redmond, E. & Wilson, Author: Seven Databases in Seven Weeks: A Guide to Modern Databases and the NoSQL Movement Edition: 1st Edition.
7. NOSQL theory and Examples by Piotr fulmanski, Simple introduction series: e-book.
8. Guy Harrison, Next Generation Database: NoSQL and big data, Apress.

Web References:

1. <http://docs.oracle.com/>
2. <http://plsql-tutorial.com/>[For PL/SQL]
3. https://www.tutorialspoint.com/mongodb_for_absolute_beginners/index.asp [MongoDB For Absolute Beginners - Tutorialspoint]
4. https://www.tutorialspoint.com/neo4j/neo4j_data_model.htm [Neo4j - Data Model (tutorialspoint.com)]

Course Outcomes:

CO1	Understand basic concepts regarding data, database systems and various data models.
CO2	Enhance the knowledge of the processes of Database Development and Administration using SQL and PL/SQL
CO3	Able to understand different types of NoSQL Databases
CO4	Architectures and common features of the basic types of NoSQL databases
CO5	Able to design Schema and implement CRUD operations, distributed data operations, and implement various column store internals

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Database System	✓				
2	Advanced Database Systems		✓			
3	Introduction to NoSQL			✓		
4	NoSQL Stores Data Models			✓		
5	Storage Architecture				✓	
6	Working and Querying (CRUD operation) with Data using MongoDB				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	3	1	-	3	-	3	3	-	3	-
CO2	3	3	3	3	3	3	3	3	1	2	3	3	2
CO3	3	2	-	1	1	-	3	-	3	-	-	2	-
CO4	3	1	2	1	3	-	3	2	3	1	-	2	1
CO5	3	3	3	3	3	3	3	3	2	1	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CAUE401 : Cloud Computing

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: Operating System Concepts and Network Technology

Methodology, Pedagogy & Andragogy: During theory lectures the emphasis will be given on the fundamentals of cloud computing. Students will be introduced basic types, architecture, service providers, mechanism, security issues and some hidden aspects of cloud computing. Students will give practical exposure in form of case study and by showing cloud infrastructure of university.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Evolution of Cloud Computing	09
2	Understanding Cloud Computing and basic types	10
3	Fundamentals of Cloud Architecture & Service Providers	11
4	Cloud Computing Mechanisms	11
5	Cloud Computing Security and Business Use	10
6	Intensive analysis of Cloud Computing with case studies	09

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Evolution of Cloud Computing

Hours 09

Introduction of Cloud Computing, Growth of technology, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

Unit II: Understanding Cloud Computing and basic types

Hours 10

Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing. Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing.

Unit III: Fundamentals of Cloud Architecture and Service Providers

Hours 11

Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture. Introduction to major Cloud Service Provider: Amazon Web Services, Google Cloud. Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terrimark, DropBox.

Unit IV: Cloud Computing Mechanisms

Hours 11

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication. Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

Unit V: Cloud Computing Security and Business Use

Hours 10

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single SignOn (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control. Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

Unit VI: Intensive analysis of Cloud Computing with case studies

Hours 09

Overview of Cloud services, Designing Solutions for the Cloud, Implement & Integrate Solutions, Emerging Markets and the Cloud.

Core Books:

1. Kevin L. Jackson, Scott Goessling: Architecting Cloud Computing Solutions: Packt Publication: 2018.
2. Thomas Erl, Zaigham Mahmood and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture, PHI, 2013.
3. S. Srinivasan: Cloud Computing Basics, Springer, 2014.

Reference Books:

1. Derrick Rountree, Ileana Castrillo: The Basics of Cloud Computing, Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing- Principles and Paradigms, John Wiley & Sons, 2011.

Web References:

1. http://whatisCloud.com/basic_concepts_and_terminology/Cloud [For basic terminology of Cloud Computing].
2. http://www.tutorialspoint.com/Cloud_Computing/ [For cloud computing lecture notes].
3. <http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloudcomputingvirtualization-building-private-iaas-guide.pdf> [For cloud computing virtualization].
4. <https://www.cs.purdue.edu/.../Any-Kim-Bhargava-MCCWorkshop.ppt> [Security issues PPTs].

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will learn basics of cloud computing.
CO2 :	Students will be familiar with various cloud architectures and services.
CO3 :	They will get the knowledge of various network mechanics.
CO4 :	Students will get various business aspects of cloud computing with security aspects.
CO5 :	They will be able to design and deploy Cloud Infrastructure.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Evolution of Cloud Computing	√				
2	Understanding Cloud Computing and basic types		√			
3	Fundamentals of Cloud Architecture & Service Providers		√			
4	Cloud Computing Mechanisms			√		
5	Cloud Computing Security and Business Use				√	
6	Intensive analysis of Cloud Computing with case studies					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	-	-	1	-	3	1	2	2	-	-	1
CO2	3	1	1	1	1	1	3	1	2	2	1	3	2
CO3	3	2	1	2	1	3	3	2	2	2	-	3	2
CO4	3	3	2	3	1	3	3	2	2	2	3	-	1
CO5	3	3	3	3	1	3	3	3	2	2	2	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE402 : Blockchain Essentials

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Group Discussions, Case Study etc.					

Pre-requisite: None

Methodology, Pedagogy & Andragogy: During the lecture sessions, a blended instructional approach will be adopted to ensure both conceptual clarity and practical relevance. Students will be introduced to various subsystems that work in an integrated manner to make blockchain technology function effectively, with emphasis on major platforms such as Bitcoin, Ethereum, and Hyperledger. The teaching methodology will involve interactive lectures, real-world case studies, and visual demonstrations to explain key blockchain components like consensus algorithms, cryptographic techniques, and distributed ledgers. From an andragogical perspective, the teaching will acknowledge the learners' prior knowledge and focus on real-world applicability, career relevance, and goal-oriented learning. This integrated approach ensures that learners not only understand the technology but also develop the skills to apply blockchain concepts in diverse application domains.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Blockchain	08
2	Functioning of Blockchain	11
3	Bitcoin and related concepts	11
4	Ethereum and related concepts	11
5	Hyperledger and related concepts	10
6	Applications of blockchain	09

Total Hours (Theory): 60
Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to blockchain

Hours 09

History of blockchain, Centralized, decentralized and distributed systems, Layers of blockchain, Importance of Blockchain.

Unit II: Functioning of blockchain

Hours 10

Cryptographic concepts- Symmetric key cryptography, Cryptographic hash function, MAC and HMAC, Asymmetric key cryptography, Diffie-Hellman key exchange, Byzantine general's problem, Merkel tree, Structure of block.

Unit III: Bitcoin and related concepts

Hours 11

What is bitcoin, Working with bitcoin, Bitcoin network, Bitcoin script, Bitcoin mining, Bitcoin wallets, Types of blockchain networks, Consensus algorithms.

Unit IV: Ethereum and related concepts

Hours 11

Design of ethereum, Ethereum blockchain, Smart contracts in ethereum, Ethereum virtual machine, Ethereum eco system.

Unit V: Hyperledger and related concepts

Hours 10

Introduction to hyperledger and hyperledger fabric, hyperledger fabric transaction flow, membership and identity management, fabric network setup, hyperledger fabric demo, hyperledger composer demo.

Unit VI: Applications of blockchain

Hours 09

Applications of blockchain in domains – Finance, Healthcare, Insurance, Supply chain management, Internet of things.

Core Books:

1. Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda: Beginning Blockchain - A Beginner's Guide to Building Blockchain Solutions: 1st Edition: APress Publication: 2018.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder: Bitcoin and Cryptocurrency Technologies - A Comprehensive Introduction: 1st Edition: Princeton University Press: 2016.
3. Tiana Laurence: Introduction to blockchain technology – the many faces of blockchain technology in the 21st century: 1st Edition: Van Heren Publishing: 2019

Reference Books:

1. S. Ashraf, S. Adarsh: Decentralized Computing Using Blockchain Technologies and Smart Contracts - merging Research and Opportunities: 1st Edition: IGI Global, 2017
2. Fabian Schar, Aleksander Berentsen: Bitcoin, Blockchain, and Cryptoassets: 1st Edition: MIT Press: 2020

Web References:

1. <https://andersbrownworth.com/cms/460/blockchain/demo> [Blockchain Demo]
2. <https://ethdocs.org/en/latest/introduction/what-is-ethereum.html> [Ethereum Introduction]
3. https://www.hyperledger.org/wp-content/uploads/2018/07/HL_Whitepaper_IntroductiontoHyperledger.pdf [Hyperledger Introduction]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	The students will learn about underlying technologies of blockchain and various applications of blockchain technology.
CO2 :	The students will understand working of blockchain in detail.
CO3 :	The students will learn about how the blockchain is used to power bitcoin.
CO4 :	The students will see how ethereum utilizes blockchain.
CO5 :	The students will recognize the role of blockchain in hyperledger platform.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Blockchain	√				
2	Functioning of Blockchain		√			
3	Bitcoin and related concepts			√		
4	Ethereum and related concepts				√	
5	Hyperledger and related concepts					√
6	Applications of blockchain	√				

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	-	1	-	-	3	-	-	-	-	1	-
CO2	-	1	2	1	1	-	3	-	-	-	-	1	-
CO3	2	2	3	-	3	-	-	-	-	-	-	2	1
CO4	3	2	3	-	3	-	-	-	-	-	-	2	1
CO5	3	1	3	1	3	-	-	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE403 : Cryptography and Network Security

Credit	Component	Instruction/ Contact Hours (Per Week)	Instruction/ Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
Experiential Learning Components : Unit Tests, Assignments, Case Study etc.					

Pre-requisite: Basic concepts of computer network and digital communication

Methodology, Pedagogy & Andragogy: During the lectures the students will learn about basic cryptographic concepts. The teacher will also present applications of cryptography with focus on network security. Further the teacher will also introduce system level security issues and their solutions.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Network Security	06
2	Symmetric and Public Key Cryptography Systems and Related Concepts	11
3	Key Distribution	10
4	Transport Level Security	11
5	Wireless and Internet Security	11
6	Advanced Security Concepts	11

Total Hours (Theory): 60

Total Contact Hours:60

Detail Syllabus:

Unit I: Introduction to Network Security

Hours 06

Network security concepts, Need for security, OSI security architecture, Security attacks, Security services, Security mechanism, Model for network security, Network security standards

Unit II: Symmetric and Public Key Cryptography Systems and Related Concepts

Hours 11

Symmetric and Asymmetric Key Cryptography, Encryption system and algorithm, Concepts of random and pseudo numbers, Operations on stream ciphers and block ciphers. Message authentication concepts, Secure hash function, Message authentication codes, Public key cryptography, Digital signature.

Unit III: Key Distribution

Hours 10

Basics of key distribution, Symmetric key distribution, Kerberos algorithm, Asymmetric key distribution, Public key infrastructure, Federated identity management, Personal identity verification

Unit IV: Transport Level Security

Hours 11

World Wide Web security, Web Security Threats, SSL Architecture, SSL Protocols, Transport layer security, Alert Codes, HTTPS, Secure Shell.

Unit V: Wireless and Internet Security

Hours 11

Wireless security: IEEE 802.11 standard, Wireless LAN security, Wireless application protocol (WAP), Transport layer security in wireless network, WAP end-to-end security, Mobile Device Security

Internet security: pretty-good-privacy, Multipurpose Internet Mail Extensions (MIME), Domain key based mail identification. IP security policy, Integrated security features in IP packet, Internet key exchange (IKE), Cryptographic suits.

Unit VI: Advanced Security Concepts

Hours 11

Intruders, Intrusion detection, Password management, Malicious software: Viruses, Worms, Backdoor, Logic Bomb, Trojan Horses, Zombie, Virus Countermeasures, Firewalls, Design principles of firewall, Characteristics of firewall, Types of firewall, Firewall location and configuration, Trusted systems.

Core Books:

1. William Stallings: Network Security Essentials: 4th Edition, Pearson Publication: 2018.
2. William Stallings: Cryptography and network security: 6th Edition, Pearson Publication: 2019

Reference Books:

1. Atul Kahate: Cryptography And Network Security: 2nd Edition, Tata McGraw Hill Education Private Limited: 2011.
2. Alphred J. Menezes: Handbook of applied cryptography: 3rd Edition: CRC Press : 2010.
3. David Pointcheval: Applied cryptography and network security: 1st Edition: Springer: 2012
4. W. Chen, H.-W.LI et al.; Applied Cryptography and Network Security, 1st Edition, Magnum Publication, 2016

Web References:

1. <https://www.edureka.co/blog/what-is-network-security/> [Introduction to Network Security]
2. <https://sectigostore.com/blog/5-differences-between-symmetric-vs-asymmetric-encryption/> [Symmetric and Asymmetric Key Difference]
3. <https://www.cloudflare.com/en-in/learning/ssl/transport-layer-security-tls/> [Transport Level Security]
4. <https://www.checkpoint.com/cyber-hub/network-security/what-is-firewall/> [Firewall]

Course Outcomes: Upon successful completion of the course, students will,

CO1	Be able to understand network security
CO2 :	Be familiar with symmetric and public key cryptography and key distribution
CO3 :	Understand transport level security
CO4 :	Be familiar with wireless and Internet security
CO5 :	Be able to understand advanced security concepts

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Network Security	√				
2	Symmetric and Public Key Cryptography Systems and Related Concepts		√			
3	Key Distribution		√			
4	Transport Level Security			√		
5	Wireless and Internet Security				√	
6	Advanced Security Concepts					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	3	2	3	1	2	2	3	2	3	3	3
CO2	3	2	3	3	3	2	3	3	2	3	2	2	3
CO3	2	2	3	3	3	2	2	2	3	2	3	3	3
CO4	2	3	3	2	3	1	2	3	2	3	2	3	2
CO5	3	3	3	3	3	2	3	3	2	2	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUP401 : On-Job Training-I

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP-2020 Classification
			CCE	SEE	CCE	SEE		
CAUP401	On-Job Training-I	Viva	-	-	75	75	06	Research Project/Dissertation

Credit	Component	Instruction /Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
06	VIVA	12	180	270	450

Pre-requisite: None

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Apply foundational IT skills (e.g., basic programming, database querying, or system configuration) to perform structured tasks in an industry setting.
CO2 :	Demonstrate effective communication and teamwork under supervision in professional IT environments.
CO3 :	Utilize industry-standard IT tools (e.g., IDEs, DBMS software) to complete assigned tasks with guidance.
CO4 :	Identify workplace practices, workflows, and professional ethics relevant to IT roles.
CO5 :	Document and reflect on foundational IT tasks through logbooks and reports to enhance learning outcomes.

- **Refer Annexure-I for more information.**

CAUR401 : Research Project-I

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP-2020 Classification
			CCE	SEE	CCE	SEE		
CAUR401	Research Project-I	Viva	-	-	75	75	06	Research Project/Dissertation

Credit	Component	Instruction /Contact Hours (Per Week)	Instruction / Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
06	VIVA	12	180	270	450

Pre-requisite: None

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Formulate a clear research question or problem statement in the IT domain using appropriate methodologies.
CO2 :	Conduct a systematic literature review to identify relevant research gaps and existing solutions.
CO3 :	Apply research tools and techniques (e.g., data analysis software, programming libraries) to design exploratory studies or experiments.
CO4 :	Demonstrate ethical research practices in data handling, citation, and adherence to intellectual property guidelines.
CO5 :	Prepare a research proposal or methodology report, articulating the research plan with clarity and precision.

- **Refer Annexure-I for more information.**

**TEACHING SCHEME &
DETAILED SYLLABUS

FOR

BCA PROGRAMME
(8th SEMESTER)
AS PER NEP 2020

EFFECTIVE FROM
ACADEMIC YEAR 2025-26**

BCA Semester-VIII

Course Code	Course Name	Theory/ Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP -2020 Classification
			CCE	SEE	CCE	SEE		
CAUC403	Full Stack Web Development	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUC404	Advanced Mobile Programming	Theory and Practical	50	50	25	25	06 (04-Theory) (02-Practical)	Major Core
CAUE404	Web Development Using Open-Source Technologies	Practical	-	-	25	25	02	Minor
CAUE405 CAUE406 CAUE407	(Select any one) 4. Python Framework 5. HTTP Web service for Enterprise Applications 6. Game Development using Unity	Practical	-	-	25	25	02	Minor
CAUP402 CAUR402	(Select any one) 3. On Job Training-II 4. Research Project-II	VIVA	-	-	75	75	06	Research Project/Dissertation
*CCE- Continuous and Comprehensive Evaluation (Assignments, Unit Tests, Sessional Test, Case Study, Attendance, Practical Test, Viva etc.) *SEE- Semester –End- Evaluation								

CAUC403 : Full Stack Web Development

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA, certificate course etc.					

Pre-requisite: Working knowledge of HTML and JavaScript

Methodology, Pedagogy & Andragogy: During the theory sessions students will understand the various JavaScript frameworks and its architecture, also able to comprehend MEAN stack application concepts. During the practical sessions students will learn how to reduce the amount of code you write to build rich user interface applications, Modularizing code, retrieving data from back-end- server and manipulate it.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to JavaScript and JavaScript Frameworks.	07
2	Directives in AngularJS	11
3	MVW-The AngularJS way	12
4	Introduction to NodeJS	10
5	Introduction to ExpressJS Framework	11
6	Working with Database	09

Total Hours (Theory): 60
Total Hours (Practical): 60
Total Contact Hours:120

Detail Syllabus:

Unit I: Introduction to JavaScript Frameworks.

Hours 07

Introduction JavaScript and its DOM concepts, Arrow function, JSON Objects, JavaScript Frameworks & Libraries, MEAN.JS introduction, Architecture of MEAN.JS.

Unit II: Introduction to AngularJS.

Hours 11

Introduction to AngularJS, Features of AngularJS, Data Binding in angularJS, working with Expressions in angularJS, Introduction to Directives, Directive Lifecycle, Using Angular JS built-in directives, creating a custom directive, Working with \$scope, \$rootScope, Lifecycle of \$scope.

Unit III: MVW-The Angular JS way.

Hours 12

MVW Architecture: Model-View-Controller and Model-View-View-Model Architecture, Introduction to AngularJS Modules – Application module & Controller modules, Attaching Properties and functions to scope, Controller in external files, AngularJS Filters, working with Angular Forms, Form events, validating Angular forms.

Unit IV : Introduction to NodeJS

Hours 10

Introduction to NodeJS, Features of NodeJS, Traditional web Server Model, Node.js process Model, Installation of NodeJS, working with Node Package Manager, Command Line options, NodeJS modules – Core Modules, Creating Local modules & Exporting Local modules, Third-party modules, Creating NodeJS web server.

Unit V: Introduction to ExpressJS Framework.

Hours 11

Introduction to ExpressJS, Installing ExpressJS using NPM, Advantages of ExpressJS, Installing Express.js, building web server, configure routes, ExpressJS middleware's, Serve Static Resources using Express.js, working with HTTP methods of ExpressJS, ExpressJS routing, Creating RESTful API.

Unit VI: Working with Database.

Hours 09

Introduction to Mongo DB, Installation of MongoDB – MongoDB Server & MongoDB Shell, MongoDB Compass, MongoDB Database, Collections, and documents, Basic operations using MongoDB – Create, Insert, Update and Delete, Access MongoDB in Node.js, setting up mongoose, Connecting MongoDB, Insert, update and delete document.

Outline of the course (Practical):

Week No.	Practical	Description
1	JavaScript function, event and basic DOM manipulation methods.	Introduction JavaScript DOM
2	AngularJS application – Auto initialization and manual initialization, Basic directives, AngularJS Expression	Basic concepts of AngularJS application
3	Working with AngularJS conditional directives	Familiar with frontend condition using AngularJS.
4	Working with Complex datatype - array and object, Working with AngularJS loop directives.	Get familiar with frontend loop with complex datatype.
5	Working with MVC and MVVM architecture – Creating module and controller	Able to identify the difference between One-way and two-way data binding methods with architecture.
6	Creating and using custom directives of AngularJS	Understand the reusability component in AngularJS.
7	Working with NodeJS – Basic NodeJS program, NPM Commands, Working with Core modules in NodeJS – Creating HTTP server, Handling files, URL & Path	Understand the NodeJS environment and its Core modules, Get familiar with Node Package Manager repository.
8	Creating custom local module in NodeJS	Able to create user defined module in NodeJS
9	Working with NodeJS third-party module – ExpressJS, Basic concept & methods.	Able to create server using ExpressJS module.
10	Working with files, middleware's of ExpressJS.	Able to access .html & its dependency file on a server.
11	Creating restful APIs in ExpressJS – GET, POST, PUT, DELETE.	Able to create restful APIs to read, Insert, Update and delete data from JSON file.
12	Working with MongoDB Command & MongoDB Atlas	Get able to create database and apply CRUD operation through MongoDB shell & MongoDB Atlas.
13	Working with Mongoose – NodeJS third-party module for Database connection.	Able to access data of MongoDB database in ExpressJS Application.
14-15	Working with MEAN Stack application	Able to create full stack application with MongoDB, ExpressJS, AngularJS and NodeJS.

Core Books:

1. Jeffry Houser : “Learn With: Angular 5, Bootstrap, and NodeJS”, Kindle Edition,
2. Shyam Seshadri Brad Green: “AngularJS – Up and Running, Brad Green”, Second Edition, O'REILLY
3. Basarat Ali Syed: “Beginning Node.js”, Apress Publication.
4. Greg Lim : “Beginning MEAN Stack (MongoDB, Express, Angular, Node.js)”, kindle Edition.

Reference Books:

1. Agus Kurniawan: “AngularJS Programming by Example 2017 Edition”, Kindle Edition.
2. Adam Freeman: “ Pro AngularJS 2017 Edition”, Apress.
3. Krasimir Tsonev: “Node.js By Example”, Packt Publishing
4. Ethan Brown: “Web Development with Node and Express”, O'REILLY

Web References:

1. <http://www.w3schools.com/angular/default.asp> [Tutorial link for AngularJS]
2. <http://www.tutorialspoint.com/angularjs/> [Tutorial link for AngularJS]
3. https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf [E-book for AngularJS]
4. <http://www.tutorialsteacher.com/nodejs/nodejs-modules> [Tutorial link for NodeJS]
5. <https://www.javatpoint.com/mean-stack-tutorial> [MEAN stack development]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Familiar with JavaScript and JavaScript frameworks.
CO2 :	Able to understand basic fundamental of AngularJS and implement Model-View-View-Model Architecture.
CO3 :	Able to create NodeJS and ExpressJS applications.
CO4 :	Able to create Restful API to access data from MongoDB.
CO5 :	Able to create Single page application and MEAN stack application.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to JavaScript and JavaScript framework	√				√
2	Introduction to AngularJS		√			√
3	MVW-The Angular JS way		√			√
4	Introduction to NodeJS			√	√	√
5	Introduction to ExpressJS			√	√	√
6	Working with Database				√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	2	-	1	-	-	-	-	1	-
CO2	1	2	2	1	2	-	1	1	-	1	1	2	1
CO3	1	-	3	1	3	-	2	1	-	-	1	2	1
CO4	2	2	3	2	3	2	2	3	-	2	2	3	3
CO5	2	3	3	2	3	2	2	3	2	2	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUC404 : Advanced Mobile Programming

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc. Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA etc.					

Pre-requisite: Object-oriented Programming Concepts

Methodology, Pedagogy & Andragogy: During the theory session illustrations emphasizing the need for Cross-platform Mobile application development will be given. Further, fundamental to intermediate concepts of programming in Flutter will also be discussed. During Practical sessions, students will be required to develop mobile applications using Dart programming language. Student shall also develop mobile applications with elegant user interface and applications that interact with popular cloud-based database.

Outline of the course (Theory):

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Cross-platform application development with Flutter & Dart	09
2	UI Designing with Flutter	11
3	State management with App creation	09
4	Leveraging Flutter packages and structuring Flutter applications	11
5	Incorporating backend data with Flutter application	10
6	Integrating Flutter Application with Firebase and State Management	10

Total Hours (Theory): 60
Total Hours (Practical): 60
Total Contact Hours:120

Detailed Syllabus:

Unit I Introduction to Cross-platform application development with Flutter & Dart **Hours 09**

Set up a new Flutter project using Android Studio. Widget tree, Interface design: pre-made Flutter Widgets. Image and Text Widgets. App Icons for iOS and Android. Add and load image assets to Flutter projects. Run Flutter apps on iOS Simulator, Android Emulator and physical devices.

Unit II: UI Designing with Flutter **Hours 11**

Hot Reload and Hot Restart, Use of Pubspec.yaml file, custom assets and fonts. An introduction to the Widget build(), layout widgets: Columns, Rows, Containers and Cards. Material icons, Icons class. Theme widgets. Refactoring widgets. Dart annotations and modifiers. Immutability of Stateless and Stateful Widgets. Update screen with the build(). Custom Flutter Widgets. Difference between final and const in Dart. Maps, enums and the ternary operator. Functions and arguments in Dart. Multi-screen Flutter apps, routes and the Navigator widget. Flutter favours composition vs. inheritance (customizing widgets).

Unit III: State management with App creation **Hours 09**

About Stateful and Stateless Widgets, callbacks. Declarative style of UI programming, Flutter widgets react to state changes. Import Dart libraries. Variables, data types and functions in Dart. Build flexible layouts. Understand the relationship between setState(), State objects and Stateful Widgets.

Unit IV: Leveraging Flutter packages and structuring Flutter applications **Hours 11**

Dart package manager, Flutter compatible packages. The structure of the pubspec.yaml file. Incorporate the audio-players package to play sound. Functions in Dart and arrow syntax. Refactor widgets, Flutter's philosophy of UI as code. The lists and conditionals in Dart. Classes and objects. Understand Object Oriented Dart. Dart Constructors for Flutter widgets. Design patterns to structure Flutter apps. Structuring and organizing Flutter apps.

Unit V: Incorporating backend data with Flutter application **Hours 10**

Asynchronous programming in Dart and use of async/await. Stateful Widget lifecycle methods, handling exceptions in dart. Null aware operators. Location data from both iOS and Android. Http package and live data from open APIs. Parse JSON data using the dart:convert library.

Unit VI: Integrating Flutter Application with Firebase and State Management **Hours 10**

State objects via the Stateful Widget. Use the TextField Widget to take user input. Pass data backwards using the Navigator widget. Hero animations in Flutter apps. Animation controller, custom animations. The use of Dart mixins. Firebase Cloud Firestore into your Flutter apps. Authentication with Firebase Auth package.

Outline of the course (Practical):

Week No.	Practical	Description
1	Setting up the development environment and understanding fundamentals of Flutter applications	Installing Flutter and configuring Android Studio for Flutter application development, understanding Flutter app directory structure, setting target device, creating and executing a simple Flutter app.
2	Widgets and widget tree	Using simple widgets to display text and images, Stateless vs. Stateful widgets, understanding the concept of widget tree.
3	More about Flutter application execution	Hot Reload, Hot Restart and Full Restart, pubspec.yaml, importance of the build() method
4	Designing UI in Flutter applications	Top-level widgets-MaterialApp, Scaffold, AppBar, Container, Drawer, Row, Column, Center Applying theme
5	Common widgets-I	Text, TextField, ElevatedButton, ListTile, RadioListTile, CheckboxListTile, DropdownList, CircleAvatar, Icon, Image, Using callbacks.
6	Commons widgets-II	ListView, GridView, Slider, DatePicker, TimePicker, Badge
7	Navigation and routing	Navigation in a multi-screen app with the help of push(), pop(), pushNamed() and named routes, passing data in navigation, using Tabs
8	Dart programming language fundamentals	Variables, constants, functions, object-oriented programming in Dart.
9	Using dependencies	Adding dependencies in pubspec.yaml, importing packages and using API.
10	Asynchronous programming	Use of await, async, Stateful Widget lifecycle methods, handling exceptions in dart.
11	Accessing location and HTTP service	Getting location of a device, fetching data from HTTP service
12	Using animation	Hero animation, Animation controller, custom animations, the use of Dart mixins.
13	Introduction to cloud-based database	Creating a Firebase project and configuring it for Flutter app, creating a Firestore database, understating collection, document, fields.
14	Database operations from Flutter app	Performing CRUD operations from a Flutter app.
15	Firebase Authentication	Configuring Firebase for authentication and using it for authentication in Flutter app.

Core Books:

4. Marco L. Napoli: Beginning Flutter: A Hands On Guide to App Development: Wrox publication: 2019.

Reference Books:

3. Eric Windmill: Flutter in Action: Edition: Manning Publication: January 2020.
4. Alessandro Biessek: Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart 2: Packt publication: September 2019

Web References:

1. <https://docs.flutter.dev/reference/tutorials>
2. <https://www.tutorialspoint.com/flutter/index.htm>
3. <https://www.javatpoint.com/flutter>
4. <https://fluttertutorial.in/>

Course Outcomes: Upon successful completion of the course, students will be

CO1 :	able to clear all object oriented programming and cross platform concepts.
CO2 :	able to learn Flutter and Dart step by step.
CO3 :	able to learn the reduce the code through native app performance, animated UI with material design and least testing.
CO4 :	able to use Firebase to authenticate the users and use the remote database.
CO5 :	able to build engaging native mobile apps for both Android and iOS.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Cross-platform application development with Flutter & Dart	√				
2	UI Designing with Flutter		√			√
3	State management with App creation					√
4	Leveraging Flutter packages and structuring Flutter applications			√		√
5	Incorporating backend data with Flutter application		√			
6	Integrating Flutter Application with Firebase and State Management		√	√	√	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	-	-	3	2	1	3	-	1	-	1	-	1
CO2	3	1	3	2	3	1	2	1	1	-	1	1	2
CO3	3	-	3	1	3	1	1	2	1	-	1	1	2
CO4	3	3	3	3	3	1	1	3	1	-	2	3	1
CO5	3	3	3	1	3	1	1	3	1	1	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE404 : Web Development using Open Source Technology

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
04	Theory	04	60	50	110
02	Practical	04	60	40	100
Experiential Learning Components (Theory): Unit Tests, Assignments, Case Study etc.					
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Pre-requisite: Basic understanding of HTML and MySQL.

Methodology, Pedagogy & Andragogy: During theory and practical sessions, students able to install & configure PHP and prerequisite software. Also, student will be emphasized to develop dynamic web applications and create web API in PHP.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Basics of HTML and JavaScript.	08	60
2	Basics of PHP.	10	
3	Handling Form Data and Database Connectivity with MYSQL	12	
4	PHP Utilities.	10	
5	PHP API client.	10	
6	AJAX With DATABASE.	10	

Total Hours (Theory): 60
Total Hours (Practical): 60
Total Contact Hours: 120

Detail Syllabus:

Unit I: Basics of HTML and Java Script

Hours 08

Basics of HTML, Introduction of JavaScript, Variables, Overview of operators, Control statements and looping statement, Overview of DOM, function declaration and calling with event.

Unit II: Basics of PHP

Hours 10

Introduction to PHP, PHP Apache Web server, Understanding of PHP.INI file Understanding of PHP .htaccess file, PHP Variable, Static & global variable, PHP Operator, Conditional Structure & Looping Structure, Array, User Defined Functions: argument function default argument, variable function, return function.

Unit III: Handling Form Data and Database Connectivity with MYSQL

Hours 12

HTML Form element & its attributes, Send Form data using GET Method & POST Method, Receive Form data using \$_GET, \$_POST & \$_REQUEST variables, PHP form validations, OOPs Concepts, PHP Data Objects (PDO), CRUD operations, Handling Errors, State Management Techniques: Concept of Session, starting session, modifying session variables, Un registering and deleting session variable, Concept of Cookies.

Unit IV: PHP utilities

Hours 10

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server, Sending mails.

Unit V: PHP API client

Hours 10

Basic API call from PHP, HTTP basics: requests, responses and using cURL, Response codes: get the HTTP status code.

Unit VI: AJAX With DATABASE

Hours 10

Basics of AJAX, PHP with AJAX, MySql with AJAX, What is JQuery AJAX, JQuery AJAX with PHP.

Outline of the course (Practical):

Week No.	Practical	Description
1	Introduction to HTML	Create basic web pages using HTML tags (headings, paragraphs, links, tables, forms).
2	JavaScript Basics	Use JavaScript for variable declaration, operators, control structures, loops.
3	JavaScript DOM & Events	Manipulate DOM elements and handle events using JavaScript functions.
4	PHP Setup & Basics	Install XAMPP/WAMP, write first PHP script, configure php.ini and .htaccess.
5	PHP Variables, Operators & Loops	Work with different types of variables and implement loops and conditions.
6	Arrays & Functions in PHP	Create indexed/associative arrays and define custom PHP functions.
7	HTML Forms & Data Submission	Build HTML forms and submit data using GET & POST methods in PHP.
8	Handling Form Data in PHP	Process form data using \$_GET, \$_POST, and \$_REQUEST, with validations.
9	PHP & MySQL CRUD with PDO	Connect to database using PDO, perform Create, Read, Update, Delete operations.
10	Sessions & Cookies	Implement session management and cookie handling in PHP.
11	OOP in PHP	Create classes, objects, constructors, and use basic OOP concepts.
12	File Upload in PHP	Upload single and multiple files using PHP forms.
13	PHP Mail & HTTP Headers	Send emails using PHP mail() and manipulate HTTP headers.
14	API Calls in PHP using cURL	Call REST APIs, fetch responses, and handle status codes using cURL.
15	AJAX & JQuery Integration	Perform AJAX operations using PHP, MySQL, and JQuery.

Core Books:

1. Vikram Vaswani, PHP: A Beginner's Guide: Indian Edition, First Edition, McGraw Hill, 2009
2. Matt Doyle, Beginning PHP 5.3, Wrox, 2010
3. Ballard and Moncur, Teach Yourself Javascript in 24 Hours, Sams Publishing, 2015

Reference Books:

1. Timothy Boronczyk, Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz, Michael K. Glass: Beginning PHP6, Apache, MySQL Web Development: Wrox , 2009.
2. Lynn Beighley and Michael Morrison, Head First PHP & MySQL, First Edition, O'Reill Publication, 2009.

Web References:

1. <https://github.com/PHPMailer/PHPMailer> [PHP Mailer Code]
2. <https://www.php.net/> [Official website of PHP]
3. <https://www.geeksforgeeks.org/php-tutorials/> [Lecture notes of PHP]
4. <https://www.w3schools.com/php/default.asp> [Lecture notes of PHP]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Gain understanding of HTML and Java Script.
CO2 :	To utilize knowledge and skills for basics of PHP and functions of PHP
CO3 :	Acquire the knowledge of array and OOP in PHP
CO4 :	Learn Form handling and utilities in PHP.
CO5 :	Be able to develop dynamic web-based application using PHP and MySQL with Ajax Call.

CAUE405 : Python Framework

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic concepts of Python Programming.

Methodology, Pedagogy & Andragogy: This course focuses on providing hands-on experience to students for design and develop web application using Django framework and MVT pattern integration.

Outline of the Course:

Week No	Practical	Description
1	Downloading and Installation: Installation of Python and Python documentation. Creating a virtual environment and installing Django for creating first web project.	Learn basic concepts virtual environment then creating and executing first web project of Django and understanding directory structure.
2	Creating first web application in web project.	Get familiar with different files (settings.py, urls.py, manage.py, models.py etc.) with their usages and then executing first application.
3	Creating multiple web applications within same web project.	Learning mappings of Django web project.
4	Django views: Writing different types of views.	Able to display the output using different types of views.
5	Creating first template and developing MVT based application.	Learn create and add a template in web project then understand the mapping of MVT based web application.
6	Integrating templates and views for developing MVT based applications.	Able to write Python code in a template language and developing MVT based applications for dynamic content generation.

7	Model and Database Setup for Django App	Setting super user credentials and using Django's admin panel for database applications
8	Creating web application with database connectivity.	Using code first approach, developing web application with database connectivity and managing it with Django Admin Panel.
9	Django Forms	Learn Django forms for taking user input from the web applications.
10	Django Forms	Developing full fledged web applications for CRUD operation implementation.
11	Django & REST APIs	Understanding Django REST Framework for building web APIs.
12	Unit Testing with Django	First, learning how to write tests with Django then executing them using Python standard library module: unittest.
13	Unit Testing with Django	Writing different test cases for various applications ensuring the reliability and correctness.
14	Admin, Security, Performance & Optimization	Applying different ways like django-debug-toolbar, Third-party services etc. for Performance Optimization.
15	Admin, Security, Performance & Optimization	Learning different types of security measures like XSS Protection, CSRF Protection, SQL injection protection etc.

Total hours: 60

Core Books:

1. Adrian Holovaty, Jacob K. Moss: The Definitive Guide to Django: Web Development Done Right
2. Jeff Forcier, Paul Bissex, Wesley Chun: Python Web Development with Django, Pearson Education, 2008.
3. Dusty Philips: Python 3 Object oriented Programming, PACKT publishing, 2010

Reference Books:

1. Adrian Holovaty, Jacob Kaplan-Moss, et al.: The Django Book, 2013
2. Nathan Yergler: Effective Django, 2015
3. Steve Holden: Python Web Programming, 1st edition, 2002.

Web References:

1. <https://www.djangoproject.com/start/> [For Django Projects]
2. <https://djangoforbeginners.com/initial-setup/> [For Django Setups]
3. <https://docs.djangoproject.com/en/2.1/intro/tutorial01/> [For Django app]

Course Outcome: Upon successful completion of the course, student will:

CO1 :	Understand Django fundamentals and use its concepts to build and deploy robust web applications and apps.
CO2 :	Learn about Django URL patterns and views and deploy Django applications
CO3 :	Learn how to configure Django for powerful databases and create the Django admin interface.
CO4 :	Learn about Django's security implications and how to create safe web applications with it.
CO5 :	To get some idea about unit testing concepts and its implementation through Django

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	2	3	2	1	1	1	3	2
CO2	2	3	3	3	3	2	3	2	1	1	1	3	2
CO3	2	3	3	3	3	2	3	3	2	1	2	3	2
CO4	3	3	3	3	3	3	3	3	2	2	3	3	2
CO5	3	3	3	3	3	3	3	3	1	2	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE406: HTTP Web service for Enterprise Applications

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Prerequisites:

Expected to have foundational knowledge from CAUC102 – Fundamentals of Web Designing, CAUC302 – Fundamentals of Visual Programming, CAUC205 – Open Source Technology and Web Application Development using .NET Framework. This skillset will ensure a smoother transition into advanced Web API concepts and practical implementations.

Methodology & Pedagogy:

In the sessions, students will actively develop Web APIs using C#, Entity Framework, and SQL Server through structured, real-world assignments. They will explore API testing and debugging using industry tools like Postman, Fiddler, and browser-based tools. JQuery and AJAX integration will enhance client-side interaction and API consumption skills. Each session fosters hands-on learning, problem solving, and application of best practices in a project-based environment.

B. Outline of the Course:

Week No	Practical	Description
1	Basic Web API with GET Method: A Web API using the GET method retrieves data from the server. The GET method does not change the server state and is used to fetch data.	Create an API endpoint that returns a simple welcome message. Test it via browser, Postman, and Fiddler to observe the HTTP request and response.
	Web API with POST Method: POST is used to submit data to be processed to a specified resource.	Develop an API endpoint that accepts name and age from the user and returns a response confirming the received data.
	GET/POST Methods with Postman: Postman is a tool for testing API requests including GET, POST, PUT, and DELETE.	Use Postman to manually send GET and POST requests to your API, and observe headers, request body, and response.

2	GET/POST Methods with Fiddler: Fiddler is a web debugging proxy tool to inspect traffic between computer and internet.	Capture and analyze Web API HTTP requests and responses using Fiddler.
	Web API Project Explaining REST Principles: REST (Representational State Transfer) is an architectural style using stateless communication and resource-based URLs.	Develop a RESTful Web API that demonstrates statelessness, a uniform interface, and resource-based routing.
	Web Services, WCF, and Web API: Compare different technologies used for building web-based communication services.	Create a comparative table highlighting architecture, protocol support, performance, and hosting model differences.
3	Web API with GET, POST, PUT, DELETE Methods: These are the primary HTTP verbs used for CRUD operations.	Implement a controller with all CRUD methods without database persistence.
	Integrate Web API with Entity Framework (Code First): EF Code First approach allows defining models in code, which EF uses to generate database schema.	Develop a Web API project with CRUD operations using EF Code First and connect to SQL Server.
	Integrate Web API with EF (Database First): EF Database First allows generating model classes from an existing database.	Build a Web API that connects to an existing database schema and provides CRUD functionalities.
4	Use ActionResult for Different Return Types: ActionResult allows an API method to return different content types like string, JSON, or XML.	Implement multiple methods that return various types of responses depending on Accept headers.
	Web API and MVC Controller: Web API and MVC are both controllers, but Web API is RESTful, while MVC serves web pages.	Create both controller types and explain differences in routing, responses, and return types.

	Use [Route] Attribute in API Controller: The [Route] attribute allows defining custom URL paths for action methods.	Apply [Route] and [HttpGet]/[HttpPost] to define and test clean and descriptive routes.
5	Action Method Naming Conventions in Web API: Method names like Get, Post, Put, Delete map to corresponding HTTP verbs.	Demonstrate method naming patterns and how they impact API behavior.
	MVC App Consuming Web API (Read-Only View): MVC application can act as a front-end client to Web API.	Use MVC controller to call Web API and display GET data in a view.
	Implement Convention-Based Routing: Routing based on URL patterns configured in WebApiConfig.	Set default routes like "api/{controller}/{id}" and test endpoint behavior.
6	Implement Attribute-Based Routing: Routing defined directly above action methods using [Route] attribute.	Use attribute routing to explicitly map methods to specific URLs.
	Custom Parameterized Routing in Web API: URL segments can be used as parameters for dynamic API behavior.	Create APIs like /api/product/{id} to fetch or process data based on route values.
	Multiple Route Templates in One Controller: Use multiple [Route] attributes on a single method for flexible API access.	Define alternate route paths that map to the same logic, improving usability.
7	Use JSON and XML Formatters: Formatters serialize data into requested formats based on Accept header.	Return different formats from API methods by testing with varied Accept headers.
	Register a Custom Media Type Formatter: Custom formatter defines how data is serialized and returned to clients.	Implement and register a formatter for a custom content type like CSV.
8	Implement HttpResponseMessage in	Use HttpResponseMessage to return meaningful status codes and error messages.

	Web API: This exception is thrown for HTTP error scenarios.	
	Exception Filter Attribute: Filters catch unhandled exceptions at a global or controller level.	Write a custom ExceptionFilter and apply it for centralized error handling.
9	Use HttpError to Return Custom Error Message: HttpError sends detailed error information in the response body.	Use HttpError inside catch blocks for structured error responses.
	Register Exception Filter Globally in WebApiConfig: Global filters apply across all controllers.	Register custom ExceptionFilter globally to handle exceptions uniformly.
10	Model Validation using Data Annotations: Validation rules defined using attributes like [Required], [Range].	Apply data annotations to model properties and return validation errors.
	Custom Validation Attributes for Complex Rules: User-defined attribute for validating complex input logic.	Implement attributes like [EmailDomain] and validate with business logic.
11	Use Custom Exception Class: Custom exception extends base Exception class for domain-specific errors.	Throw and catch these exceptions for controlled error reporting.
	Handle ModelState Errors in Web API: Model State contains validation error information.	Check ModelState.IsValid and return all validation errors if false.
12	Backend Validation using Try-Catch and HttpError: Exception handling using try-catch ensures robust error response.	Wrap logic in try-catch and return error details using HttpError.
	Global Error Logging with Custom Exception Filter: Logs unhandled exceptions	Extend ExceptionFilter to log exception details using tools like ILogger.

	centrally using logging frameworks.	
13	Consume Web API GET using jQuery AJAX: AJAX allows asynchronous web requests using JavaScript.	Use jQuery's \$.ajax or \$.get to fetch data and show it in an HTML table.
	Consume Web API POST using jQuery AJAX: Use AJAX to send form data to Web API without page reload.	Collect form inputs and use \$.post or \$.ajax with type 'POST'.
14	Perform PUT and DELETE Operations via AJAX: PUT updates data, DELETE removes it. AJAX can perform both.	Create buttons triggering AJAX requests to update or delete records.
	Display API JSON Response in HTML Format: JSON is the default data exchange format.	Fetch JSON data from API and parse/display it using jQuery DOM manipulation.
15	AJAX Form Submission with Validation Feedback: Combines client-side validation with AJAX submission.	Validate form, show errors or success, and post data to API via jQuery.
	Use jQuery to Dynamically Populate DropDown from Web API: Populate UI controls with data from API dynamically.	On page load, call API to get options and populate <select> using jQuery.
Total hours: 60		

Core Books:

1. Jamie Kurtz, Brian Wortman: ASP.NET Web API 2: Building REST Services with .NET Framework, Apress
2. Glenn Block, Pablo Cibraro: Professional ASP.NET Web API, Wrox
3. Adam Freeman: Pro ASP.NET MVC 5, Apress
4. Dino Esposito: Programming Microsoft ASP.NET MVC, Microsoft Press
5. Jon P. Smith: Entity Framework Core in Action, Manning Publications

Reference Books:

1. Leonard Richardson, Mike Amundsen, Sam Ruby: RESTful Web APIs, O'Reilly Media
2. Dino Esposito: Microsoft ASP.NET and AJAX: Architecting Web Applications, Microsoft Press
3. Badrinarayanan Lakshmiraghavan: Pro ASP.NET Web API Security, Apress

Web References:

1. <https://learn.microsoft.com/en-us/aspnet/web-api/> [ASP.NET Web API (Legacy) Documentation]
2. <https://learn.microsoft.com/en-us/aspnet/core/web-api/> [ASP.NET Core Web API (Latest Version)]
3. <https://learn.microsoft.com/en-us/ef/> [Entity Framework Documentation (EF Core & EF 6)]
4. <https://learn.microsoft.com/en-us/training/modules/build-web-api-aspnet-core/> [Microsoft Learn – Web API Beginner's Learning Path]
5. <https://www.tutorialsteacher.com/webapi> [TutorialsTeacher – Web API Tutorials]
6. <https://www.dotnettricks.com/learn/webapi> [Dot Net Tricks – ASP.NET Web API]
7. <https://www.c-sharpcorner.com/technologies/web-api> [C# Corner – ASP.NET Web API Articles]

Course Outcome: Upon successful completion of the course, student will:

CO1 :	Understand the basics of Web API, REST principles, and testing using Postman and Fiddler.
CO2 :	Develop CRUD operations in Web API with and without Entity Framework integration.
CO3 :	Configure routing in Web API using convention-based, attribute-based, and parameterized routes
CO4 :	Implement formatters, filters, and exception handling for structured Web API responses
CO5 :	Integrate jQuery and AJAX to consume Web API services with client-side validation

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	3	3	1	3	-	-	2	2	2	2	2	2
CO2	2	1	2	2	3	2	2	3	-	3	2	3	3
CO3	2	2	2	2	3	2	2	2	-	1	2	3	2
CO4	3	3	3	3	2	3	2	2	-	2	3	3	2
CO5	2	3	3	1	3	2	2	3	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUE407 : Game Development Using Unity

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
02	Practical	04	60	40	100
Experiential Learning Components (Practical): Lab Assignments, Practical Test, VIVA					

Prerequisites: Basic concepts of object oriented programming language - c#.

Methodology, Pedagogy & Andragogy: This course focuses on experiential learning using the Unity game engine. Students will engage in hands-on activities to design and develop 2D games using Unity and basic C# scripting. Emphasis is on visual development, animation, game logic, and interactive content creation.

C. Outline of the Course:

Week No	Practical	Description
1	Unity setup	Install Unity and Visual Studio; Explore Unity Interface; Create a 2D Project; Add GameObjects.
2	C# Basics	Introduction to C# syntax: variables, data types, functions, and basic movement script.
3	Object Manipulation	Manipulate GameObjects using Transform; Work with Vector2/Vector3; Group and prefab objects
4	Physics Setup	Rigidbody2D and Collider2D setup; Physics Materials and gravity simulation
5	Player Control	Implement player movement and jump with keyboard input; Clamp player position

6	Sprite Handling	Import sprite sheets; Setup sorting layers; Create parallax backgrounds.
7	Animation	Create basic animations with Animator; Transitions between states; Trigger with script
8	Collision & Scoring	Detect collisions and triggers; Create collectible objects; Implement scoring system.
9	Audio & UI	Add background music and sound effects; Design UI elements like score and health.
10	Game Loop	Develop basic game states (start/play/pause/over); Manage game loop and UI controls.
11	GameObject Lifecycle	Unity GameObject lifecycle; Script order (Awake, Start, Update); Optimize asset usage
12	Game Build	Build and export game (Windows/WebGL); Create splash screen and credits screen
13	Project Planning	Mini project planning and design approval
14	Project Development	Mini project development and internal testing
15	Final Submission	Final game polishing, presentation, and viva

Total hours: 60

Core Books:

1. John Horton: Beginning C# Game Programming, Packt Publishing, 2016.
2. Harrison Ferrone: Learning C# by developing Games with Unity 2021, Packt Publishing, 5th Edition.
3. Jesse Freeman: Introduction to Game Development using C# and Unity, Apress, 2014.

Reference Books:

1. Eric Lengyel: Foundations of Game Engine Development – Vol 1: Mathematics, 2017
2. Alan Thorn: Unity Game Development Cookbook, Packt Publishing.
3. Jonathan Weinberger: Unity 2021 Mobile Game Development, Unity certification associate guide.

Web References:

1. <https://learn.unity.com/> [For Unity tutorials and learning paths]
2. <https://docs.unity3d.com/Manual/index.html> [For Unity Documentation]
3. <https://www.raywenderlich.com/unity> [For Unity game development guide and articles]
4. <https://geeksforgeeks.org/unity/> [For beginner tutorials for Unity]

Course Outcome: Upon successful completion of the course, student will:

CO1 :	Understanding the Unity editor interface and core game development workflow.
CO2 :	Develop basic 2D games using GameObjects, physics and C# scripting
CO3 :	Animate and control objects with logic and input
CO4 :	Design Game UI, scoring systems and simple audio integration
CO5 :	Export a playable game prototype and present it with confidence

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	3	2	2	2	2	2	2	2	3	3
CO2	3	3	3	3	2	2	2	2	2	2	2	3	3
CO3	3	3	3	3	2	2	2	2	2	2	2	3	3
CO4	2	2	2	3	2	2	2	2	2	2	2	2	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CAUP402 : On-Job Training-II

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP-2020 Classification
			CCE	SEE	CCE	SEE		
CAUP402	On-Job Training-II	Viva	-	-	75	75	06	Research Project/Dissertation

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
06	VIVA	12	180	270	450

Pre-requisite: None

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Apply advanced IT skills (e.g., system integration, full module development, or research methodologies) to solve industry-relevant problems independently or in teams.
CO2 :	Lead or contribute significantly to team-based IT projects, demonstrating advanced collaboration and problem-solving skills.
CO3 :	Optimize and implement IT solutions using advanced tools or techniques (e.g., cloud platforms, APIs, or emerging frameworks).
CO4 :	Adhere to professional standards, including project management, confidentiality, and ethical considerations in IT environments.
CO5 :	Deliver comprehensive project documentation and presentations, showcasing employability and technical expertise.

- **Refer Annexure-I for more information.**

CAUR402 : Research Project-II

Course Code	Course Name	Theory / Practical / Others	Marks (Theory)		Marks (Practical)		Credit	NEP-2020 Classification
			CCE	SEE	CCE	SEE		
CAUR402	Research Project-II	Viva	-	-	75	75	06	Research Project/Dissertation

Credit	Component	Instruction/Contact Hours (Per Week)	Instruction /Contact Hours (Per Course)	Experiential Learning Hours (Per Course)	Total Notional Hours
06	VIVA	12	180	270	450

Pre-requisite: None

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Execute a research project or dissertation, applying methodologies developed in Research Internship-I to address the research question.
CO2 :	Analyze research data or results using appropriate IT tools (e.g., Python, R, statistical software) to derive meaningful conclusions.
CO3 :	Communicate research findings effectively through a dissertation/thesis and oral presentation, adhering to academic standards.
CO4 :	Ensure originality and ethical conduct in research outputs, including compliance with plagiarism policies and intellectual property rights.
CO5 :	Evaluate the significance and potential impact of research findings on IT practices or interdisciplinary applications.

- **Refer Annexure-I for more information.**

Annexure - I

Guidelines for Research Internship/On-Job Training (OJT) for BCA Programme

1.0 Introduction

In alignment with the National Education Policy (NEP) 2020 and the University Grants Commission (UGC) Guidelines for Internship/Research Internship/On-Job Training (OJT), the Bachelor of Computer Applications (BCA) programme at CMPICA, CHARUSAT, integrates practical training and research opportunities in Semesters VII and VIII. This document outlines the structure, objectives, and operational framework for:

- **On-the-Job Training (OJT):** For BCA Honours Honours, students to enhance employability through industry exposure.
- **Research Internship:** For BCA Honours Honours with Research, students to develop research aptitude and innovation in computer applications.

These guidelines ensure that the students are equipped for professional careers or advanced studies, contributing to NEP 2020's vision of a research-oriented, skill-driven education system.

2.0 Objectives

2.1 OJT Objectives

1. Enhance employability through practical IT skills (e.g., programming, web development).
2. Bridge classroom learning with workplace practices.
3. Develop professional competencies like teamwork and decision-making.
4. Expose students to emerging technologies and industry trends.
5. Encourage entrepreneurial thinking and job readiness.

2.2 Research Internship Objectives

1. Develop research aptitude using tools and methodologies in computer applications.
 2. Enable students to solve complex IT problems through inquiry and analysis.
 3. Enhance analytical skills (e.g., data analysis, model development).
 4. Promote ethical research practices and awareness of intellectual property rights.
 5. Prepare students for postgraduate research or doctoral studies.
-

3.0 Categories

As per UGC guidelines, the research internship/OJT is classified into two types for BCA students:

1. **OJT for Enhancing Employability:** Practical training in IT industries (e.g., software firms, start-ups) or in-house at CHARUSAT to build job-ready skills (BCA Honours).
2. **Research Internship for Developing Research Aptitude:** Research projects/dissertations under faculty or industry mentors, including multidisciplinary topics (BCA Honours with Research).

4.0 Structure

The internship structure for BCA students applies to Semesters VII and VIII with the following general requirements:

- **Credit Allocation:** 12 credits (6 credits/semester, 15-week semester); 1 credit = 30 hours.
- **Eligibility:** Completion of Semesters I-VI (132 credits) and prerequisite courses.
- **Flexibility:** Summer/winter breaks can extend hours beyond the 6-week period without affecting credits
- **Portal:** CMPICA digital portal with API integration (to be developed)
- **Mode:** Physical, digital, or hybrid

Aspect	OJT (BCA Honours)	Research Internship (BCA Honours with Research)
Total Duration	Semesters VII – 6 Credit Course Semesters VIII – 6 Credit Course 360 hours (180 hours/semester, including a 6-week intensive period at 30 hours/week)	Semesters VII – 6 Credit Course Semesters VIII – 6 Credit Course 360 hours (Semester VII: methodology, 180 hours over 6 weeks; Semester VIII: dissertation/thesis, 180 hours over 6 weeks)
Specifics	Hands-on training in software development, database management, networking; includes in-house options via CHARUSAT IT Cell	Research methodology (Semester VII) and dissertation/thesis (Semester VIII); may include multidisciplinary topics

Collaboration	MOUs with IT firms (local/MNCs) and in-house CHARUSAT IT Cell	Partnerships with research labs (e.g., DA-IICT) and CHARUSAT Institutions (e.g., Depstar, PDPIAS)
Learning Outcomes	- Technical proficiency (e.g., coding, networking). - Problem-solving. - Teamwork. - Professional ethics etc.	- Research techniques (e.g., data analysis, tool usage). - Complex problem-solving. - Communication of findings. - Ethical conduct etc.

5.0 Roles and Responsibilities

5.1 Role of Internship Providing Organization (IPO)

- **OJT:** IT firms, startups, service providers, or in-house CHARUSAT IT Cell offering practical training and mentorship.
- **Research Internship:** HEIs, research labs, IT companies, or in-house CHARUSAT units providing research opportunities and resources.
- Issue completion certificates for both tracks.

5.2 Role of Nodal Officer

- Coordinate with IPOs (external and in-house), sign MOUs, and maintain the CMPICA internship portal.
- Register students, supervisors, and mentors for both OJT and Research Internship.
- Oversee both tracks, ensuring smooth execution and addressing stakeholder concerns.

5.3 Role of Supervisor

- **OJT:**
 - Monitor attendance and task completion (external or CHARUSAT-based supervisor).
 - Evaluate performance and certify completion.
- **Research Internship:**
 - Oversee research progress (e.g., milestones, methodology adherence) by faculty or research mentor.
 - Assess reports and conduct viva-voce.

5.4 Role of Mentor

- **OJT:**
 - Guide students in technical tasks and validate skill acquisition (industry or CHARUSAT mentor).
- **Research Internship:**

- Provide research guidance, validate outcomes, and certify reports (faculty or interdisciplinary mentor).

6.0 Mechanism

1. **Application:** Students apply via the *CMPICA portal or directly to IPOs (including CHARUSAT IT Cell)* with mentor approval.
2. **Selection:** IPOs select based on their criteria; digital or in-house alternatives provided if needed.
3. **Execution:**
 - **OJT:** Practical training (Semesters VII and VIII, 180 hours each), including in-house training through CHARUSAT IT Cell or similar units.
 - **Research Internship:** Methodology coursework (Semester VII) and dissertation (Semester VIII, 360 hours total), with potential multidisciplinary topics explored via consultation with *in-house CHARUSAT organizations (e.g., engineering, pharmacy, or sciences)*.
 - Fortnightly progress reports for both tracks.
4. **Group Projects:** Allowed (2-5 students) (e.g., OJT: app development; Research: AI-healthcare integration).
5. **Digital Backup:** Virtual training or simulation-based research if physical options are unavailable.

7.0 Evaluation

Internships will be evaluated as follows:

Component	CCE (Internal) - 50 Marks	SEE (Final Exam) - 50 Marks
Weekly Logs & Attendance	14 Marks (20% of CCE)	-
Mid-Term Report & Evaluation	23 Marks (30% of CCE)	-
Final Report	38 Marks (50% of CCE)	50 Marks (Report: OJT Training Report / Research Dissertation)
Presentation & Viva-Voce	-	25 Marks (Assessed by a committee with one internal and one external examiner)

- **Assessment Criteria:**
 - **OJT:** Skill acquisition, report quality, attendance, and project outcomes.
 - **Research Internship:** Innovativeness, originality, research skills, and significance of findings.
- **Research Specifics:** Report includes an originality certificate (no plagiarism), signed by student and mentor.

- **Passing Criteria:** Minimum 36% in each component (*CHARUSAT norms as per NEP 2020*).
-

8.0 Technical Specifications for Report

- **Content** (*detailed template for the final report will be provided to the students*):
 - **OJT:** Introduction, tasks performed, skills learned, and outcomes.
 - **Research Internship:** Research question, methodology, results, and conclusions.
 - **Format:** A4 size, 12 pt. Times New Roman, 1.5-line spacing, 1-inch margins, spiral-bound, printed on both sides.
 - **Submission:** Two hard copies (one for FCA - CMPICA, one for student) + digital copy, 10 days before viva-voce.
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